Northern Nut Growers Association Report of the Proceedings at the 44th Annual Meeting

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44th Annual Report

OF THE

Northern Nut Growers Association

Incorporated

AFFILIATED WITH THE AMERICAN HORTICULTURAL SOCIETY

* * * * *

Annual Meeting at

ROCHESTER, NEW YORK

August 31—September 1, 1953

[Illustration: NORTHERN NUT GROWERS ASSOC.

ROCHESTER N.Y.-1953]

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Officers for 1953-54

President Richard B. Best, Eldred, Illinois
Vice-President Gilbert Becker, Climax, Michigan
Secretary Spencer B. Chase, Knoxville, Tennessee
Treasurer William S. Clarke, Jr., State College, Pennsylvania
Directors Dr. L. H. MacDaniels, Ithaca, New York
Dr. William Rohrbacher, Iowa City, Iowa
Dean of the Association Dr. W. C. Deming, Litchfield, Connecticut

EXECUTIVE APPOINTMENTS 1953-54

Program Committee:

Dr. Lloyd L. Dowell, Royal Oakes, Dr. J. W. McKay, Roy D. Anthony, J. G. McDaniel, Lewis E. Theiss, W. B. Ward.

Local Arrangements:

Mrs. Herbert Krone, R. P. Allaman, John Rick, Elwood B. Miller, Victor Brook.

Place of Meeting Committee:

To explore meeting places for three years, Michigan and Connecticut as possible places for the 1955 and 1956 annual conventions. W. M. Beckert, R. P. Allaman, Carl Prell, Lloyd L. Dowell.

Publication Committee:

Professor George L. Slate, Dr. L. H. MacDaniels.

Varieties and Contests Committee:

J. C. McDaniel, Dr. L. H. MacDaniels, Sylvester M. Shessler, H. F. Stoke, Royal Oakes.

Standards and Judging Committee: (Section of Varieties Committee)

Dr. L. H. MacDaniels, Dr. H. L. Crane, Louis Gerardi, Spencer B. Chase, Professor Paul E. Machovina.

Survey and Research Committee:
H. F. Stoke (With all state and foreign vice-presidents).

Exhibits Committee:
Dr. L. H. MacDaniels, Fayette Etter, H. F. Stoke, Royal Oakes, J. F. Wilkinson, G. J. Korn.

Understock Committee:
J. C. McDaniel, Albert B. Ferguson, Dr. Aubrey Richards, Louis Gerardi, Dr. A. S. Colby, Max Hardy, Gilbert L. Smith.

Auditing Committee:
Raymond E. Silvis, Sterling A. Smith, Edward W. Pape.

Legal Advisor:
Sargent H. Wellman, Esq.

Finance Committee:
Carl F. Prell, Ford Wallick, Sterling A. Smith.

Necrology Committee:
Mrs. H. L. Crane, Mrs. C. A. Reed, Mrs. Wm. J. Wilson.

Nominating Committee:
(Elected at Rochester, N. Y.) Paul E. Machovina, Raymond Silvis, George Salzer, Dr. H. L. Crane, Ira M. Kyhl.

Membership Committee:
Gilbert Becker, Raymond E. Silvis, Edward W. Pape, Gordon Pulliam, Hon. Paul C. Daniels, Max B. Hardy.

Publicity Committee:
Paul E. Machovina, Wm. J. Wilson, Carl F. Prell, Frank M. Kintzel.

Change of Name Committee:
Elwood B. Miller, John Davidson, Dr. J. W. McKay, Dr. H. L. Crane.

State and Foreign Vice-Presidents

Alabama Edward L. Hiles, Loxley
Alberta A. L. Young, Brooks
Arkansas W. D. Wylie, Univ. of Ark., Fayetteville
Belgium R. Vanderwaeren, Bierbeekstraat, 310, Korbeek-Lo
British Columbia,
Canada J. U. Gellatly, Box 19, Westbank
California Thos. R. Haig, M.D., 3021 Highland Ave., Carlesbad
Colorado J. E. Forbes, Julesburg
Connecticut A. M. Huntington, Stanerigg Farms, Bethel
Delaware Lewis Wilkins, Route 1, Newark
Denmark Count F. M. Knuth, Knuthenborg, Bandholm
District of
Columbia Ed. L. Ford, 3634 Austin St., S.E. Washington 20
Florida C. A. Avant, 960 N. W. 10th Ave., Miami
Georgia William J. Wilson, North Anderson Ave., Fort Valley
Hawaii John F. Cross, P. O. Box 1720, Hilo
Hong Kong P. W. Wang, 6 Des Voeux Rd., Central
Idaho Lynn Dryden, Peck
Illinois Royal Oakes, Bluffs (Scott County)
Indiana Edw. W. Pape, Rt. 2, Marion
Iowa Ira M. Kyhl, Box 236, Sabula
Kansas Dr. Clyde Gray, 1045 Central Ave., Horton
Kentucky Dr. C. A. Moss, Williamsburg
Louisiana Dr. Harald E. Hammar, 608 Court House, Shreveport

Maryland Blaine McCollum, White Hall
Massachusetts S. Lothrop Davenport, 24 Creeper Hill Rd., North Grafton
Michigan Gilbert Becker, Climax
Minnesota R. E. Hodgson, Southeastern Exp. Station, Waseca
Mississippi James R. Meyer, Delta Branch Exp. Station, Stoneville
Missouri Ralph Richterkessing, Route 1, Saint Charles
Montana Russel H. Ford, Dixon
Nebraska Harvey W. Hess, Box 209, Hebron
New Hampshire Matthew Lahti, Locust Lane Farm, Wolfeboro
New Jersey Mrs. Alan R. Buckwalter, Route 1, Flemington
New Mexico Rev. Titus Gehring, P. O. Box 177, Lumberton
New York Stephen Bernath, Route No. 3, Poughkeepsie
North Carolina Dr. R. T. Dunstan, Greensboro College, Greensboro
North Dakota Homer L. Bradley, Long Lake Refuge, Moffit
Ohio Christ Pataky Jr., 592 Hickory Lane, Route 4, Mansfield
Oklahoma A. G. Hirschi, 414 North Robinson, Oklahoma City
Ontario, Canada Elton E. Papple, Cainsville
Oregon Harry L. Pearcy, Route 2, Box 190, Salem
Pennsylvania R. P. Allaman, Route 86, Harrisburg
Prince Edward Is. Canada Robert Snazelle, Forest Nursery, Route 5, Charlottetown
Rhode Island Philip Allen, 178 Dorance St., Providence
South Carolina John T. Bregger, P. O. Box 1018, Clemson
South Dakota Herman Richter, Madison
Tennessee W. Jobe Robinson, Route 7, Jackson
Texas Kaufman Florida, Box 154, Rotan
Utah Harlan D. Petterson, 2076 Jefferson Ave., Ogden

Vermont A. W. Aldrich, R. F. D. 2, Box 266, Springfield
Virginia H. R. Gibbs, Linden
Washington H. Lynn Tuttle, Clarkston
West Virginia Wilbert M. Frye, Pleasant Dale
Wisconsin C. F. Ladwig, 2221 St. Lawrence, Beloit

CONSTITUTION
of the
NORTHERN NUT GROWERS ASSOCIATION, INCORPORATED
(As adopted September 13, 1948)

NAME

ARTICLE I. This Society shall be known as the Northern Nut Growers Association, Incorporated. It is strictly a non-profit organization.

PURPOSES

ARTICLE II. The purposes of this Association shall be to promote interest in the nut bearing plants; scientific research in their breeding and culture; standardization of varietal names; the dissemination of information concerning the above and such other purposes as may advance the culture of nut bearing plants, particularly in the North Temperate Zone.

MEMBERS

ARTICLE III. Membership in this Association shall be open to all persons interested in supporting the purposes of the Association. Classes of members are as follows: Annual members, Contributing members, Life members, Honorary members, and Perpetual members. Applications for membership in the Association shall be presented to the secretary or the treasurer in writing, accompanied by the required dues.

OFFICERS

ARTICLE IV. The elected officers of this Association shall consist of a President, a Vice-president, a Secretary and a Treasurer or a combined Secretary-treasurer as the Association may designate.

BOARD OF DIRECTORS

Article V. The Board of Directors shall consist of six members of the Association who shall be the officers of the Association and the two preceding elected presidents. If the offices of Secretary and Treasurer are combined, the three past presidents shall serve on the Board of Directors.

There shall be a State Vice-president for each state, dependency, or country represented in the membership of the Association, who shall be appointed by the President.

AMENDMENTS TO THE CONSTITUTION

ARTICLE VI. This constitution may be amended by a two-thirds vote of the members present at any annual meeting, notice of such amendment having been read at the previous annual meeting, or copy of the proposed amendments having been mailed by the Secretary, or by any member to each member thirty days before the date of the annual meeting.

BY-LAWS

(Revised and adopted at Norris, Tennessee, September 13, 1948)

SECTION I.—MEMBERSHIP

Classes of membership are defined as follows:

ARTICLE I. ANNUAL MEMBERS. Persons who are interested in the purposes of the Association who pay annual dues of Three Dollars ($3.00).

ARTICLE II. Contributing members. Persons who are interested in the purposes of the Association who pay annual dues of Ten Dollars ($10.00) or more.

ARTICLE III. LIFE MEMBERS. Persons who are interested in the purposes of the Association who contribute Seventy-Five Dollars ($75.00) to its support and who shall, after such contribution, pay no annual dues.

ARTICLE IV. HONORARY MEMBERS. Those whom the Association has elected as honorary members in recognition of their achievements in the special fields of the Association and who shall pay no dues.

ARTICLE V. PERPETUAL MEMBERS. "Perpetual" membership is eligible to any one who leaves at least five hundred dollars to the

Association and such membership on payment of said sum to the Association shall entitle the name of the deceased to be forever enrolled in the list of members as "Perpetual" with the words "In Memoriam" added thereto. Funds received therefor shall be invested by the Treasurer in interest bearing securities legal for trust funds in the District of Columbia. Only the interest shall be expended by the Association. When such funds are in the treasury the Treasurer shall be bonded. Provided: that in the event the Association becomes defunct or dissolves, then, in that event, the Treasurer shall turn over any funds held in his hands for this purpose for such uses, individuals or companies that the donor may designate at the time he makes the bequest of the donation.

SECTION II.—DUTIES OF OFFICERS

ARTICLE I. The President shall preside at all meetings of the Association and Board of Directors, and may call meetings of the Board of Directors when he believes it to be the best interests of the Association. He shall appoint the State Vice-presidents; the standing committees, except, the Nominating Committee, and such special committees as the Association may authorize.

ARTICLE II. Vice-president. In the absence of the President, the Vice-president shall perform the duties of the President.

ARTICLE III. Secretary. The Secretary shall be the active executive officer of the Association. He shall conduct the correspondence relating to the Association's interests, assist in obtaining memberships and otherwise actively forward the interests of the Association, and report to the Annual Meeting and from time to time to meetings of the Board of Directors as they may request.

ARTICLE IV. Treasurer. The Treasurer shall receive and record memberships, receive and account for all moneys of the Association and shall pay all bills approved by the President or the Secretary. He shall give such security as the Board of Directors may require or may legally be required, shall invest life memberships or other funds as the Board of Directors may direct, subject to legal restrictions and in accordance with the law, and shall submit a verified account of receipts and disbursements to the Annual meeting and such current accounts as the Board of Directors may from time to time require. Before the final business session of the Annual Meeting of the Association, the accounts of the Treasurer shall be submitted for examination to the Auditing Committee appointed by the President at the opening session of the Annual Meeting.

ARTICLE V. The Board of Directors shall manage the affairs of the association between meetings. Four members, including at least two elected officers, shall be considered a quorum.

SECTION III.—ELECTIONS

ARTICLE I. The Officers shall be elected at the Annual Meeting and hold office for one year beginning immediately following the close of the Annual Meeting.

ARTICLE II. The Nominating Committee shall present a slate of officers on the first day of the Annual Meeting and the election shall take place at the closing session. Nominations for any office may be presented from the floor at the time the slate is presented or immediately preceding the election.

ARTICLE III. For the purpose of nominating officers for the year 1949 and thereafter, a committee of five members shall be elected annually at the preceding Annual Meeting.

ARTICLE IV. A quorum at a regularly called Annual Meeting shall be fifteen (15) members and must include at least two of the elected officers.

ARTICLE V. All classes of members whose dues are paid shall be eligible to vote and hold office.

SECTION IV.—FINANCIAL MATTERS

ARTICLE I. The fiscal year of the Association shall extend from October 1st through the following September 30th. All annual memberships shall begin October 1st.

ARTICLE II. The names of all members whose dues have not been paid by
January 1st shall be dropped from the rolls of the Society. Notices of
non-payment of dues shall be mailed to delinquent members on or about
December 1st.

ARTICLE III. The Annual Report shall be sent to only those members who have paid their dues for the current year. Members whose dues have not been paid by January 1st shall be considered delinquent. They will not be entitled to receive the publication or other benefits of the Association until dues are paid.

SECTION V.—MEETINGS

ARTICLE I. The place and time of the Annual Meeting shall be selected by the membership in session or, in the event of no selection being made at this time, the Board of Directors shall choose the place and time for the holding of the annual convention. Such other meetings as may seem desirable may be called by the President and Board of Directors.

SECTION VI.—PUBLICATIONS

ARTICLE I. The Association shall publish a report each fiscal year and such other publications as may be authorized by the Association.

ARTICLE II. The publishing of the report shall be the responsibility of the Committee on Publications.

SECTION VII.—AWARDS

ARTICLE I. The Association may provide suitable awards for outstanding contributions to the cultivation of nut bearing plants and suitable recognition for meritorious exhibits as may be appropriate.

SECTION VIII.—STANDING COMMITTEES

As soon as practical after the Annual Meeting of the Association, the President shall appoint the following standing committees:

1. Membership 2. Auditing 3. Publications 4. Survey 5. Program 6. Research 7. Exhibit 8. Varieties and Contests

SECTION IX.—REGIONAL GROUPS AND AFFILIATED SOCIETIES

ARTICLE I. The Association shall encourage the formation of regional groups of its members, who may elect their own officers and organize their own local field days and other programs. They may publish their proceedings and selected papers in the yearbooks of the parent society subject to review of the Association's Committee on Publications.

ARTICLE II. Any independent regional association of nut growers may affiliate with the Northern Nut Growers Association provided one-fourth of its members are also members of the Northern Nut Growers Association. Such affiliated societies shall pay an annual affiliation fee of $3.00 to the Northern Nut Growers Association. Papers presented at the meetings of the regional society may be published in the proceedings of the parent society subject to review of the Association's Committee on Publications.

SECTION X.—AMENDMENTS TO BY-LAWS

ARTICLE I. These by-laws may be amended at any Annual Meeting by a two-thirds vote of the members present provided such amendments shall have been submitted to the membership in writing at least thirty days prior to that meeting.

Proceedings

44th Annual Meeting

Northern Nut Growers Association

Rochester, New York

August 31—September 1, 1953

MONDAY MORNING SESSION

PRESIDENT BEST: We are opening this 44th Annual Meeting of the Northern Nut Growers Association with this historic gavel which was made from wood grown in the Thomas Littlepage pecan grove near Washington, D. C. Opening each session with this gavel has been a custom of this organization for many, many years.

We are very anxious to have you folks meet some of the men who have made
our meeting possible here at Rochester. I would first like to introduce
Mr. W. Stephen Thomas, Director of the Rochester Museum of Arts and Sciences. Mr. Thomas.

MR. THOMAS: Thank you, Mr. Best.

We are always glad to welcome groups such as yours. You represent a unique organization to us with interests not in our field. We are a public institution, and are glad to have you here.

I feel there are many things of interest in this museum and in our program to interest you, because you are horticulturists and people interested in the out-of-doors.

This museum is owned by the City of Rochester. By the way, there are only about 12 museums throughout the country that are supported as we are. We get 98 per cent of our funds from the City of Rochester. It is not endowed. It is the people's museum. In the exhibit upstairs are three dimensional models showing the evolution of the Genesee Valley in New York from early times to the present. Here you will see a beautiful panorama of what it looked like two hundred million years ago right where we are sitting and standing now when the seas overlay the area during the

Devonian and Silurian times. We have reconstructed the little sea creatures that lived in the rocks in their natural colors.

Another exhibit is the Indian story, primitive man, not just before the white man came, but going back 1500 years. On the top floor you may see how the pioneer man worked here as a woodcutter and running flour mills and how the city came about. The whole story of our region is in the museum.

But more important than these exhibits is what we do through the educational system; adult lectures, and so forth. That is just a little background of our work. I know you have your important business at hand, but I hope you will have a little time to view the exhibits. We want to help you in any way we can. If there is anything we can do, don't fail to ask.

PRESIDENT BEST: Thank you, Mr. Thomas. Most of you met Mr. Wilbur Wright last night out at the park. He is going to make an address of welcome from the City of Rochester and from the parks. Mr. Wright.

MR. WRIGHT: Ladies and gentlemen, on behalf of the City of Rochester and the Park Department, we want to welcome you to The Friendly City. We want you to feel that Rochester has its hand out for a wide open welcome for anything we can do to make you happy while you are here.

The parks are particularly interested in the fact that you have chosen Rochester as your conference city for 1953. The parks, as you know, are a good deal like the museum. They are botanical collections in the heart of the city, the money coming from the city; the taxpayers pay the bill. We have a tremendous botanical collection here, and are known the country over for our lilac and other collections.

We have, in the past two years, appointed Bernard Harkness to take charge of our plant collections, with the title of taxonomist. It took quite a bit of backing to get Civil Service to break down and make such a title. There wasn't such a title in the State of New York, and they couldn't understand why they should give it.

Mr. Grant is another good Cornellian coming along as Assistant Superintendent of Parks, and he is, again, looking after the maintenance and upkeep of the various plant materials that we have.

We have a very large organization here, the Parks Division includes the cemeteries, 90,000 street trees, 56 playgrounds, and about 2,000 acres of parks. Our peak employment is 756 people. All-in-all we have a tremendous amount of interest in our parks, and they are increasing. We are exchanging plants with about 25 foreign countries right now, and we expect to expand that now with the various facilities we are setting up at our new herbarium, which you visited last night.

We are proud of Rochester, and the park system. We are doing our best to continue the excellent work of Dunbar, Laney, and Slavin who built up the park collections. Our aim is to increase the collections, and make the park system better for the people to enjoy. We hope you have a fine time while you are here. Thank you.

PRESIDENT BEST: Dr. MacDaniels, ex-president of our Association will give our organization's response.

DR. MACDANIELS: Chairman Best, Director Thomas and Director Wright, I don't know whether I am particularly well qualified for this particular assignment, but I am certainly very happy to express the thanks of the Northern Nut Growers Association for the excellent cooperation in arranging the facilities which we have found here in Rochester. Few of us can recall any situation in which the Association has been helped all along the way, as they have been here, and we feel most welcome in this truly friendly city.

Before the meeting I thought I was going to be able to claim a sort of paternal interest in the training of Director Wright in that he studied just prior to the war in the Department of Floriculture and Ornamental Horticulture at Cornell University where I am stationed. Although we saw a good deal of him after the war, he came directly here, so I can't say that I knew him "way back when" he was an undergraduate student. Still we do have a proprietary interest in all Cornellians, and we like to see the home team make good as has certainly been the case here.

Fortunately, Ithaca is close enough to Rochester, so that our classes can come to the Rochester parks on field trips where we have always received the most friendly cooperation and help just as the Northern Nut Growers is receiving today. I assure you that we are most grateful.

PRESIDENT BEST: We will proceed with the business of the organization.
On the Resolutions Committee which will give us resolutions for adoption at our final night session, I appoint Mr. Davidson, Mr. Allaman, Mr. Oakes and Mr. Snyder.

The next item of business is the election of a Nominating Committee. This committee is to nominate the officers which will be elected at our next annual meeting. Nominations are now in order.

DR. MCKAY: I nominate Mr. Machovina.

MR. DAVIDSON: Mr. Silvis.

DR. CRANE: I'd like to nominate Mr. Salzer.

MR. DAVIDSON: I think Dr. Crane ought to be nominated.

MR. STOKE: I nominate Mr. Kyle from Iowa.

DR. DOWELL: I move nominations be closed.

PRESIDENT BEST: Is there a second to Dr. Dowell's motion that nominations be closed? (Motion seconded and passed.)

PRESIDENT BEST: Nominations are closed. Those in favor of this list, Mr.
Kyle, Dr. Crane, Mr. George Salzer, Mr. Silvis, Mr. Machovina, for
Nominating Committee for next year make it known by saying "Aye."
(Chorus of "ayes") Opposed? (None.)

PRESIDENT BEST: May we have the report of the Program Committee. They have been at work, we can see that. The evidence is on every hand. Dr. McKay?

DR. MCKAY: The program you have in your hands represents the work of the Program Committee. The work of the Program Committee is done prior to the meeting and I want to say that this year I really did have fine cooperation from the members and from the members of the committee in responding to requests for numbers on the program. That always makes the work of a committee easy. Because of this fine cooperation I can say truthfully that the effort on my part was relatively small.

As all of you know, we now have a larger group of people to draw from for our programs than formerly. We always go back, of course, to our tried and true members who, year after year, give us numbers for the program, but we also like to give the new members a chance and recruit from new sources whenever possible. I haven't analyzed the program enough to know exactly how many new members are listed on the program this year, but I think you will find a few, and as the organization continues to grow, it will be desirable to use these new sources of information for items on the program as much as we can.

PRESIDENT BEST: That's fine. I don't think we can emphasize that too much, this new-member proposition.

We are ready now for the report of the Secretary, Mr. Chase.

MR. CHASE: About the only report that I have to make is one that was prepared by Mr. Carl Prell, and I don't see why in the world he didn't give it, since it's such a fine job of getting together the information on membership. I am going to try to sum this up for you, in order that you will know the progress we made in the membership drive to which so many of you contributed.

On the books as of today we have 1013 paid up members. (Applause.) In addition to that, we have 15 more who will begin membership the beginning of our fiscal year, and in addition to that, there are ten more too new to be acknowledged yet. So we are in pretty good shape on membership.

New members total is 455, which is, I think, just about double from last year, if my memory serves me correctly. Leader in the members by state is our good, old friend Ohio with 126. They produced 42 new members this year. Second on the list is Illinois, with 102, and they came up with 38 new members.

DR. MACDANIEL: I think this is the first year we have had a hundred members from any state.

MR. CHASE: Pennsylvania has 83, with 26 new members. New York 78, with 27 new members. Indiana 70 with 31 new members. Michigan 58 with 28 new members. That covers the top six states.

During the year we lost 71 members. That breaks down to five deceased, 12 resigned and 54 that we haven't heard from. Out of the 12 that resigned, seven were one-year old and only 5 older members. Now, of the 54 not heard from, 40 were one-year members and 14 were older members. Total loss is 71, actually. We had 14 reinstatements this year.

Does anyone have a question on membership? There are quite a few folks in the Association who are really working hard to get new members, and a great number have come up with at least one. But, actually, I believe, Carl, it's a very small percentage of the membership that's really working, is that correct?

MR. PRELL: I am afraid so.

MR. CHASE: And the 71 lost, you considered about normal, didn't you? We have to figure on losing about 10 per cent. Well, we can't afford to lose a hundred.

I don't have too much to report as Secretary, except we might briefly review this hectic year since the little sub-zero walnut story appeared in the Farm Journal. In June a year ago I received a request for an article on the hardy English walnut. I handled it as a routine request and sent it to the Farm Journal. Of course, Joe McDaniel was secretary, and I referred all the interested readers to him for further information. The first batch of mail hit Joe right after our meeting in Rockport, and he had 1500 inquiries within two weeks. I forgot to warn him that this might be coming up, and he went ahead and handled about 1500 of these inquiries, and then I don't know what happened to him, he started sending them down to me. Between myself, my secretary, my wife, and my boy we handled the other 4,000, and they are still, as Joe says, actually coming in.

To handle that, took some of our funds as you see under "promotion business", in the treasurer's report. The mimeographing was gratis, also the assembling and mailing, but the postage we had to pay for.

So all we have to show for that is about how many members, Carl?

MR. PRELL: I will say 200.

MR. CHASE: That's about right. As these inquiries came in we compiled lists of names and sent them to Mr. Best. Then Mr. Best mimeographed a letter and some other material, along with an application folder and followed up these inquiries except the last 500. So we hit them once with a three-page information sheet from the Secretary's office, then Mr. Best at least once again with a follow-up letter, and out of almost 5,000 we get about 200 members, which is pretty good. And there are a lot of other folks I know would join if somebody would contact them.

MR. CHASE: So ever since last October I don't know what side is up so far as N.N.G.A. is concerned. I don't pretend it hasn't taken a good deal of effort and a lot of time from some things that I should have done, but I enjoyed doing it for the Association, and I have no regrets. The only thing I am sorry about is that we didn't get 500 instead of just 200 members.

PRESIDENT BEST: Thank you. Spencer. (Applause.) That's a fine report.

May we hear from our Auditing Committee, Mr. Silvis.

MR. SILVIS: The report of the Treasurer, which I have just had an opportunity of inspecting, is the most professional document I have ever had the pleasure of examining as a member of this Auditing Committee on which I have been several times. And I think a testimonial is due our Treasurer, Mr. Carl Prell, who has combined the rare talents of bookkeeping and comparative reporting.

The Auditing Committee, composed of Sterling Smith, who was not able to be here, Mr. Pape of Indiana, and myself, accept this report of Carl Prell on behalf of this Northern Nut Growers Association,

PRESIDENT BEST: Let's have the Treasurer's report. Mr. Prell

Report of the Treasurer

CARL PRELL, *South Bend, Indiana*

At the beginning of this fiscal year it seemed likely to your Board of Directors that the Association's investment in government bonds would have to be converted into cash to meet the year's expenses. There was barely enough money in the treasury to pay for the 42nd Annual Report, which should have been billed the preceding year.

Normally, the treasurer collects only enough money to pay for one report, plus the year's operating expenses. The problem this year was to pay operating expenses and to discharge our obligation on two reports.

Anticipating this problem, and in an effort to correct recurring deficits, your Board made plans back in 1951 for a drive to increase membership. Some momentum was gained by the end of last year, which carried into this year with increasing force. The result was the substantial gain in membership reported upon by your Secretary—with a substantial increase in revenue.

Membership drives, of course, are a mixed blessing. They may produce more dues; but they certainly cost money. Our promotion expenditure jumped from practically nothing in 1951 to $115 in 1952 to $620 in 1953. However, our dues collection from new members thus gained, were more than $1200 in 1953 alone—twice as much as was spent. And it is important to note that an expenditure to gain a member is made only *once*, whereas the member's dues continue year after year.

In any event, increased membership was a factor in keeping us from cashing our reserves.

Another important factor was the very generous response of the membership to a plea for Sustaining and Contributing dues. Thirty percent of our old members responded with $10.00 payments for Contributing Memberships or $5.00 payments for Sustaining Memberships. This help was needed. It is deserving of special mention in this report.

One other factor contributed to successful operation this year, as it has in other years. This factor does not show up in figures in a financial statement for the simple reason that the figures are modestly withheld from the treasurer. I refer to the out-of-pocket and unreported expenditures of officers and committeemen, which expenditures sometimes are sizeable. Certainly they were this year. The fact that such contributions were made should be noted.

The sum total of all this is a financial showing for the year that may be considered satisfactory. Our debts are all paid. No bonds were cashed. Nothing was borrowed. And we have money in the bank.

At this time last year we had a cash balance of $1313.78. Today our balance is $303.70. We spent $1,000 more than we took in. But we paid for two Annual Reports. The lesser report cost $1200. If this were subtracted from this year's business, where it does not belong, our cash balance would be $1500.00. In short, on this year's business—even with all its unusual expenses for promotion—our income was more than our disbursements—by $200.00. This reverses the deficit trend of recent years.

RECEIPTS

Membership Dues $3,638.05
Sale of Annual Reports 394.00
Advertising in Nutshell 110.00
Contributions 39.00
Interest on Government Bonds 37.50

TOTAL $4,218.55

DISBURSEMENTS

42nd Annual Report (Urbana) $1,205.53
Printing (1000 copies) $1,050.00
Reporting (addi. billing) 97.05
Postage & Addressing 58.48
43rd Annual Report (Rockport) $1,760.72
Printing (1200 copies) $1,477.42
Envelopes, 800 19.60
Reporting 100.00
Postage & Addressing 163.70
The Nutshell
Printing, 4 issues 353.11
American Fruit Grower 39.00
73 Subscriptions at 50c 36.50

2 Subscriptions at 75c 1.50

1 Subscription at 1.00 1.00

Association Promotion 620.47

Application folder, printing 11,200 164.28

Stationery for Sub-Zero and V. P. campaigns 337.24

Mimeo Sub-Zero and follow-up, 1500 59.00

Postage, Things of Science 59.95

Secretary's Fee, 50c per member 517.50

1952-53 Fee to date 506.50

Balance of 1951-52 11.00

Stationery and Supplies 268.22

Secretary's Expense 315.87

Treasurer's Expense 143.21

Dues, American Horticultural Society 5.00

TOTAL $5,228.63

Cash on deposit, First Bank, South Bend $ 303.70

Disbursements 5,228.63

$5,532.33

On hand, August 18, 1952 $1,313.78

Receipts 4,218.55

$5,532.33

U. S. Bonds in Safety Deposit Box $3,000.00

MR. PRELL: I am going to close right now with this information which the Association, I think, should have. The membership promotion consisted of a campaign called the "Vice-president's Campaign" sparkplugged by Mr.

Best. Thousands of letters were sent out through the vice-president's and from the president's office to the membership. You may have received some of them. In addition to that, thousands of other letters were sent out to people who had responded to a story that appeared in Farm Journal wanting to know about the Association. I can't calculate how many went out, I have never been told, but I would guess about 5,000 of them. And those all went out from Mr. Best's office. In addition, our addressograph plate system was not in very good shape, due to the fact the organization was too poor to keep it up. Mr. Best supplied addressograph plates for the whole list.

I wrote to Mr. Best on April 27th, as I wanted all the bad news, and I wrote to some other people. I said, "You have not yet rendered a bill for postage on your mailing. Will you please make your request?" And he answered,

"I was surprised you asked about the postage charge from here. It has been my intent from the beginning of the campaign to carry the postage charge myself."

Thank you very much.

PRESIDENT BEST: Carl, I know you have done a lot of hard work, and I'd like to say for the organization that we do appreciate what you have done for us.

I see Mr. Slate has come in back here. Mr. Slate have you a word from the Publications Committee?

MR. SLATE: I have no formal report. The part of the Publications Committee with which I am concerned is the proceedings. The speed with which that job was done depends upon how fast the papers come in and the transcript of the proceedings finished. The transcript is rather complicated and a lot of things are said that shouldn't go into the report. It takes a lot of work with the blue pencil to boil the material down to something that's useful and worth paying a printing bill for.

One other thing that I should mention is the cost of mailing. I don't know whether that has been mentioned previously or not. We had a little difficulty with the Post Office Department. Carl Prell can tell you about that.

PRESIDENT BEST: Yes, he did. I have heard only good reports of your fine job. I think we all agree that it was a scholarly production.

Do we have anything from the Survey Committee?

Blossoming Habits of the Persian Walnut

H. F. STOKE, *Roanoke, Va.*

The Survey Committee, as its project for the current year, has undertaken a study of the blossoming habits of the Persian walnut. The prime object of this study is to solve the problem of pollination, so that the planter may be reasonably sure of a satisfactory crop, whether his planting be a single tree or an orchard.

While this study has dealt exclusively with the Persian species, *Juglans regia*, the habits and principles involved apply equally to all walnut species.

In most plants the reproductive function inheres in a single bisexual flower, consisting of both male and female elements. In walnuts, as well as most other nuts, the male and female functions are performed by unisexual flowers of very different type and appearance.

Both the male or staminate flower and the female or pistillate flower spring from buds that are formed in the axils at the base of leaves of the previous season's growth. In the Persian walnut they may be detected as early as July. The staminate bud that forms the pollen-producing catkin of the next season, can be distinguished by its checkered appearance, something like a tiny pine cone. They occur in the axils of the lower leaves of the shoot of the current season.

The pistillate bud, which produces the nut, occurs at or near the tip of the growth of the current season. It can usually be distinguished from leaf buds by its larger size and plumpness.

When these blossom buds develop the following season, the male or staminate blossom assumes the form of a catkin, which elongates rapidly a few days before maturity. As the pollen is shed, beginning at the stem end, the pale yellow-green of the bursted pollen capsule turns dark or black, proceeding to the tip of the catkin. This change readily shows that pollen is shedding, which may be confirmed by touching such a catkin with the tip of the finger, and noting the yellow pollen that adheres, or rises in a tiny cloud.

Making note of the date when a given variety begins shedding pollen, and the date when all catkins on the tree have opened, gives the period during which that variety is effective as a pollinizer.

The female, or pistillate flower, does not, like the catkin, spring directly from the wood of last season's growth, but occurs at the end of the new growth of the current year, being preceded by a number of leaves which nourish the young nut to maturity.

The pistillate blossom assumes the form of one or more tiny nutlets with little sharp-pointed tips. When the blossom has become receptive to pollen, each tip has separated into two separate pistils which spread apart and present fresh, slightly sticky surfaces, which are known as stigmas. This is the time that pollination can take place, which period continues until the stigmas have lost their freshness and stickiness. This period marks the time during which pollination can occur.

In many cases Persian walnut trees remain barren when planted alone, not because of incompatibility between the pollen and the pistillate flower, but because pollen shedding and receptivity do not occur at the same time. Sometimes pollen shedding is over before pistils are receptive. Such blooming is termed protandrous by botanists. In about an equal number of cases the pistils lose their receptivity before pollen is shed. Such blooming is termed protogynous. There are quite a number of varieties, however, that mature both types of blossoms simultaneously, in which the variety is self-fertile and will produce crops, even if isolated from other trees of the species. Of these Hanson and Bedford are representative. On some other trees there is some overlapping of the shedding and receptive periods, enough to produce partial, but not full crops.

Warm weather hastens blooming; cool cloudy weather retards it. A warm spell may start blossoming early, but if broken by a cool wave the period of bloom may be greatly extended.

A southern exposure with a light soil will cause a variety to blossom earlier by some days than in the same locality in heavy soil. The blossoming period is generally shorter in the North than in the South.

Climax, Michigan reports blooming beginning May 10 and ending May 31, a period of 21 days. Millerton, N. Y., and Massillon, Ohio, report the same.

Urbana, Ill., reports blooming beginning May 5 and ending June 1, a period of 27 days.

At Roanoke, Va., the period begins April 9 and ends May 10, running 31 days.

At Greensboro, North Carolina, the season began April 2 and ended May 5, a total of 33 days.

The report of Mr. Royal Oakes of Bluffs, Ill., is unique in the shortness of the blossoming period, both of individual varieties and as a whole. Blossoming began April 29 and ended (estimated) May 13, a period of only 14 days. The reason may partly lie in the weather and partly because the planting is on high bluffs overlooking the broad Illinois River valley, affording excellent air drainage.

One major difficulty the Committee encountered in tabulating the reports was the fact that so few of the same varieties were being grown by the various reporters, making it difficult and sometimes impossible to synchronize the blossoming period of the various varieties from different places with sufficient accuracy. Because of this, two tables have been prepared.

Table Number 1 shows named varieties, for the most part.

Table Number 2 shows varieties that are being propagated asexually, but have not yet been given variety names. Seedlings not propagated by budding or grafting, if recognized have been omitted because of individual variability.

Each table consists of five vertical columns, earliest to the right, successively later towards the left.

Varieties above the dividing line are shedding pollen at the time varieties in the same column below the line are receptive. A variety like Hanson, appearing in the same column both above and below the line, is self-pollinizing. Varieties appearing in more than one column indicate a long blooming period.

MR. BECKER: The Crath No. 1 I have been able to propagate is what Mr.
Neilson gave to me as Crath No. 1—I guess he called it Crath.

MR. STOKE: We may have to discard a number of other Crath No. 1's, because the variation between them indicates a mixture of several clones with the same name.

PRESIDENT BEST: The next thing to consider is this resolution which is given word for word on page 26 of our 1952 annual report.

TABLE 1.

Broadview Bijur Carpathian D Beckman Parnell
Lancaster Cutleaf S Bayer Bedford Watt
McDermid Fort Custer Beckman Caesar
[symbol: male] Hanson Breslau Etter
Hilltop Caesar Eureka
Ill. No. 23 Eureka Lake
Mundt Grande S-24
Metcalfe James
Lancaster
Littlepage
McKinster

Caesar Beckman Bijur Bayer Burtner
McKinster Bijur Cutleaf Bedford Dewey
Breslau Colby Broadview Etter
Fort Custer Fickes No. 22 Etter Eureka
Carpathian D Hanson Fort Custer Franquette
Colby Hilltop Grande Mayette
Cutleaf S Jacobs Ill. No. 23 L-2
[symbol: female] Fickes No. 22 James Lake
Hansen Lake Lancaster
Jacobs Lancaster Mundt
Keener S-24 S-24
McDermid Metcalfe Metcalfe

TABLE 2.

Early Medium early Midseason Medium late Late

S-6 S-6 S-12 S-22 S-17
[symbol: male] S-12 S-12 S-22 S-24 S-57
S-33 S-66 S-24 S-25
Littlepage S-XD S-25 S-29
S-29 S-32
S-33 S-41
S-35 S-45
S-66 S-66
S-XD

S-5 S-5 S-6 S-6 S-7
[symbol: female] S-25 S-17 S-17 S-12 S-12
S-29 S-22 S-22 S-17 S-33
S-32 S-24 S-32 S-35
S-35 S-25 S-38 S-46
S-66 S-29 S-41 S-57
S-35 S-45 Littlepage
S-38 S-46
S-41 S-48
S-45 S-XD
S-48 Littlepage
S-XD
Littlepage

It was made by Mr. Dowell of the Ohio group, although it is of interest to every state that has an affiliated group or a chapter. Last year the matter was referred to the Board for their consideration. The Board carefully considered the resolution of the Ohio group, and the spirit of the Dowell resolution, was approved.

The matter was finally left with a committee made up of J. C. McDaniel, Mr. Carl Prell and Mr. Machovina. It is now in order to hear from these men the changes necessary in our by-laws to create the right atmosphere for the formation and operation of our state organizations, because we do want to encourage them. After hearing the proposals, a motion will be in order that they be approved. If approved, a year from now we will vote on the amendments to the by-laws. Mr. McDaniel will give the report of the committee.

DR. MACDANIEL: Your committee agreed on the suggested revision of Section IX, which covers chapters and affiliations. If this meets with the approval of the members here, final action must be deferred until the 45th annual meeting. Proposed amended Section IX of the by-laws reads as follows:

"Section IX.—Chapters and Affiliations.

> "Article 1. The Association shall encourage the formation of regional groups of its members into chapters, which may elect their own officers and organize local field days and other programs. Such chapters need not limit their membership to members of the parent organization. They may publish their proceedings and selected papers in the proceedings of the parent society subject to review of the Association's Committee on Publications.

> "Article 2. Any independent association or society interested in nut tree culture may affiliate with the Northern Nut Growers Association by payment of an annual affiliation fee of $5.00. Selected papers presented at the meetings of such an affiliated society may be published in the proceedings of this society subject to review of the Association's Committee on Publications."

It is the thought of this special committee that the new wording simplifies and clarifies these two articles in regard to organization of chapters or sections, and second, affiliation of independent organizations. We thought we shouldn't limit it to affiliation of nut growers associations as such but to extend affiliation possibility to any society from garden club on up which is interested in nut growing. And under Article 1, whether we call this state group a chapter or a section doesn't seem to matter materially. I believe the Michigan group prefers to be called a chapter. The Ohio group, I think, will take it either way. At the present I think we have called that the Ohio Section of the Northern Nut Growers Association.

Any discussion? I submit this proposed amendment, Mr. President, to be printed in the 44th annual report for action at next year's meeting, the 45th annual meeting of this Association.

PRESIDENT BEST: Do we have a motion to accept the report? (The motion to accept was made by Dr. Crane, seconded and passed without dissent.)

PRESIDENT BEST: I was in hopes that we wouldn't have time for the next feature, but since some of the committees haven't reported yet, I am afraid that we have. That is the address by the president.

President's Address

RICHARD B. BEST, *Eldred, Ill.*

The task of setting down ideas for the reflection of the NNGA fills me with consternation. My scanty rills of thinking are inadequate.

You remember the old Arabian tale of the poor student who was shut up in an enchanted room in the bosom of the earth. You remember how the earth opened only once each year. The student was waited upon by demons and spirits who furnished deep and dark knowledge. When the door opened, the student emerged, loaded with great lore and pertinent facts. Like this Arabian student, by delving into antiquity and our old annual reports of the NNGA, I have put together some thoughts from men living and dead.

Irving says: this pilfering disposition which some of us have may be implanted in us for a good reason. Maybe through us pilferers or borrowers, Heaven takes care of the seeds of knowledge and wisdom from age to age. The worthwhile thoughts which some of our early members gave us may be purloined by me and made to sparkle again in today's light, even though the early members' general idea is obsolete.

So, just as nature has provided for the distribution of her plant varieties through the maws of birds and animals, so it may be that Heaven has provided for the fine thoughts of our old members to be caught by us predatory individuals and made to bear fruit again in this new day. Really this is one way we exist and go forward in our organization.

A crop of "tares" which we read about in the scripture enriches the soil for the next crop. As a forest dies, a new crop of trees spring up. Even a dead tree gives rise to a whole creation of countless bacteria and fungi.

So on "ad infinitum." Members who have talked and studied our problems in the past have made possible our work here today. So, likewise, our words will sleep with the others from whom we have borrowed. So, to escape with a good conscience, to avoid having fingers pointed at me, of hearing cries of—"you stole this from me," I will try to give credit where credit is due.

Otherwise, I might be, figuratively speaking, stripped of my material here piece by piece, and I would finally stand before you with hardly a loin cloth of an idea which I could call my own.

There is a popular appeal to the nut business which most of us are susceptible to,—like wanting to produce large nuts,—and of seeing the first nut,—and to again gather nuts like we did as children. Ask a man how large a nut he found and he will lie as he will about a fish he has just caught.

Then, there is the romantic visionary who would transform the whole universe into a sort of fairyland nut grove—where there are no insects, diseases, or squirrels,—and where the nuts fall polished into open bags.

Then, there are those of us—and I am one—who reasoning that the "groves were God's first temples," flee to a twilight hill top or to a forest shade, and, as Mr. Stokes said, "Sit humbly at the feet of the great mother of us all. There is wisdom and healing in the shadow of her wings."

We need this philosophical attitude to generate encouragement and inspiration to withstand the hard knocks that we have had—and will have coming. But, the NNGA must be more realistic and really do some grappling.

Read the experiences which all our reports are filled with. Mr. G. A. Miller on page 99 of the 1940 report handles this matter of success and failure very well. We live on our successes and not on our failures. Nut culture is pioneering, and it is well to be fully aware of the possibility of failure so that we may be steeled for it when it comes. Failure makes our successes sweeter.

Abraham Lincoln's life was a series of failures. Thomas Edison usually failed. Plant breeders at our stations nearly always fail. But, once in a while they succeed. In the nut business, if we succeed 1 in 10,000 times, success may be cheap at that.

Dr. MacDaniels stated so many important aspects of the NNGA that I want to list his outline here and then simply hang some thoughts on the skeleton of his report. For your own enjoyment and understanding, please read again Dr. MacDaniels's address "The Forward Look," which is found on page 27 of our 1952 report. I just mention his subjects and comment on them for emphasis.

1. *Variety Evaluation*

The Ohio Nut Growers did a fine job of getting this job of evaluation in the groove. Read about it on page 29 of the 1946 report. How many of us here have wasted years on varieties that good evaluation might have discarded, before we started to plant the nut.

2. *Judging Standards*

Which covers such things as—

(a) Sealing of nuts (b) Recovery of halves (c) Size, quality, etc.

Evaluation and judging include all those fine things we look for in a nut.

3. *The Naming of Varieties*

Many of us have the same tree growing but calling it a different name.

4. *Securing New Varieties*

And getting them into as many channels. Mr. Wilkinson started several "Chief" pecan trees last year and gave them all away. Chief is a fine new variety of pecan. If we had a few more Ford Wilkinsons in this organization, "God bless him," we wouldn't have so many problems.

5. *The Work of Individuals*

There is always the possibility of finding the "perfect nut," so we need to continue our search through the earth for better varieties.

Scientific techniques must be applied before we breed better nuts.

We simply cannot have nuts put in our coffins expecting to continue our work in the next world; so we need to do the next best thing to this—that is of instructing our sons in the little we have found and where our different varieties are planted. Then, some of our sons will start where we left off.

Private research is playing a more important part in the world today. Private research, using the large amount of basic research available, could accomplish wonders in the nut world.

I am deeply grieved when I see vast estates which have had a fortune spent in plantings that had little practical value. The men who spent this money would gladly have furnished the land, labor, capital and management for a nut breeding program had we been there to have sold them on nut trees.

As Mr. Churchill said—"Too little and too late."

But members of the NNGA forget "what might have been." New estates are developing and younger men are wondering how they can immortalize their lives and work. Men pass away; their names perish from record and recollection; their history is only a tale and their tombstone becomes a ruin, but a good nut tree bearing a man's name, gives that name immortality.

6. *Work of the Experiment Stations.*

In the most practical vein, our basic research and most of our actual breeding must still be done by our Experiment Stations.

Any nut project is a long-time program and it lends itself best to an Experiment Station which is not set up on "a three score and ten" basis like we as individuals are. Stations also have trained workers and information at hand.

7. *The Real Aim of the NNGA*

Is better living on the farm and the improvement of the garden and farmstead. Almost every farm and home, especially in the great corn belt, needs shelter, shade trees, and the beautifying effect of trees.

Psalm 19:1-4 says: "The heavens declare the glory of God; and the firmament showeth his handiwork. Day unto day uttereth speech and night unto night showeth knowledge. Their voice goes out through all the earth, and their words to the end of the world."

8. *Nut Growing as a Hobby.*

For people who like to see things growing, there are few projects that yield more genuine satisfaction than hardy-adapted varieties of nut trees.

Few people know what a heart nut or chestnuts are, and most have never cracked a butternut. Most of us have never tasted a good persimmon, and the paw-paw is practically unknown. We of the NNGA have something to offer our members.

9. *Keep our Organization Solvent and Functioning.*

All costs have increased. Our strength lies in our letters, reports and information which we send to our membership. To keep this information coming through letters and our annual report takes money.

10. *How to Finance the NNGA.* Dr. MacDaniels makes the following suggestions after stating that we have reached the point where increasing the dues will not give us more income, because of loss of membership.

(a) Increase our number of members.
(b) Provide different types of membership to encourage contributions.

(c) Gifts.
(d) Special fund raising projects.

Increasing the membership seems to have the most promise in the future.

We are now at the cross roads. Do we want a strong, hard-hitting organization, capable of doing these things which we know NNGA can do, or do we want to ease down the other road to a whimpering senile existence, in the plant society world?

We have increased our membership 40% the past year, and after hurriedly congratulating ourselves, let us hurry on to this problem of setting ourselves another goal for 1954.

What shall our new goal be? 30%, 50%, 100%—let us think "high." It is easier to come down than to go higher.

I'd like to have someone's idea in the audience here. Does anybody have some figure we should hit at? 10%, 20%, 25%?

MR. KINTZEL: I think since we have a thousand members, for each of the thousand to bring in one more member and make it a hundred per cent.

PRESIDENT BEST: Now, the high man is going to win here, you know. Is there anyone that can raise this man, that can say that we increase our membership more than a hundred per cent? If not shall we take that as a goal?

We have now reached the point in this discussion of goals where we can make the following profound deduction—"Nothing is worth two cents until it is sold."

Our church bells ring to tell the world that something good is being offered and that the church has something to sell. Everything in this world must be sold. The NNGA is competing not only with the resistance people offer it, but also against every other human activity. People buy what they think will give them the most satisfaction.

We are living in a cold-blooded society and people are not going to choke with emotion when we mention the old hollow tree where the possums hatched, or the wide spreading chestnut. People may not even want to join our NNGA. This is a free country and people can just sit in the sun on the bare ground if they want to. They may not want trees and can eat grape nuts if they want. We know they need the hobby—the shade—the beauty—the protection or even nuts which nut trees will bring them.

Because we know people need these fine things, then we must ask them to join the NNGA. If everybody knew what you know about the NNGA, we would have a membership of 100,000 members. But they do not.

This is what we mean by selling our organization. Indirectly as we sell our memberships to help other people, we help our organization.

Finally, let me suggest that we build up a backlog of ideas here at Rochester to add to what we have on increasing our membership.

Give your ideas to our Vice President, Mr. George Salzer and his publicity committee and you will be helping to solve our NNGA problems.

MR. DAVIDSON: It seems to me the most successful thing that was done during the past year, as far as the raising of membership was concerned, was done by Mr. Chase, when he wrote that article on Persian walnuts that survived sub-zero temperatures. That had a tremendous impact on the public imagination, and it got a tremendous number of inquiries. I think it had more effect than even the work of individual members, so I would suggest that anybody who has an idea that can be sent to a magazine that would have a public appeal should do that thing.

MR. MACHOVINA: The thought strikes me that in addition to the goal for new members, we should also work to keep these members we have picked up this year. That means the older members should contact the newer members, help them, give them trees. Otherwise, a lot can be lost quickly.

PRESIDENT BEST: In this world no matter whether we are selling seed corn—if you will pardon that little plug—or you are running a restaurant or any form of human activity, you can figure each year about a 10 per cent loss in your clientelle or your customers, whatever you want to call them, and we of N.N.G.A. are no exception to that rule. We do have to keep working for that replacement; otherwise, in 10 or 12 years we are going to be out of business entirely.

MR. KERR: I am a Spanish War veteran. At the national convention in Portland, Oregon, in 1938, one of my comrades showed me a walnut tree that he planted before he went to the Philippines during that war. It was on the banks of the Willamette River where he had planted three nuts. Two were so near the river that a log boom had torn them out, but one was left. It was 80 feet high, four feet in diameter, and on one occasion had produced almost a ton of very good nuts.

I told that to the science editor of the Associated Press, and he put a little article in the local paper, but no picture. If he had a picture with that article, everybody would have read it. I think we need more publicity on these old trees that are bearing nuts. I live in Plymouth, Mass., where the Pilgrims settled. In their settlement papers they mentioned the groves of walnuts and other wild nuts in the territory. We found a low-branched walnut 5 feet in diameter and over 450 years old.

MR. BROOK: Let me suggest Mr. Kerr write such an article for such a magazine, because he is just the person.

MR. KERR: I have already written a few articles for several men's clubs, and I am writing another article.

MR. BROOK: When are you releasing it?

MR. KERR: Pretty soon. The Northern Nut Growers will get a copy.

PRESIDENT BEST: The report of the Place of Meeting Committee, Mr. Allaman?

MR. ALLAMAN: A year ago this summer I went to Lancaster to visit the Franklin Marshall College to see if we could hold our convention there next year. They considered it quite an honor to have us there, feeling that it is an educational feature. They can furnish us a nice auditorium with a cafeteria nearby. Sleeping quarters would be scattered throughout the grounds. They can furnish our meals and a banquet. About ten days ago I checked to see if things were still all right, and they said, "Come ahead," so I am suggesting that the Association hold its next annual convention at Lancaster, Pa. at Franklin Marshall College.

Our field trip at Lancaster would be to Mr. W. W. Posey's orchard. He has by far the biggest planting in the state with trees of various ages and many different varieties. He entertained the Pennsylvania group a year ago. He has a nice pavilion up on the hill, where we can have our lunch. We had a most enjoyable time, and he is delighted to have us. Mr. Posey is owner of the Posey Iron Works in Lancaster.

For 1955 we were thinking about Michigan, probably Michigan State College. For the following year we were wondering about Connecticut. Just the place in Connecticut has not been given any thought.

Session adjourned.

MONDAY AFTERNOON SESSION

PRESIDENT BEST: In the morning session Mr. Allaman proposed that we hold our next convention in Lancaster, Pa. Is there a motion to that effect?

MR. RICK: I move that our next meeting, 1954, be held in Lancaster County, Pennsylvania. That is supposed to be one of the garden counties in the country. It has stood first and second in production. I don't know what it stands now, but it wouldn't be far from the top. It would be most interesting to all our members, I am sure, to pay Lancaster County a visit. Motion seconded by Mr. Ellis and passed.

PRESIDENT BEST: According to the by-laws, we get a report of the Nominating Committee at this time. We have our election at the last business session.

MR. SLATE: The Nominating Committee, consisting of Max Hardy as chairman
and some others, including myself, presents the following candidates:
For President, R. B. Best. For vice-president, Gilbert Becker; for
Treasurer, W. S. Clark; and for Secretary Spencer Chase.

PRESIDENT BEST: Are there other nominations? (No response.) You will have a chance for further nominations at our last business session.

PRESIDENT BEST: Now, then, we have an opportunity to hear from a group of what we might call authorities in their various fields. We have quite an assortment. The only way I know of to express it is to say we have the wise men out of the east and the wild men out of the west. I think we first might hear from Mr. Kyhl of Sabula, Iowa. Mr. Kyhl, you come up and give us your version of the nut business.

About Nuts

IRA M. KYHL, *Sabula, Iowa*

What we all have in mind at this time is nuts and more nuts. One way to get them is to plant more nut trees. Why not start a campaign in this direction? Where I live in the midwest the black walnut is at home and likewise the hickory, hazel, etc. Farmers may be reluctant to set aside acreage for this purpose but they could be planted along fence rows around the entire farm and would produce shade for livestock, an abundance of marketable nuts, and later a fortune in saw logs. The average size farm of 160 acres could support a great many black walnuts if planted along fence rows which ordinarily grow up to brush and weeds. Seedlings are cheap or one could buy 2 or 3 bushels of Thomas nuts and raise their own. One could also plant hickories, heartnuts, filberts and chestnuts if variety is desired along fence rows, but the main thing is to get this work started. We could no doubt get cooperation from County Agents and Conservation Departments because of wind breaks and erosion control. Farmers who could be induced to do this work would no doubt become nut enthusiasts in due time.

I feel that at this time it may be in line to pay a slight tribute to our friend the squirrel. I wonder how many of us gave a thought as to who was responsible for all of our wild trees, such as black walnuts, butternuts, hickories, hazels and so forth, and how they came about. The answer is simple, the squirrels, of course. They have been planting nuts for centuries and without their good work in the past, there would be very few wild nut trees.

The squirrel has been wrongfully condemned for his apparently good work and has even been cussed a little for living on the efforts of his own labor, and due to my appreciation of his good work, I have grafted or rather topworked some of the trees he planted to Persian walnuts, pecans, etc., so that he may have more of a variety of nuts. Someday I expect to have some of the largest and fattest squirrels in America. I cover some of the choice varieties with stove pipe. They seem to take the hint and don't bother the nuts. One more thing, there does not seem to be enough nuts to go around, that is, enough for both the squirrels and ourselves. So let's plant more trees so that the squirrels can't possibly eat them all and when we have done that, then let's plant a lot more.

We now have many species of nuts and many varieties of each species, many of which have proven hardy in cold climates. It is very encouraging to note the good work that is being done to produce better and more varieties. One very fine nut that doesn't seem to have had much work done on it is the hard shell almond. It does very well for me, is self-pollinating, bears very heavily, and can be grafted on peach stocks with good results. I have also had very good success with Persian walnuts, heartnuts, filberts, chestnuts, hickories, pecans, hazels and black walnuts.

Natural Variation Observed in Shagbark Hickory, *Carya ovata* (Mill.) K. Koch. in Central New York

DAVID H. CALDWELL, *N. Y. State College of Forestry, Syracuse, N. Y.*

The shagbark hickory has been extremely important in the economy of the United States during its period of early development. The handles of the axes which leveled extensive forested areas in Colonial days were frequently made from sturdy hickory wood. The nuts furnished food for man in the form of oil or nutmeats and often hogs were fattened on hickory nuts, beechnuts and chestnuts. As settlement progressed, the demand on hickory as wood for wagon parts increased while the use of the thick-shelled nuts for food decreased except by the country boy or girl who wandered from tree to tree in the fall collecting nuts for cracking by the fireside in the wintertime.

The author remembers bounding out of bed as a child in the fall before dawn on the nights when there had been a frost or a heavy wind, in an effort to beat the squirrels in the race to obtain the rich harvest of hickory nuts to be found lying beneath the fine old trees near Herkimer, N. Y. By some coincidence, both the boys and the squirrels knew of the same trees which were most sought after for their crops of nuts. It was at this time that the variability of hickory nuts was first observed. Thus it was that the nuts of certain trees were never gathered, while the grass beneath other favorite trees was gleaned carefully for all fallen nuts.

The present investigation of the shagbark began in the fall of 1949 and continued through the summer of 1953. It was initiated with the previous knowledge of the extreme variability to be observed between the nuts of individual shagbark hickory trees and was conducted for the purpose of determining whether or not that variability was also expressed through other features of the tree such as buds, leaves, bark or form. Consequently, a systematic study was begun of individual trees totaling 158 found mostly in Onondaga County, New York plus the edges of surrounding counties. The trees were observed throughout the growing season so that the various tree parts could be observed for comparison. It was a preconceived idea by the author that there might be several or more distinct subdivisions into which individuals of shagbark might be placed through the use of macroscopic characters.

Observations were made over a period of three growing seasons on the following characters of each tree:

(1) bark (2) buds and twigs (3) leaves (4) flowers (5) fruit

Each character was observed more than once for each tree as a check on possible yearly variation for specific characteristics in the trees from which data was collected.

The generalized description for shagbark hickory is as follows:

SIZE—a tree ranging at maturity from 50 to 100+ feet in height, generally 2 to 3 feet in diameter and very occasionally reaching 4 feet in diameter.

BARK—usually under 3/4 of an inch in thickness, occasionally up to 1 inch thick with a characteristic light or smoky-gray color when dry and breaking up into long plates or strips loosely attached to the trunk near the middle of the plate.

BUDS—terminal buds usually 3/8 to 3/4 of an inch long, subglobose to narrowly ovate, with 8-10 imbricate scales, the outermost of which are a blackish brown with dark brown tomentum, and a short mucronate or attenuate apex, inner scales light brown with longer lanate pubescence and apex acute to obtuse;

lateral buds smaller, about 1/4 of an inch with tightly appressed scales.

TWIGS—angled or rounded, reddish brown to yellowish brown, or gray, turning more or less gray with age; pubescent the first year.

LEAVES—compound—ranging from 3-7 ovate to oblong lanceolate leaflets, usually 5, terminal leaflet as large or larger than the first two laterals, usually 4-8" long, generally glabrous on both surfaces but with a finely serrate, ciliate margin; total leaf size ranging from 8-15" for mature leaves.

FLOWERS—(a) *female*—occurring in 2 to several flowered spikes, with a one-celled ovary, about 1/4 to 1/3 inch long covered with tomentum; flowers rusty to yellowish green in color; stigma with two stigmatic lobes; bracts much longer than the lateral bractlets. (b) *male*—in three parted or branched aments, each flower usually containing 4 stamens with a 2 or a 3 lobed calyx; aments 3-4" long with glandular hairs.

FRUIT—oval, globose or pear shaped, consisting of a woody husk 1/4 to 1/2" thick breaking usually along 4 lines of suture exposing a flattened nut generally 4 ridged, smooth or slightly roughened; usually white or cream in color, seed sweet with 2 cotyledons.

RESULTS OF FIELD OBSERVATIONS

BARK—Most of the shagbark hickory trees observed were found to have a smoke-gray, shaggy bark from 20 years of age to maturity. However, among the 158 individual hickory trees observed, there were found 7 trees which had a bark much more blackish than the normal shagbark type and with closely furrowed bark consisting of inter-lacing scaly ridges more similar in character to that of *Carya ovalis* (Wangenh.) Sarg.

The trees found growing under timberland conditions rather than as open field or hedgerow trees did not have the characteristic shaggy bark except for the upper trunk which had been exposed to the weather conditions of the forest canopy. Where the trunks of the trees were somewhat protected from direct rays of the sun and force of the wind, the bark was smooth, gray and but slightly plated with none of the shagginess typical of open field grown shagbark.

BUDS AND TWIGS—The buds of shagbark were observed to divide themselves into two general groups based upon terminal bud shapes and two more groups based upon the sizes of the attenuated apex of the outermost bud scales. In all cases the bud scales were observed to be pubescent though the degree of pubescence varied considerably in the outer scales only.

The two general bud shapes were globose-ovate and narrowly elliptical. The broadly oval (Fig. 1a) type of buds were smaller, generally under 1/2 inch in length while the elliptical (Fig. 1b) type of buds were usually over 1/2 inch in length.

[Illustration: Fig. 1a Fig. 1b

Shagbark Hickory Terminal Buds (1-X)]

The long attenuated apex on the outer bud scales of the elliptical type of buds is evident in Figs. 1b and 2b.

[Illustration: Fig. 2a Fig. 2b

Shagbark Hickory Terminal Bud Scales (1-X)]

The number of lateral buds at one position varied considerably with the usual number being one (Fig. 3a) bud located just above the lobed leaf scar. On exceedingly vigorous sprout growth, or on very vigorous terminal growth twigs, it was found that 2, 3, 4 and occasionally 5 superposed buds might occur (Fig. 3b).

Twigs of shagbark varied considerably both in the rapidity of growth and in color. Frequently the color seemed to be associated with the incident rays of the sun and orientation of the twig on the branch seemed to largely control color.

Twigs upon the same tree would vary from gray to reddish brown to yellowish brown or tan. The majority of observed trees had a reddish brown as the predominant color. Terminal shoot growth of as much as 40 inches was observed and as little as 2-3 inches in very slowly growing mature trees.

[Illustration: Fig. 3a Fig. 3b

Shagbark Hickory Lateral Buds (1-X)]

[Illustration: Fig. 4a Fig. 4b

Shagbark Hickory Leaves (1/3X)]

The degree of pubescence on the surface of the twigs varied considerably and was found to frequently follow group location patterns. Thus nearly all of the individuals growing in one field might be found with dense pubescence on the twigs while a similar group several miles away might have, for all practical purposes, no pubescence on the twigs. In general, the most rapidly growing trees (or twigs) had the least amount of pubescence on the twigs.

LEAVES—There was extreme variability found with the leaves of the 158 individual trees observed. All trees were found to have compound leaves, but the leaflet numbers varied greatly. The typical number for shagbark is 5, but 3 to 7 were found; three leaflets were common, 5 were abundant and 7 leaflets were rare. Six cases of leaves with 7 leaflets were obtained from the vast number of leaves checked on the 158 trees; thus the frequency of occurrence is quite low for the group as a whole. Where 7 leaflets were observed, 5 of the leaves were normal pinnately compound leaves (Fig. 4a), while one leaf consisted of five palmately arranged leaflets plus two normal pinnately compound leaflets (Fig. 4b). The leaflets on each tree were fairly uniform in shape but the shape of leaflets between trees varied considerably. Thus one tree might have 5 leaflets quite broadly ovate to obovate in shape while another equally valid shagbark would be found with narrowly elliptical to lanceolate leaflets similar to those of red hickory (oval pignut hickory), *Carya ovalis* (Wangenh.) Sarg.

[Illustration: Fig. 5a Fig. 5b Fig. 5c

Shagbark Hickory Fruits (1/2X)]

The margins of the leaflets were generally finely serrate and disposed to be ciliate—i.e. with a fringe of hairs along the serrate margins. The presence of cilia tend to differentiate shagbark hickory from red hickory in the field. This feature is a consistently good one if a hand lens is available but the degree of ciliation varies considerably from tree to tree and during different parts of the growing season. The presence of cilia on the margin of the leaflets should not be used as a means of differentiating shagbark from shellbark hickory, *Carya lacinoisa* (Michx. f.) Loud., since shellbark also has a ciliate margin on the leaflets.

FLOWERS—The female flowers of shagbark are found on short 1 to 5 flowered spikes produced on the current season's growth. Most of the flowers are around 1/3" long, sessile and covered with a tawny tomentum. Each flower tends to have two yellowish green stigmatic lobes but three-lobed stigmas may be found and one case of a 4-lobed stigma was observed. Various amounts of an amber, or yellow scurfy, substance was also observed on the new flowers. The male flowers occur on 3 parted, slender, glandular-hairy aments from the basal portion of the current season's growth. The aments are usually 3-4 inches long with individual flowers consisting of 4 stamens with their surrounding bract and calyx lobes. The anthers are yellowish or greenish yellow. Occasionally a two branched ament may be found but this seems to occur when one branch of the ament has failed to develop due to an injury of some sort. One case of an unbranched ament was observed.

Both female and male flowers are found to be mature after the leaves have grown to nearly their fully expanded mature size. There are more male aments to be found on the lower branches than female spikes of flowers, which would tend to aid in cross pollination of the flowers by wind action. In general the stigmatic lobes are not quite mature at the time that the bulk of the pollen is being shed, yet individual trees, at a considerable distance from another pollen bearing shagbark tree, will bear considerable quantities of nuts indicating self fertility.

FRUIT—The husk of the shagbark is extremely variable in size, shape, thickness and opening habits. In general the husk consists of 4 segments which split along 4 sutures and fall apart at maturity dropping the nut to the ground. In many cases the husk falls to the ground with the nut and does not break apart until it reaches the ground. A few of the trees examined had husks which were not quite deciduous to the base and were retained on the tree until after the nut had been released. One tree among the 158 examined consistently had a 5 parted husk.

The husks varied considerably in thickness, the dried measurements ranging from 1/8 to 1/2 inch with the bulk of the measurements averaging around 1/4" thick. Two trees had husks so thin as to be more typical of red hickory while only 6 trees had husks 1/2 inch thick or more.

The overall shape of the husk around the nut ranged from globose (Fig. 5a) to ovoid (Fig. 5b) to obovate (Fig. 5c).

It would seem that the shape of the nut enclosed within the husk might be predetermined by examination of the husk itself. The obovate husk shape could most frequently be depended on to produce either elliptical or obovate nuts but this was not an absolute certainty. The thickness of the husk effectively concealed the true shape of the nut beneath; the thinnest husks most nearly conforming to the true nut shape.

The size of the mature shagbark hickory nut and husk ranged from as small as one inch in a tree which had a seed barely 3/8" wide to as large as 2-1/4 inches. The size of husk and nut is variable and adjacent trees which may have developed from the same parent seldom have similar nuts in the area examined.

The nut itself exhibited the greatest variability of all features examined on the test trees. These trees exhibited striking dissimilarities in:

(1) nut size (2) nut shape (3) shell color (4) thickness of shell (5) sweetness or palatability of nutmeat.

One tree was discovered with a nut which might have caused a taxonomist to coin the name *Carya ovata* var. *microcarpa* due to the very small dimensions of about 3/8 x 3/8 x 3/4 inches in width, thickness and depth. Even the squirrels of the area did not feel that this tree deserved their attention The largest nut obtained had overall dimensions of 1 x 3/4 x 1 inches in width, thickness and depth. The majority of average sized nuts were roughly 3/4 x 1/2 x 3/4 inches.

The nut shapes have fallen into a general pattern which include the following normal types:

Type A—The normal 4 angled nut, nearly rectangular in cross section (Fig. 6a).

Type B—An elliptical form, nearly oval in cross section (Fig. 6b).

Type C—A smooth oval nut, oval or elliptical in cross section (Fig. 6c).

Type D—An obovate nut, oval to angled in cross section (Fig. 6d).

Type E—A fat globose nut, broadly oval to orbicular in cross section (Fig. 6e).

[Illustration: Type A Type B Type C Type D Type E

Fig. 6a Fig. 6b Fig. 6c Fig. 6d Fig. 6e

Normal Fruit Forms of Shagbark Hickory (1X)]

[Illustration: Type F Type G Type H

Fig. 6f Fig. 6g Fig. 6h

Abnormal Fruit Forms of Shagbark Hickory (1X)]

In addition to the afore mentioned 5 normal types, three abnormal types were encountered:

Type F—A smooth or angled nut, triangular in cross section—found in the same trees as normal nut forms (Fig. 6f).

Type G—A smooth or angled nut square in cross section—found on the same trees as normal nut forms (Fig. 6g).

Type H—A Siamese twin form occurring very rarely on the same trees as other normal forms (Fig. 6h).

Type A was the commonest form of nut found in the Onondaga County area. It roughly exceeded Types B, D and E by a 2:1 ratio. Type C exceeded Types B, D and E with a ratio of about 7:5 in frequency of occurrence. Types B and D were the two most easily cracked nut forms when using a hammer and anvil for a cracking device. It should be noted at this time that *all* of the abnormal fruit types were found in *conjunction with* normal fruit types. Thus, one individual tree used as a collection might produce *both* a normal nut type (A, B, C, D or E) *and* an abnormal nut type (F, G or H). Occasionally a few nuts in a collection from one tree might be classed as a *second* normal type. This was rare however (5 cases) and only occurred in "borderline trees" which were then classified and recorded as per the dominant nut type for the tree. It should be noted here that the nut type did *not* vary from year to year for the trees examined. Also the frequency of nut crops varied considerably; less than 1/4 of the sample trees produced nuts each year. Most of the trees produced crops in alternate years, and a very few have not fruited in the third year following a heavy nut crop.

The 158 trees examined provided the following distribution by fruit types:

Fruit Type		Number of Individual Trees	Number of abnormal types found in conjunction with normal types		
			F	G	H
Normal	A	54	5	2	1
	B	23	2	1	
	C	36	1	2	
	D	21	7	1	
	E	24	4	1	
		158 collections			
Abnormal	F	15	15		
	G	8	8		
	H	4	4		
		27 collections			

Shell color of the nuts varied between a brownish white and a pinkish white color when fully dried. From the trees used as a sample, there were

14 which might be classed in the brownish white categories, and the remainder (144) as pinkish white or creamy white. Types B and C were the ones which most frequently were found with the brownish white nutshell color. Type A was typically pinkish or creamy white in color.

Nutshell thickness varied somewhat. In all but 2 cases, the nuts were too hard to crack with the teeth. The thin-shelled ones are *comparatively* thin only, being about like paper-shelled pecans with the shell thinnest on the sides of the nut. It is not suggested that these two thin-shelled nuts be exploited as paper-shelled shagbarks since they are poorly formed nuts and of small size. One of the two trees might be a hybrid since it does not have a ciliate leaflet margin although the buds, bark and leaves are typical of shagbark hickory. The minimum shell thickness observed for the side of the nut was 1/2 a millimeter (0.5 mm.) and the thickest was 2.0 millimeters. As previously stated, nut types B and D (the elliptical and obovate nut forms) were the easiest to crack. Nut type A was the most difficult and had generally the thickest shell.

The seed coat color range was from a light tan to a bronze color. The seed itself was in all cases sweet although certain of the nuts had a more pleasing taste than others. The nuts eventually became rancid though 3 years of storage in a heated room did not cause the bulk of the test samples to change in flavor. This is unlike the pecan which, stored in the same room with the hickory nuts, became rancid by the following year after collection.

Summary of Observations

The following observations concerning shagbark hickory may be made from this study:

(1) The buds of shagbark fall into 2 classes based on bud shape, (1) globose-ovate and (2) elliptical, the latter being the largest bud as a rule.

(2) The buds of shagbark fall into 2 classes based upon the length of the attenuated apex of the outer bud scales. The elliptical form of bud consistently had the longest drawn out apex.

(3) Normal buds of shagbark occur singly on the twigs above the lobed leaf scar; however, 2, 3 or 4 superposed buds may occur on very fast grown sprouts or terminal shoots of vigorously growing trees.

(4) The twigs of shagbark are pubescent but range in degrees from almost none to densely pubescent. The fastest grown twigs are apt to be the least pubescent.

(5) Leaves are compound with 3-5 leaflets commonly found and 7 leaflets rarely found.

(6) Leaflet shapes varied from tree to tree being ovate, obovate or elliptical.

(7) Leaflet margins with one exception were more or less ciliate.

(8) Most female flowers of shagbark have 2 stigmatic lobes, however, 3 stigmatic lobes resulting in triangular nuts are not uncommon.

(9) The typical male ament is three branched but one and two branched aments have been observed.

(10) The husk of shagbark varies in thickness from 1/8 inch to 1/2 inch in thickness when dry. The usual husk is 4-parted but one tree bore 5-parted husks consistently.

The average husk thickness is around 1/4 inch.

(12) There are three general fruit shapes, (1) globose, (2) ovoid and (3) obovate.

(13) There are at least 5 general types of normal nut shapes for Onondaga County, N. Y. as listed in the text of this paper.

(14) Three abnormal nut types were also encountered growing concurrently with the normal types.

(15) Nutshell color varied from brownish to creamy white. The darker colors were generally associated with the elliptical, oval or obovate nut forms.

(16) Nutshell thickness varied between 1/2 and 2 millimeters; the more angled the nut, the thicker the shell.

(17) All of the hickory nuts tested had sweet, edible seeds. The seed coats varied from a light tan to a bronze in color.

Conclusions

Within the single species of nut tree called shagbark hickory, *Carya ovata* (Mill.) K. Koch., in central New York, there exists a great degree of diversity. However, in spite of these differences, the examined sample trees may be placed without a question in their proper genus and species and the author would venture the opinion that the advisability of placing variety names on portions of the species is a doubtful and hazardous procedure until much more is known concerning the species than is known at present.

MR. PAPE: This paper is the result of the fact that some of us down in Indiana are losing 75 to 95 per cent of our hickory crop each year by the curculio, and what we are trying to do is work up a little interest with this paper, so at the conclusion of this we can get a discussion started and learn the experiences of other people. Maybe you will be able to help us down in Indiana.

The Control of the Hickory Weevil (Curculio caryae)

EDWARD W. PAPE, *Marion, Indiana*

It is our thought that if some effort were made to bring to this assembly, a digest of what has been done to control the Hickory weevil, we might arouse enough interest to carry on some experiments.

If, at the conclusion of this paper, we can get enough discussion, we will be able to avail ourselves of the knowledge and experiences of others who have made attempts to control this pest, it would be to our advantage.

The Pecan weevil of the south and the Hickory weevil are identical and we learn the following from the experiments carried out by G. F. Moznette, Bureau of Entomology, U.S.D.A.

Pecan weevil damage is of two types—(1) that resulting from attack before the shell-hardening period in July and August and causing all affected nuts to drop, and (2) that resulting from attack after kernel formation and usually causing the shuck of infested nuts to stick tight to the shell instead of opening normally. Weevil-injured nuts of the second type contain grubs which destroy the kernels, or they contain holes about one-eighth inch in diameter which mature grubs have bored and through which they escaped after destroying the kernels. The first type of damage often passes unnoticed and is due to the feeding of early emerging weevils, which puncture the immature nuts with their long lancelike beaks to feed on the juices within. Since all nuts punctured in this way before the shell-hardening period drop to the ground, the entire crop may be lost if weevils are abundant and the crop is light. Such damage may be heavy even when a large crop is attacked. The second type of damage is generally noticeable at harvest-time in October and November, and in seasons when large numbers of weevils have been present practically the entire crop may be wormy at harvest.

Since the weevils do not feed very much on the outer surface of developing pecan nuts, stomach poisons applied to trees have been of little practical value in control. In 1944, however, laboratory tests showed that DDT could kill the adults, and that it was worthy of field trial.

Field tests were made at Fort Valley, Georgia, with DDT and the conclusions drawn from these tests show that the effectiveness of two applications of DDT at the rate of 6 pounds of a 50-percent wettable powder to 100 gallons of water in reducing harvest infestations to 1 percent gives rise to the hope that this treatment, applied for several seasons, will eliminate a pecan weevil infestation in an orchard, or will reduce it to such an extent that spraying every year will not be necessary.

The time of the first application of DDT cannot be based on the time of the first drop of nuts, because other pecan insects also cause the nuts to drop during July and August. However, pecan growers who wish to make the effort can time the first application accurately by spreading a sheet on the ground beneath an infested tree and lightly jarring the branches to dislodge the weevils. When the weevils are disturbed they fall and "play possum" and can be easily collected. When a minimum of six weevils can be taken by jarring the branches on any one tree, it is time to make the first application.

While the above will probably give an indication as to what can be done, using DDT to control the Hickory weevil, for those who have large plantings and can afford the expensive spraying equipment necessary, it will be necessary to look farther for control methods for the small orchard, where expensive equipment is not feasible.

The following is part of a letter from Dr. C. C. Compton, Entomologist for the Julius Hyman and Company.

"It is our thought that since DIELDRIN is so highly toxic to Curculionids it might be possible to take advantage of the habits of this insect and control it by spraying the soil surface. The larval stage of this insect leaves the nuts and enters the soil sometime in the fall. It is believed that the larvae penetrate the soil rather deeply, to a depth of perhaps a foot or more and remain in the soil over winter. In the spring or early summer the larvae transform to adults and emerge to lay their eggs. In some regions at least the adults do not emerge until the second year after the larvae enter the ground.

It is our thought that if DIELDRIN was applied to the surface of the soil that many of the larvae would be killed upon entering the soil or would be killed at some later time when the adults emerge.

Dr. C. L. Fluke at the University of Wisconsin has been working for the past several years with DIELDRIN applied as an orchard spray for the control of plum curculio. In Dr. Fluke's work he applied the DIELDRIN to the orchard floor or cover. He has had some very promising results. Dr.

Fluke has used two application rates, namely, six pounds and three pounds of DIELDRIN per acre. Since he obtained a high degree of control at the three pound level, it would seem worthwhile to investigate the possibilities of applying even a lower rate, say one and a half pounds per acre. In Dr. Fluke's work he applied the DIELDRIN to the soil in the orchard using a DIELDRIN emulsifiable concentrate containing one and a half pounds per gallon.

DIELDRIN is now available impregnated on a 30-60 mesh attapulgus clay. Such formulations are now available containing 5%, 10%, and 15% DIELDRIN. The DIELDRIN granules would appear to have certain advantages over liquid sprays where the grove has considerable ground cover. A high percentage of the insecticide is retained by the cover and does not reach the soil. The 30-60 mesh granules have the advantage of penetrating even the densest cover and their application results in a maximum deposit of the insecticide on the soil surface.

Groves or orchards under cultivation can be sprayed or treated with the granules. In either case it is advisable to disc the insecticide into the soil following application.

The granules are free flowing and can be applied quite readily with any fertilizer or distributor.

Without any field experience to go by it would seem that a 5% 30-60 mesh DIELDRIN granule formulation would be most convenient to use. By using a 5% DIELDRIN granule material you would obtain a dosage of 1-1/2 pound of actual DIELDRIN per acre by applying 30 pounds of granules per acre. Likewise, 60 pounds of the granules per acre would give a dosage of 3 pound of DIELDRIN. On the basis of work done with DIELDRIN for the control of the Japanese beetle, 3 pounds of DIELDRIN per acre will control this insect for more than 5 years. While it is not safe to assume that we could expect the same results in the case of the Hickory weevil, it does give us something to go by."

It seems likely that the foregoing will create some interest and that by the time of the next annual meeting we should have the results from the use of DIELDRIN to control the Hickory weevil.

MR. PAPE: It is my thought now if we could get a little discussion here concerning what some of you have been doing to control this pest, we might get somewhere, or at least get enough suggestions or get enough parties interested to carry on some experiments in different parts of the country.

MR. SILVIS: What company makes Dieldrin?

MR. PAPE: Julius Hyman Company is the one that sent the most literature and Shell Corporation local agents handle it. Also in Indiana the Farm Bureau Cooperative store handles it. The cost in small quantities is two pounds for 85 cents.

MR. KYHL: Is Dieldrin poison?

MR. PAPE: It's poison like all of these modern sprays, but it isn't as dangerous as Parathion.

MR. STOKE: In Virginia I have had no experience with DDT, except with chestnut, and it takes three sprays at two-week intervals to control the pest.

DR. GRAVES: What time of the year?

MR. STOKE: Apply the last spray about two weeks before the nuts ripen. That means, with us, starting in late July. You have to figure it for your own region.

MR. GRAVATT: There is literature available from the Bureau of Entomology in Washington on spraying to control pecan and chestnut weevils. They have done quite a bit of research on it.

MR. STOKE: If this ground treatment is effective, I'd like to try it. It's a lot easier.

MR. PAPE: That would be very nice if you would repeat the work in Virginia. I know that the Pecan Growers will work on the problem in the South. If we could get work done in the Central States, it would be an advantage for all of us.

MR. STOKE: In my area the control of the pest is complicated by the presence of the chinquapin.

PRESIDENT BEST: We have a surprise feature this afternoon. Dr. Graves of the Connecticut Experiment station, who, as you know, is the father of a lot of this work on chestnuts, has consented to discuss with us certain new procedures that he used in grafting chestnuts.

DR. GRAVES: We have worked with this method of inarching blighted chestnuts so long and found it so successful that I felt it my duty to tell you people something about it. It's really a method of cure for the blight on Oriental chestnuts and their hybrids. I have not found it to work well on the American chestnut.

Now, suppose we have our tree, with a blighted area on the trunk. I am assuming that the blight starts near the base of the tree as it usually does.

When you see it, you cut it out with a sharp knife removing the bark to the wood. Blighted trees send up shoots from the base, below the blighted bark. So you take one of these shoots, sharpen it at the top and insert this sharpened tip under the healthy bark at the top of the blighted area. The shoot should be a little longer than the blighted area so that you can get a spring to the shoot as you push its tip in between the bark and the trunk. Even if it goes up above and breaks the bark a little bit, it doesn't matter. This inarched shoot renews the connection between the leaves and the roots across the blighted area.

You know the leaves make the food of the tree, which goes down in the bark to the roots. The reason blight kills these trees is that it begins to girdle and sometimes does girdle the tree and destroys the connection between the leaves and the roots, so the roots eventually die. But by this method of inarching you restore that connection between leaves and roots.

Now, you'd be surprised to see how well that's worked with us. We tried it first in 1937. We have been doing it now for 16 years. Every spring we take our trees that show the blight, our hybrids and Oriental chestnuts, and inarch, and the whole thing doesn't take more than a few minutes. Then after our shoot is inarched here, we tie it with old-fashioned string. The tips of the inarched grafts should be covered with grafting wax or paraffin.

The scion will probably send out shoots which should be removed. And another thing, cut the string when you know the graft has taken above.

If the blighted area is higher up in the tree, you can use bridge graft. This, you can see, is a kind of bridge grafting. But in bridge grafting, the scion must be anchored in the bark both above and below the lesion.

As I say, we have cured our hybrids. There doesn't need to be anybody losing a Chinese chestnut tree ever, using this method. No sense in it. You can usually do this grafting in the spring about April when the leaves and the buds are beginning to show their green.

Any questions?

MR. DAVIDSON: You say you paint the wood?

DR. GRAVES: Yes, with any ordinary paint. There is a tree wood paint, I know, that's better, but we use ordinary paint.

Meeting adjourned at 4:50 o'clock, p.m.

MONDAY EVENING SESSION

"What's Your Problem?"—Round Table Discussion

Moderator: J. W. McKay

Panel: J. C. McDaniel, D. C. Snyder, Jesse D. Diller, Stephen Bernath

DR. MCKAY: In these panel discussions the moderator usually lays a little background as an introduction to the subject of the evening. This title came from a conversation with Dr. Crane. We were talking about people asking questions about their problems, and decided to have a panel discussion. Right there we chose the title, "What's your problem?"

All of us have problems to deal with in every walk of life. We run up against difficulties, and usually much of our time is taken up in solving or coping with them. At Beltsville we answer a great many letters, and a great many people ask us about their problems. In answering problems, we push the industry forward, because we remove something that is holding it back.

Sometimes the answer to a problem is found by trying to analyze our successes. In growing nut trees we may have an unusually good crop on a particular variety or tree. The question is, why does that tree bear well that particular year, and very frequently it is difficult to understand why. It is very difficult, for example, in the case of one success, to repeat that success, because the second time you try to do it, something else comes in, and you probably have a failure and, well, you don't know why. It is frequently very difficult to analyze our successes. Another way of stating it is, of course, that our successes are often Nature's gift, and we do not know the factors and the forces that go into that gift.

I want to digress here just a little bit by quoting one thing that Mr. Best said. I wish, by the way, that we could incorporate some of his homey philosophy into some of our minutes so as to really benefit by some of his remarks. I was impressed this morning by his statement in dealing with a "fairyland of nuts," you remember that language he used, "no diseases, no insects, no failures."

DR. MACDANIEL: No squirrels.

DR. MCKAY: Wouldn't that be wonderful? I was impressed with another thing he said; I wrote it down here: "People are not going to break down with emotion when we talk about the old hollow tree down in the corner of the pasture where the 'possums were hatched." Language like that, to me, strikes a very harmonious note. I want to continue this digression a little to consider the fact that all of us, in a sense, are hobbyists in nut growing.

Hobbies in this day and age are coming to be more important all the time, because of the fast pace at which we live. We need to leave our regular duties once in a while and get out in the garden or the forest where we can observe nature and get away from some of the stresses and strains that modern living places upon us.

On this trip we were taking a hike along the north rim of Grand Canyon in an organized nature walk. The trees, bushes and flowers were all labelled right down the walk, and we came to this little poem on a regulation label. It goes like this:

"If you keep your nose to the grindstone rough,
And you keep it there long enough,
You will cease to hear the birds that sing,
And the brooks that babble in early spring,
And finally all that your world will compose,
Will be the grindstone and your poor, old nose."

So that little poem strikes a real note about a person's hobby. You keep your nose to the grindstone, you will forget all these things. You need to get away from the grindstone once in a while. I don't mean to neglect your work, but I do mean to at least take a little time to go out and look at your trees and forget your troubles and relax and get away from the stresses and strains that modern living places on us.

Going back now to the subject; we asked you, through the Nutshell and through members of the Program Committee, to send in the biggest problem in connection with whatever nut species you grow.

The seventeen replies received included eight problems as follows: (1) brooming disease of walnut; (2) early vegetating particularly of Carpathian walnut and frost damage resulting therefrom; (3) delayed fruiting of chestnut seedlings; (4) season too short for ripening of fruit; (5) squirrels get the nuts; (6) failure of hicans to set fruit; (7) grafting problems under which are grouped all asexual propagation and cuttings; and, finally, (8) getting hickory trees established.

This is a rather low number, but I think out of those eight problems submitted you have a good representation of some of the things about which members of this Association talk when they come to meetings. I will first ask the audience if there is any one who would like to ask a particular question at this time.

MR. BECKER: At the Weber planting at Rockport, Indiana last year, we saw no nuts on the trees. I would like to know what is the cause for those trees not bearing.

DR. MACDANIEL: I would think that failure to bear was caused by a combination of things; lack of soil fertility, in the first place, soil physical conditions, probably insect damage and diseases like anthracnose keeping the trees from being vigorous, overcrowding now, with many of them, and perhaps to some extent genetic, some varieties that just naturally don't fruit very heavily.

DR. McKAY: Any others in the audience care to comment on that question?

MR. STOKE: Weather conditions, freezing may have caused it.

DR. MACDANIELS: My impression was that the trees were starving to death. Cutting down the competition with the weeds and feeding them nitrate would help.

DR. MCKAY: I think most members felt there that the trees were probably crowding each other.

MR. BECKER: They had never borne, had they?

MR. WILKINSON: I don't like to comment on it. My opinion is it's due to the undergrowth under the trees. Keeping the circulation of the air to the roots of the trees has an effect on its non-bearing. Up until they quit cultivating and pasturing the orchard, it bore, but after they quit, production stopped. There is a two- or three-year growth of grass and weeds, a mat on the ground, and I think it's a lack of air to the roots of the trees.

DR. MCKAY: Mr. Wilkinson, I heard the question raised as to whether the orchard had ever produced heavily or not. Can you answer that?

MR. WILKINSON: Yes, it certainly did for several years. As long as it was cared for, it was a heavy producer.

DR. MCKAY: How long ago was that, could you say?

MR. WILKINSON: That's been eight years and farther back. Nothing has been done for it in the past eight years.

MR. BEST: May I make a comment? Last year in our part of the country, which is a little bit west of the orchard we are talking about, we had almost zero weather in November before the leaves were off of the trees, and I felt that that took all the buds off our trees. We didn't have any nuts even on varieties that would bear every year. There are hardly any. And I think that cold freeze in the fall before the buds really got ready for it did a lot of damage.

DR. MACDANIEL: I believe that was a factor in the light walnut crop in that area last year, though some trees did bear.

DR. MCKAY: Of course, I think many of us fail to realize that a tree is a thing that's confined to one spot, and when it fills the ground with feeding roots and mines the soil of all nutrients near it, it's stymied, so to speak, until we give it some more food. Isn't that right, Dr. Crane.

DR. CRANE: That's right.

DR. MCKAY: And trees, when they reach out as far as they can and can't get any more food and no more leaves are allowed to fall on the ground, nature doesn't add any nutrients anymore, naturally, those trees are in a bad way and will continue to be until fertilizer is applied in some way.

A MEMBER: I'd like to trim my Persian walnuts so I can walk under the lowest limb. Does that have an adverse effect upon the bearing of the tree?

MR. BERNATH: I don't think that's good, to trim them too high. I think the lower the tree the better it handles all along. Take the fruit growers, they aim to have the trees as low as possible to make picking and spraying easy. If you prune a tree, especially the Persian walnut, too much, it will have a bad effect on the tree.

DR. McKAY: What about the effect on bearing?

MR. BERNATH: You won't have fruit for several years until the tree recovers what it has lost in foliage.

MR. KINTZEL: I have followed the orchards in California, and I have noticed they follow the practice of leaving the lower branches on the trees, and I have noticed that the lower branches have a lot of nuts on them, also. The branches are hanging down on the ground.

MR. KERR: In France and Germany, they are crazy to get wood, and so they cut off all the low limbs. I have a Persian walnut that is beside the walk, and they cut off the low limbs because they hit the sidewalk. This year I got a tremendous crop.

MR. SILVIS: I think this man has a tree that he wants to walk under. Under these circumstances he can cut off the low limbs. I agree with Mr. Bernath, however, that it will reduce the crop for two or three years.

MR. BERNATH: Yes, but he should start pruning when the tree is young. A tree is just like a child: you have to start to train them while they are young.

MR. STOKE: You must consider the tree at all ages. In the young tree Mr. Bernath is right, it will produce sooner if you leave all the leaves on. But we must consider the mature tree. The branches that are low to the ground have to have the sunlight and if they do not get it they become practically barren during later years. If the lower branches are cut back when they are young and the tree headed higher, the Persian walnut will have a trunk, say, 10 feet to 14 feet to the first limb, but these will produce walnuts ultimately. I think the gentleman is right in having the tree pruned high enough to walk under, and he will get more nuts in the long run than if he lets the lower limbs develop and then eventually cut them down.

DR. MACDANIEL: We had an example of that with that huge black walnut tree with black walnuts starting out 30 feet in the air arching down and touching the ground. But you wouldn't want to do that immediately with a young tree, take all the branches up so high.

A MEMBER: Do you have any control for the stink bug on filberts?

DR. MCKAY: We haven't worked with the control of stink bug, because it is what might be classed as one of our minor problems. The damage is not so great but what we can overlook it at the present time.

DR. CRANE: In pecan and almond growing in California the effective control measure for stink bug is the elimination of the host plants on which the stink bug breeds. Peach growers have the same problem. Stink bug will, if allowed to multiply in a peach orchard, ruin the peaches, making injuries very similar to that caused by the plum curculio. The only satisfactory method of control of stink bug injury is to eliminate the host plants on which they live, such as most legume plants, blackberry briars and other brambles. In an orchard, in a grass sod, stink bug is no problem, but where we have soy beans or cow peas or something like that growing in the orchard, or we have blackberry briars or wild raspberries nearby, stink bug is a bad problem.

DR. GRAVATT: I have filbert trees, and the stink bug gets practically all the nuts. The entomologists looked into the situation, and the condition that Dr. Crane mentioned was borne out. If there are blackberries around, it will be quite a problem to control the stink bugs.

DR. MCKAY: Now I am going to take up the problems that have been sent in by mail. The one dealing with early vegetating and frost damage to Persian walnuts was sent in by the most people.

Mr. Snyder lives in a fairly cold country. I am going to ask him to give us his ideas on this problem and what might be done with it.

MR. SNYDER: I am not qualified to discuss that problem, because we can't do anything much with Carpathian walnuts. We do have some grafted this year, and we will have one, in particular, Carpathian, No. 5—I don't know where it got that number—Crath No. 5, I believe it is, on a young grafted black walnut tree which is ripening up almost ahead of the black walnut, and both have made a remarkable growth. But so far as the spring is concerned, I don't know how they will come out.

DR. MCKAY: Mr. Bernath, what are your views? You live in a fairly cold area. You propagate Persian walnuts. What is your opinion of this problem?

MR. BERNATH: Well, there is a way to help that situation. After the ground freezes, keep that ground frozen. That will delay the growth of that tree, if you have the time and patience to keep the ground frozen.

MR. SNYDER: I don't believe it.

MR. BERNATH: Yes, it will.

DR. MCKAY: It seems to me we have a difference of opinion here between Mr. Snyder and Mr. Bernath. The question is this: During a warm spell in the spring will a tree with frozen roots grow up here in the air. That's the question.

(There was a chorus of "yes"es from the audience).

MR. STOKE: I would say that one good solution is to select late vegetating varieties. Mr. Oakes in a report to me on the blooming habits of Persian walnuts, stated that the variety Schaeffer did not start growth until the 29th of April. That is almost four weeks later than most other varieties. And I know from the tabulations that I have made that some varieties are weeks ahead of others. So let's select the late varieties that are good and worthwhile and plant those.

In my section our latest spring frost averages the 20th of April, and yet I have several varieties that do not bloom until after the first of May. That's the ideal condition.

DR. MCKAY: That's true, Mr. Stoke, but here is another point to consider. Persian walnuts have a short cold requirement, you know that. Hence, in February or early March or any time, even in January, when we have a warm spell of a week or ten days or even shorter, sap will rise in the trees, and they will start to grow.

MR. STOKE: Not all. In plants of some varieties new growth will hardly start.

DR. MCKAY: Perhaps you may have varieties that will not start, but the tendency is to start.

MR. STOKE: If you have one with that early tendency, cut it out.

DR. MCKAY: I'd like to get back to this opinion here on the question of frozen ground, dormant roots and the effect it has on the top of the tree. Now, how about our academicians over here, Dr. MacDaniels or Dr. Crane. Let's hear from one of you.

DR. MACDANIELS: It is my opinion that with a walnut tree of good size the frozen ground would have little or no effect on the buds starting growth. The twigs and the trunk would warm up to the temperature of the air, and when that happens growth occurs. Water is available from that in the trunk and the deeper roots. This would happen regardless of how the surface roots were treated.

DR. MACDANIEL: Or whether the tree had any roots on at the time.

DR. MACDANIELS: The best solution to the frost damage problem is to find trees which vegetate late enough to avoid the spring frosts. Somewhere on this terrestrial globe there must be some, because I remember years ago J. F. Jones sent out some Persian walnuts of Chinese origin. I planted three, and they did not start any growth until about the first of July, and they were still growing strong when frost hit them in the fall. Now, somewhere in between these extremes, somewhere in the climatic analogue of our region we will find Persian walnuts which will have a delayed vegetating period, and that will be the final answer. At least, I think so.

DR. MACDANIEL: I'd like to ask a question. In F-2 hybrid walnuts do you find much segregation of those for later initiation of vegetation?

DR. MCKAY: Yes, we find in those seedlings in some cases the tendency to vegetate very early and others very late. The most striking case that I know is an F-1 hybrid which is a very, very late starter in the spring. It is perfectly dormant when the other young walnuts are in practically full leaf.

We do not have any offspring from that particular one yet, but it gives us some hope that from this hybrid we may get something later.

MR. BECKER: With us I don't think this early vegetation means anything. We are in Michigan. Dark, cold weather continues until about the middle of May, when frost ends, and then all of a sudden spring breaks loose, everything comes out, and we don't have any setback, as a rule, from then on. So early vegetation matters little, means nothing, the way I feel about it.

DR. CRANE: Mr. Moderator, you ought to point out that most of the United States isn't Michigan. If we had climatic conditions that Michigan has, we wouldn't have that problem, but this problem becomes much more acute, for example, as you go south.

The north knows nothing about cold injury, absolutely nothing. If you want to see cold injury, you go south. I told Dr. George Potter that twelve years ago. He was born and raised in Wisconsin and spent 17 years in the mountains of New Hampshire. I told him he never saw any winter injury, and he said, "Why, I never heard such a wild statement in my life." Well, that was because of the fact he had never seen it. He has been in the South now for 12 years, and he says, "You made a very truthful statement." He has seen the injury.

In Oregon in 1950 or '51 we had a fall freeze. The temperature was measured by the Experiment Station in Eastern Oregon, where they are trying to grow some fruit and nut trees so they will have something else to eat besides sage brush. They had extensive plantings of walnuts, Mayette, Franquette and all of those hardy varieties, and along with them they had some Carpathians. The temperature there in the fall dropped to 36 degrees below zero, and all of their walnuts of these other strains were killed to the ground, but the Carpathian came through uninjured. In the spring of the year however it warmed up, the Carpathians leafed out and were about ready to bloom when there was a sharp freeze, and the Carpathians sure got it in the neck. So what difference does it make whether you lose the trees in the winter or you lose them in the spring? You have lost them just the same. I think we ought to hear Spencer Chase cite the history of their big collection of Carpathians in Tennessee Valley Authority. I understand from him that they have never fruited any Carpathians down there at all. It's not winter hardiness, it's this early foliation. So we have got a lot of areas that are vastly different from that peninsula in Michigan which the Good Lord designed to make a favored country in a lot of respects.

DR. MCKAY: I recognize Mr. Devitt, who is here from Canada and is well qualified to discuss Reverend Crath's work there.

MR. DEVITT: It is interesting to me to hear of this early budding and late fruiting. Along the north shore of Lake Ontario and down through the Niagara Peninsula our climate is quite consistent. There was only one year when we had a late frost—it was on May 19th. That was in the year 1936. Every other year since they have bloomed every year.

MR. STOKE: I'd like to speak of a tree Mr. Crath sent me. The tree was bearing in Toronto 20 years ago. With me it winter kills sometime in the winter each year, I don't know when. In some years it has been killed back to 5-year-old wood, and this spring I found it was all dead. This tree comes out of dormancy as soon as the sun gets warm. It's hardy in Toronto but not hardy in Virginia.

DR. MCKAY: I think you can all see why this problem is one of the most acute ones we have to deal with today. This variation over the country in the behavior of this so-called hardy strain of walnut is of great interest now to people everywhere. People are believing that it can be grown, and there are still problems we have not solved. I would like to have just a brief statement from Spencer Chase on the performance of Carpathian varieties at Norris, Tennessee.

MR. CHASE: We reported this, I think, at our Beltsville meeting several years ago. Trees we had at Norris are Carpathian types secured from the Wisconsin Horticultural Society about 1940. After two years in the nursery they were planted, and last year, 1952, was the first year that they bore any nuts. But that was simply because we did not have a late frost last year. This year, they were all frosted again. So we have, in the South, from Virginia and Tennessee to a little farther southward, a problem of early vegetation of English walnuts. We should encourage everyone to watch for any late vegetating kinds for trial in the South.

MR. STOKE: Dr. Dunstan reported two walnut trees in North Carolina, where the season is about ten days earlier than at my place in Virginia that blossomed after the first of May. I am going to investigate these trees further.

DR. MCKAY: We have about five minutes that we might devote to some other problem. Nearly all of us do grafting work of one sort or another. Do I have a question from the floor on grafting?

MR. MACHOVINA: With cleft grafts or splice grafts held with grafting rubbers, do you have to cut the rubbers?

DR. MACDANIEL: If it's a chestnut and you have it waxed, I think the answer is yes.

MR. MACHOVINA: The wax is a hot wax and the rubber does not disintegrate very quickly.

DR. MACDANIEL: Probably you will have to cut it on species in which the growth bulges up between the turns of the rubber. This is true of chestnuts in particular, possibly persimmons, walnuts probably not quite so much trouble. Let's hear from one of the nurserymen.

MR. BERNATH: I think the best way is after the union is firm enough, to cut the rubber with a sharp knife.

MR. STOKE: I'd make one qualification. I said I didn't think you had to cut rubber. I think that's true with grafting above ground. Underneath ground, with moisture around it, it should be cut.

MR. BERNATH: If you leave the rubber on and bury it, that lasts for years. Even above ground you find it sometimes.

MR. PATAKY: If you get a fast-growing callus, you have to cut the rubber band, but if it is rather slow you don't. I do a lot of budding with roses. I don't cut the rubber bands off, because they will eventually drop off. If you graft a black walnut or Persian, you will have to cut it or it will girdle the graft.

MR. STOKE: It doesn't do it for me.

A MEMBER: Has anybody done work with polyethylene film in grafting?

MR. BECKER: I hesitate to tell you my experiment. I don't think much of it. I used polyethylene bags on chestnuts early in the season, and practically every one grew, but everything else that was out in the hot sun boiled. In the hot weather of June the grafts actually cook in the bags.

MR. MACHOVINA: Did you use a bag over the whole graft, or just a tube around it?

MR. BECKER: A bag over the whole thing. I have a few Carpathian grafts that grew well. I think I have better luck with hot wax than anything else.

DR. McKAY: Our time is up. I want to thank the panel, although we didn't work you too hard. The panel is adjourned.

PRESIDENT BEST: Dr. Gravatt will show a film entitled: "It Bringeth Forth Much Fruit."

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DR. GRAVATT: Mr. President, ladies and gentlemen, you all are familiar with the fact that the chestnut blight is loose in Europe. It was reported in Italy in 1938, and it spread rather rapidly in Italy. It had been there many years before they found it. In spite of our numerous warnings to get them to watch for it, they let it get away. It has spread into Switzerland, caused a great deal of damage there with no hope of saving the larger chestnuts there or in Italy. It's spreading into Yugoslavia. They are making very energetic efforts to control the disease in Yugoslavia, trying to delay it as much as possible. It happens the forest pathologist who handles this work is a young lady, and she has got the forester and other people interested to try to hold it back as long as possible.

The threat of the chestnut blight to the entire chestnut growth in all of Southern Europe helped to bring about the organization of an International Chestnut Council and Congress. This is made-up of delegates from a number of the European countries, Spain, Portugal, France, Switzerland, Yugoslavia, Greece, Turkey, Japan and the United States. They have been meeting every other year, first for two years in succession, but the plan now is to meet every other year. They had a meeting in Spain and Portugal this past June, and the State Department paid my expenses over, and so forth, to attend as a delegate from the United States at this international meeting.

The meeting was very enjoyable. They have a very fine system there. They hire big buses to take you around over the country. Your hotel is all arranged for in advance, and you go sightseeing to the orchards and utilization plants. We have meetings just here and there along the way where we stop a half day or a day.

The next meeting will probably be held in about two years. They have decided now that the meetings will be more in the way of conferences, because the last three meetings have been partly sightseeing to observe chestnut orchards and laboratories.

The possibility of holding the meeting in the United States has been discussed by the delegates there. But it involves a lot of expense and the delegates were of the opinion that there would be a very small meeting in the United States, because the countries over there simply couldn't afford the expense of sending them over here.

The problem in Italy is very serious, because they have something over a million acres of grafted chestnut orchards, all of which they are probably going to lose, and something like a million acres of coppice growth that is going to be damaged but not such a severe loss. In connection with the work in Italy I suggested the production of a movie film that could be shown to the Italian people showing the chestnut industry and also the chestnut blight. This was to be shown in different parts of Italy to arouse more interest in watching out for the disease. They have more opportunity there of slowing up the disease if they will work hard at it, but they are not doing too much.

As some of you know when a lot of different people and agencies work on a movie film there must be all sorts of compromises. This was done by a temperamental Italian director, and other people had parts in it, so what you see is a compromise. They made 30 copies in Italian. H has been shown in many moving picture houses, and it is also on the loan basis to the United States. There are extensive film loan libraries, located in different parts of Italy, so any high school, college forestry group can borrow films showing different operations, many of them prepared in the United States and part of them in Italy.

DR. MACDANIEL: What about this so-called Korean chestnut? Is it actually a third species?

DR. GRAVATT: I don't think so, We had quite a bit of argument on this question, because in Spain where I found chestnut blight on chestnuts brought from Japan, we found the name Korean chestnut. Sometimes the Korean chestnut looks more like a Jap, sometimes it looks more like a Chinese, and usually it's sort of a blend between the two. We prefer to recognize these two species and call the Korean a natural hybrid. Both species are grown in pure form in Korea, and they intercross readily, and we do not regard it as a new species.

MR. WILSON: Are the Italian enough aware of their problem so that they will have developed an Asiatic chestnut in time to replace their present orchards, so that there will not be an interim?

DR. GRAVATT: There will be a big interim. That's an opportunity in this country to get the market before the Italians ever come back, I think.

(The film, "It Bringeth Forth Much Fruit" was shown.)

The International Chestnut Commission and the Chestnut Blight Problem in Europe, 1953

G. FLIPPO GRAVATT, *Senior Pathologist, U. S. Plant Industry Station, Beltsville, Maryland*

The International Chestnut Commission was organized under the auspices of the Food and Agriculture Organization of the United Nations. The aim of the Commission is to promote international cooperation in the study of all scientific, technical, and economic questions relating to chestnut growing. The main problem facing all chestnut culture in Europe is the rapid spread of the chestnut blight. France, Italy, Japan, Portugal, Spain, Switzerland, Turkey, United States and Yugoslavia are members of the organization. A representative from the F.A.O. in Rome serves as Secretary of the Commission. An international conference on chestnut problems was held in France in 1950, the first meeting of the Commission was held in Italy and Switzerland in 1951 and the second in Spain and Portugal, June 18-30, 1953. The average attendance at the meetings was 50 to 60 persons. I have attended all three conferences as the representative from the U.S. Departments of State and Agriculture.

The International Chestnut Commission meetings differ from the meetings of the Northern Nut Growers Association in many ways. Our Northern Nut Growers Association meets annually for 2-1/2 days while the meetings of the International Chestnut Commission last from 10 to 12 days but not every year. In Europe the members travel mostly in a large tourist bus, which carries the party for hundreds of miles, visiting nurseries, orchards, chestnut utilization plants and not neglecting the scenic parts of the route. All lodging and meals are carefully arranged for in advance. The group in Europe is made up quite largely of Federal and State professional workers, University professors, and representatives from the chestnut utilization industries.

Among the places which the delegates visited in Spain in 1953 was the Agricultural Experiment Station at La Coruna, where the Phytophthora ink disease of the chestnut has been studied extensively. They also visited the Experiment Station at Pontevedra, where new methods of propagating chestnuts are being studied. At Bilboa and at Villa Presente Nursery, Santander, we inspected plantings of Asiatic chestnuts; I found chestnut blight present on several trees at both locations and recommended immediate removal of the diseased trees. Fortunately, the Asiatic chestnuts are some distance from any native European chestnuts at each place and, according to the local foresters, the blight has not spread to the distant stands of native chestnut. Some years ago the Spanish authorities imported seed from Asia; chestnut blight probably was brought in on these nuts. All infected trees that are found are being destroyed, but a thorough inspection and eradication program is needed to control the disease before it spreads into the native European chestnut stands, from which the disease probably would spread into Portugal and southwestern France.

In Portugal we inspected many very fine chestnut orchards. These orchards are composed of grafted varieties, with only 3 or 4 varieties in each locality or region. Because of this there is a more standard nut product in most of Portugal than in the other European countries where mixtures of local varieties are frequently grown. A very large portion of the European chestnut orchards in Portugal are made up of seedling trees, topworked with local selections. In Portugal most of the orchards are located on the lower slopes and various crops are grown among the trees. In most other European countries the orchards are on rougher mountain land which is grazed.

In Portugal the State Road Department has established a number of roadside plantings of chestnut. These plantings are very productive. The State Road Department sells the nut crop to the highest bidder and uses the funds for additional roadside tree plantings.

In northern Portugal authorities have conducted a large-scale program to control the Phytophthora ink disease of chestnut by the following treatment: The soil is removed from the base of the tree and larger roots. The base and roots are sprayed with a sticker compound and then dusted with copper oxide and copper sulfate before the soil is replaced. Treatment is repeated every 5 to 7 years. Government officials secured the cooperation of owners of chestnut stands in treating practically all trees over large areas. Although this treatment for the Phytophthora ink disease was originally worked out by the Spanish pathologists at La Coruna, it has not been used extensively in Spain. The Phytophthora root disease is damaging chestnut orchards throughout southern Europe. In 1950 I noted that this disease was causing severe damage even in Asia Minor. In the southern part of the United States this same disease (here called Phytophthora root rot) caused heavy losses at lower elevations.

The 1953 Chestnut Commission meeting terminated on June 30 at the famous Palace Hotel at Bussaco, Portugal, where the Under Secretary of Agriculture gave the delegates an official farewell dinner. No definite plans were made for the next meeting of the Commission. It was the general opinion that a meeting in the United States would be poorly attended because of the expense of sending the delegates from Europe.

After the conclusion of the meeting, the U. S. Foreign Agriculture Services sponsored my trip to France, Italy, Switzerland, and Yugoslavia, to consult with Federal and local authorities on their chestnut blight problems. This disease was found in Genoa, Italy, in 1938; later it was determined that the disease was present at that time in other localities in Italy. The blight is spreading rapidly and is almost completely destroying the orchard and larger forest trees of European chestnut in Italy in localities where the disease has been present for some time. The blight occurs in many areas in northern Italy and as far south as Naples. The young chestnut coppice is not so seriously affected, but the losses caused by the blight will make growing coppice on a 10- to 20-year rotation basis less profitable than formerly.

The chestnut blight is abundantly present on the east slopes of the mountains along the French-Italian border; although it has not yet been found in France, its distribution in adjoining Italy makes it highly probable that advance spot infections are already present in France. The blight has spread into Tessin Province in southeastern Switzerland where it is destroying many of the orchards and forest trees. A large chestnut extract plant in this Province uses wood in making tannin for leather manufacturers. However, this plant, as well as some of the extract plants in northern Italy, is unable to utilize the chestnut wood as fast as the blight is killing chestnut trees.

In Yugoslavia, chestnut blight is spreading rapidly in the orchards and native growth along the Italian border. Authorities are actively cutting out all advance spot infections, to delay or possibly stop its spread across their country. In Yugoslavia, chestnut stands frequently are widely separated, a natural advantage in delaying the spread of the blight.

Chestnut blight has been controlled in western North America, where chestnut orchards and plantings are not numerous. Scattered infections have been found during the last 30 years in California, Oregon, Washington, and British Columbia; infected trees have been removed. Strict State Quarantine regulations have been enforced, to prevent chestnut blight from spreading to the West Coast.

The chestnut blight fungus is attacking three of the important oaks of Europe. The typical fanlike mycelial growth can be observed in the bark of infected oaks. In 1953 in Yugoslavia I observed vigorous young durmast oak (*Quercus petraea*) being killed by the blight. In Italy I found the disease killing pubescent oaks (*Q. pubescens*) and causing minor injury to the holly oak (*Q. ilex*). Before we can estimate the probable damage to these European oaks, we need more information on the effects of this disease on oaks of various ages and under various environmental conditions. In the United States the post oak (*Quercus stellata*) is the only oak species that has been seriously damaged by the blight.

Thus, the blight is threatening not only the native chestnut forest growth and orchards of Europe, but also the oaks. A steady extension of the blight throughout Italy can be expected. Advance infections in Yugoslavia are being cut out but how long the disease can be held back depends on future efforts along this line. Delay work in Yugoslavia also delays the time of loss of the chestnut and damage to the oak growth of Greece and Turkey. The inspection and eradication work being carried out in Spain may result in the elimination of this threat to the chestnuts and oaks in Spain, Portugal and southwest France. However, there is the possibility of the blight occurring anywhere in Europe. People working with chestnut should be on the alert to find and eradicate the first infections.

The film entitled "It Bringeth Forth Much Fruit", shown here today, was prepared at my suggestion by the U. S. Foreign Agricultural Services at Rome. It is being used to aid local authorities in Italy in attempts to delay the spread of the chestnut blight.

The Italian authorities, with assistance from the United States Foreign Agricultural Service, have purchased blight-resistant chestnuts in this country for planting in Italy. These resistant chestnuts are doing very well in

Italy so far. However, the development of a new orchard industry with the Chinese chestnut and its hybrids in Italy will be a slow process. It is expected that shipments of chestnuts from Italy to this country, which is now going on at a rate of 15 to 18 million pounds per year, will gradually decrease.

DR. GRAVATT: I will talk on while they are fixing this next film.

Much of the trouble in Italy is that so many of the chestnut orchards are overgrazed, sadly overgrazed, and as these chestnut orchards are killed by the blight, the land is going back into this overgrazed condition, which leads to serious erosions. Italy needs all the water that can be saved. The mountains are eroded down to the rock in many areas and when you get to the rock, you can never bring the soil back. It's a serious problem to meet because of the tremendous over-population. Every little twig of wood is used. As these chestnut orchards are killed it's going to be a very difficult problem to plant them again because the land is overgrazed. Protecting the plantings against sheep and the goats is quite a problem.

(The film, "The Filbert Valleys," was shown.)

MR. STOKE: I noticed them grafting chestnut trees several feet from the ground. Why are they doing that?

DR. GRAVATT: They are doing it in order to develop a quick supply of scion wood. But the procedure is bad. It is much better to graft close to the ground, and mound it up with dirt. The blight gets in below the graft if the graft is high on the trunk. They have had success grafting below the ground level and find they may get a shoot six feet high the first year.

DR. MACDANIEL: How about the incompatibility in the graft? Does that show up much?

DR. GRAVATT: We don't know yet, because they always get a certain number of failures. I looked over quite a lot of grafting of Chinese chestnuts on Japanese-European hybrids, and they are thriving. After four years they are already regular trees with big crops on them.

TUESDAY MORNING SESSION

PRESIDENT BEST: Our first paper is "Rooting Chestnuts from Softwood Cuttings" by Roger W. Pease.

Rooting Chestnuts from Softwood Cuttings

ROGER W. PEASE, *West Virginia Agricultural Experiment Station, Morgantown, W. Va.*

Some 15 or 20 years ago the West Virginia Agricultural Experiment Station undertook to develop, if possible, blight resistant chestnuts from American chestnut stock. With the passage of time the approach to the problem has changed. During the early days little thought was given to procedures for propagation, but recently the emphasis has shifted toward methods for propagation when and if there are found hardy, timber-type, blight-immune chestnuts of any species.

The practicability of budding or grafting chestnuts is debatable. We are leaving budding and grafting to experienced workers throughout the country and are endeavoring to develop a method for rooting chestnuts from softwood cuttings. Results so far are encouraging, but the work is still in the experimental stage. We do not advise anyone to start rooting chestnuts on a commercial basis, but we hope that further experimental work will be done by interested agencies.

To give complete details of several years' work would take more time than is feasible here. Circular 87, *Growing American Holly from Cuttings—Cold Frame Method*, obtainable from the Mailing Room, West Virginia Agricultural Experiment Station, Morgantown, West Virginia, gives construction details of a suitable bottom-heated cold frame. However, with chestnuts, natural shade was not used and half of the sunlight was excluded. An article in the October issue of *The National Horticultural Magazine*—"Rooting Chestnuts from Cuttings"—outlines procedure and results through 1952.

In this paper I will present a resume of our experiences and observations. Our facilities were limited so that the number of cuttings set in each case was very small. Percentages of failure or success should be taken as indicative only.

In the propagation experiments, preliminary observations were made by placing softwood cuttings in a bottom-heated cold frame at intervals during the growing season. The soil medium was two thirds washed sand and one third peat moss. Daily watering was by a hand hose. The root-inducing substance was indole-butyric acid crystals in a talc based mixture, one to one hundred. The results were completely negative.

The next season a small cold room was constructed in which conditions thought to be desirable could be maintained. Air temperature was kept at approximately 65 deg. F., fog nozzles were operated continuously except for an occasional airing of the cold room, and about 200 foot candles of white fluorescent light were delivered upon the rooting surface. The rooting medium was white, washed, building sand placed over one half inch of sphagnum moss. The moss, in turn, had been laid in a rooting bench with a hardware cloth bottom exposed to the air. The interior air circulation was maintained by an electric fan operating day and night. The soil temperature was held at 70 deg. F.

Cuttings were taken at intervals throughout the season and their basal sections soaked in a water-based solution of indole-butyric acid crystals at concentrations varying around 60 parts per million. During a 70-day period roots were formed on cuttings taken in June, July, and August. Among the successful cases the poorest result was 66-2/3%, and the best was 100%.

The young plants were fed nutrient solution and later transplanted to a light, sandy soil within a bottom-heated cold frame. Some roots were dead at the time of transplanting, burned, perhaps, by the nutrient solution. The soil temperature within the cold frame was maintained at 70 deg. F. until late in the fall, and then the plants were hardened by reducing the water content of the soil medium and lowering the temperature. All of the plants were dead when they were inspected in March.

The next year a bottom-heated cold frame was equipped with fog nozzles. The soil medium was white, washed, building sand. Softwood cuttings, treated the same as the previous year, were inserted on August 20. Cuttings from juvenile American chestnut seedling trees, juvenile Chinese trees, and mature Chinese trees were used. Within a 70 day period heavy root systems were formed on 54-6/11% of the cuttings from the juvenile Chinese trees, 50% from mature Chinese trees, and 20% from juvenile American trees. No nutrient solution was applied, the young plants were transplanted to a sandy soil in another cold frame, were hardened as during the previous year, but the soil medium was not allowed to freeze during the winter. In April the plants showed well-formed terminal buds starting to swell and turn green. Some were transplanted into pots and placed in the greenhouse; others were transplanted into a light soil in a lath house. All died subsequent to transplanting. Inspection of the roots showed severe breakage. It was concluded that repeated transplanting had been fatal, and that in the future cuttings would be rooted in plant bands or pots and transplanted only once.

It is too early in the current season for accurate results to be recorded. However, modifications have been tried and observations made. These are presented here in outline.

Type of cutting:

a. Cuttings with soft, growing tips will apparently root more quickly than hardened shoots, but the leaves tend to turn brown and the plant dies. Conversely, cuttings from short, lateral growth, well-hardened, will retain their leaves better and eventually show a higher percentage of success.

b. Cuttings made from the basal and intermediary sections of long shoots show a greater death incidence than do well-hardened, terminal sections. Both types root satisfactorily.

c. Apparently sucker shoots and water sprouts are useless.

Time of taking cuttings:

a. Cuttings taken in late May, with soft growing tips, rooted quickly—some within two weeks. On the other hand, their foliage darkened quickly, and death followed. Short, lateral shoots, well-hardened, were not available in May.

b. As the season progressed, the percentage of rooted cuttings with healthy foliage apparently rose, at least through July, but roots were formed more slowly by the late season cuttings.

Condition of parent tree:

Apparently tree vigor as indicated by healthy, dark green foliage, is more important than vigor as indicated by the length of current season's growth. In Morgantown this has been one of the driest seasons on record. Cuttings from trees with pale or brown foliage, or with foliage tending to be brittle from lack of water soon lost their leaves. Whether this was caused by the condition of the parent tree or of the individual cutting is not apparent. It is too early to determine whether or not the drought will cause a general lowering of rooting percentages this year.

Root formation:

Cuttings may or may not callus. Roots seldom if ever spring from the extreme base of a cutting. Well above the base the stem enlarges, turns white, cracks, and sends out roots. Often the bottom inch of the cutting is black and dead, with a healthy and vigorous root system above the blackened portion.

Plant bands and pots:

Plant bands are apparently preferable to small pots. The slope of the pots tends to pack the soil medium and interfere with aeration. Bands or pots less than three inches in diameter tends to cramp the rapidly growing roots.

Cold room vs. cold frame:

Last year higher percentages of success were obtained in the cold room than in the bottom-heated cold frame. This year the cold frame was definitely superior. Because construction and operation of a suitable cold room is expensive, we do not plan to continue its use in chestnut work.

Fog nozzles:

In the cold frame, fog nozzles operating during eight hours each day are apparently more effective than nozzles operating continuously.

Auxin:

No success has been attained with indole-butyric acid crystals in a talc-based powder or with untreated cuttings.

Formula for preparing auxin:

The auxin solution is prepared as recommended by G. H. Poesch in the Ohio Agricultural Experiment Station Bimonthly Bulletin, 191, April, 1938. One gram of indole-butyric acid crystals is dissolved in 125 cc. of 95% alcohol. Then 125 cc. of distilled water is added. This makes a stock solution of four thousand parts to a million in strength. The stock may be cut to the desired strength with distilled water. For late August cuttings, well-hardened, 80 parts per million is not too strong. For early June cuttings, forty parts per million appears to be adequate. The softer the cuttings, the weaker should be the solution.

Algae:

In both the cold frame and the cold room the growth of algae is a problem. The sand medium becomes crusted, with subsequent interference with aeration. The algae sometimes creeps up the stems of cuttings, coats the leaves, and covers terminal buds. Starting each season with completely clean sand and equipment will not prevent the appearance of algae over a long season of continuous operation. On August 20 of this year the interior of the cold frame, including all of the plants, was well dusted with tri-basic copper sulphate, according to manufacturer's directions. To date no effect is noticeable either on the algae or on the plants.

The various observations reported here should be verified by further tests. They are offered merely as aids to anyone planning to experiment with rooting chestnuts. When sufficient data and experience have been gained, a complete Station circular will be published.

PRESIDENT BEST: If you have any questions, please save them until later. It's been suggested that we hear from Dr. Jesse D. Diller next, and that will give our good work horse, Dr. Crane, a chance to build up again for us, because we are going to work him mighty hard.

DR. DILLER: I'd like to have the title of my paper changed to, "Evaluating Chestnuts Grown Under Forest Conditions."

Evaluating Chestnuts Grown under Forest Conditions

JESSE D. DILLER, *Pathologist, Division of Forest Pathology, Bureau of Plant Industry, Soils, and Agricultural Engineering, U. S. Department of Agriculture, Beltsville, Maryland*

During the 49-year period since chestnut blight was first reported from New York City, the U. S. Department of Agriculture has made more than 500 importations of chestnut seeds and scions, including nearly every species of chestnut in the world, as well as some closely related chinkapins and Castanopsis species. As early as 1909 the Department initiated chestnut breeding work. It was known that few, if any, of the chestnut, or related species, possess the timber-type characteristics of our American chestnut. It was also known that, in general, the Asiatic species show great natural resistance to the blight. But little, or nothing, was known about their site requirements.

In 1927 the U. S. Division of Forest Pathology began breeding chestnuts to produce timber-type trees. The chestnut breeding work was expanded and has been carried on actively to date. From 1927 to 1930, the Division conducted an extensive exploration in search of orchard and timber-type chestnut in China, Korea, and Japan, and imported over 250 bushels of chestnut seed, representing four species.

During the early 1930's the Division of Forest Pathology distributed thousands of chestnut seedlings, grown from the imported chestnut seed. The planting stock was made available to interested Federal and State agencies, as well as to owners of farm woodlands, located in 32 Eastern States. The cooperators were asked to establish small experimental forest plantings with the trees furnished them. It was believed that such wide distribution of the many kinds would readily demonstrate which ones possess the desired timber-tree form, or possessed the ability to bear large crops of nuts suitable to wildlife; and would furnish valuable information on their site requirements.

As we now know, most of these early cooperative experimental forest plantings were doomed to failure because often the chestnut trees were planted on dry, grassy areas having infertile, shallow soil. Another serious contributing factor in poor establishment was the severe general droughts that occurred over most of the eastern half of the United States in the early thirties. But despite these heavy losses, a few plantations succeeded, in part, and from these limited areas, and from a few earlier plantations that succeeded, valuable information on their general site requirements was obtained; however, we still lacked information on specific differences in behavior between the progeny, as fast-growing forest trees or nut producers in the forest.

From these early plantings we learned that (1) Asiatic chestnuts and hybrids are more likely to develop into forest trees when planted on cool, moist, fertile situations; (2) in their silvicultural characteristics they are more nearly like our native yellow-poplar, northern red oak, and white ash, than like our American chestnut and native chinkapins; (3) with respect to tolerance of shade, they are much like our northern red oak; and (4) neither the Chinese nor the Japanese chestnut has quite the same forest-type growth as that of our native American chestnut.

With this background of experience, the U. S. Division of Forest Pathology from 1936 to 1939 established a series of 21 climatic test plots on above-average sites on Federal- and State-owned forest land in eight Eastern States. Fortunately, we still had available suitable planting stock of the many kinds of chestnut, chinkapins, and hybrids for conducting such an extensive test. At this point we should also mention that from 1947 to date, the Division of Forest Pathology, in cooperation with the Connecticut Agricultural Experiment Station, also established 11 hybrid test plots in Arkansas, Connecticut, Illinois, Michigan, Ohio, Pennsylvania, South Carolina, Tennessee, and West Virginia. In 1930 the Brooklyn Botanic Garden also began breeding blight-resistant chestnuts of timber type, and in 1947 transferred this project to the Connecticut Agricultural Experiment Station.

The 21 climatic test plots ranged from one to two acres each, and were planted with more than 20 progenies represented, as well as forest-tree chinkapin and some hybrids. Nearly all of the 21 climatic test plots were fenced against deer and domestic livestock. The 11 hybrid test plots, approximately 1/4 acre each, were planted with 100 hybrids (50 furnished by each of the two agencies), and 50 Chinese chestnuts—P.I. 58602, the most outstanding Chinese chestnut from the forestry standpoint, thus far discovered. The climatic test plots were established on freshly cleared forest sites, with trees randomized, and planted 8 feet apart. In the hybrid test plots, the seedlings were planted under forest growth and the overstory trees were girdled; the seedlings were randomized in these plots, with spacing of 10 by 10 feet.

The 1- to 6-year period of testing for the hybrid chestnut, and the 14- to 17-year period of testing of the chestnuts planted in the climatic test plots are too short for final judgment of performance; however, certain characteristics are appearing with reference to blight resistance, winter hardiness, timber-tree form, early fruiting, and rate of growth. The present paper does not attempt to summarize all of the data obtained from all these climatic plots but rather to point out some striking results obtained from several widely separated climatic plots. Results from the hybrid test plots are not included in this discussion.

Discussion

A performance rating of 28 chestnuts, chestnut hybrids, and forest-tree chinkapins, tested in forest plantings for 12 to 13 years in Indiana, Iowa, North Carolina, and Pennsylvania showed that certain kinds always produce better trees than others. P.I. 58602 is the best Chinese chestnut tested thus far, as determined by performance in the above-mentioned test plots and in several plantations established in 1926. In the Middle Western States, all

Japanese chestnuts, Henry (forest-tree) chinkapins, and the "ever-blooming" Sequin chestnuts have shown poor growth or have died. They do not appear to be winter hardy.

On the basis of these findings, the Division of Forest Pathology since 1946, has made available to Federal and State agencies only one introduction of Chinese chestnut—P.I. 58602—for planting as forest trees. They were distributed in lots of 50 trees, and used to establish 1/4-acre demonstration forest plots. All are located on public-owned land on favorable forest sites where Asiatic chestnuts would be expected to do well. The underplanting-and-girdling method was recommended in the establishment of all the plots.

Chinese chestnut P.I. 58602, because of its superiority in performance as a forest tree, is now also being used extensively at Beltsville, Maryland, in hybridizing work. Nearly all of the Japanese chestnut, Henry chinkapin, and Sequin chestnuts, as well as inferior hybrids in the climatic test plots during the past several years have died a natural death or have been destroyed. They have been replaced with Chinese chestnut (P.I. 58602) replants—thus gradually converting the climatic test plots into future Chinese chestnut "seed" plots of the very best Chinese chestnuts.

During the spring of 1953 several nurserymen members of the Northern Nut Growers' Association furnished the Division of Forest Pathology a total of 2,600 Chinese chestnut seedlings for tests to determine their suitability for forest planting. These and 600 seedlings of Chinese chestnut P.I. 58602 are now being tested for performance, in randomized plots, on favorable forest sites in North Carolina, Ohio, and Illinois.

Conclusions

Of 28 Asiatic chestnuts, forest-tree chinkapins, and hybrids grown in 21 climatic test plots in the eastern United States under forest conditions, only certain Chinese and hybrid chestnuts show promise of becoming satisfactory timber-type trees. The best Chinese chestnut discovered thus far is P.I. 58602—a seed importation made by the U. S. Department of Agriculture in 1924, from Nanking, China. Foresters, as well as farm woodland owners, interested in growing Asiatic chestnuts as timber trees, should accept only planting stock that, through performance under forest conditions, is known to develop into straight, single-stemmed trees.

PRESIDENT BEST: I think that Dr. Crane has his panel ready.

DR. CRANE: Mr. President, before I start, I have a few slides here to illustrate a couple of points before we call the panel to the rostrum. (Several slides were shown illustrating sunscald injury to the Southwest side of high headed Chinese chestnut tree trunks.)

DR. CRANE: On this panel, I want to get representatives from the various states. Mr. Wilson, from Georgia. Mr. Stoke from Virginia. Mr. Silvis from Ohio. Mr. Allaman from Pennsylvania. There is another good man down there who grows a lot of chestnuts, by the name of Gibbs.

Now, there seems to be a lot of disagreement in regard to the Chinese chestnut in two or three respects. One is the problem of named varieties versus seedlings. Another big problem is hardiness, how hardy they are, these Chinese chestnuts. Where can we grow them and where are they going to fail? A third question is the ability of the Chinese chestnut to compete with other vegetation as Dr. Diller has discussed. I think we ought to settle some of these questions for once and maybe for all, or at least for this meeting, through a discussion. Nurserymen and others have emphasized that chestnuts, to be successful in the United States and hardy, should come from North China, at the Great Wall or beyond. Others don't agree, claiming that chestnuts in China are grown from the extreme south to the extreme north and that we ought to do the same in this country also.

MR. STOKE: I haven't enough knowledge on it to express an opinion. I planted a good many seeds I got from the Yokahama Nursery Company, and the nuts were rather inferior as to size. They were healthy and hardy, but I don't know where they came from. I presume they came from Korea, but I am not sure. The size and productivity wasn't too high of that seedling stock I secured there.

DR. CRANE: What do you folks think? Anyone in the audience that has an idea?

MR. PATAKY: At our fall meeting in the Ohio group we had two bushels of chestnuts from Sterling Smith. As far as I know the seed is Korean chestnut, which is obviously a Chinese variety. He had three bushels last fall and they looked identically like the American chestnut. Mr. Stoke said the quality wasn't so good in what he had. That might be true, but I tested a lot of these chestnuts from Sterling Smith, and compared them with American chestnuts. They were just as good or better than the American.

MR. CALDWELL: I spent about a year in China travelling pretty well throughout the country. I believe you will find the better seed sources in the southern part. China is like Southern Florida or warmer for part of the year and yet in the other six months it would be colder than it is right here in Rochester.

They have timber trees, some as big as 50 or 60 feet high and two or three feet in diameter. In the warmer area you find better seed by far. What Dr. Diller describes as No. 58602 is not just one tree, but a whole collection of trees from a certain area where the trees have proven their resistance not only to cold but to frost injury in the spring or in the fall, which is even more important than the straight cold hardiness. Some people have mistaken ideas about the value of seed from trees in the northern part of China above the Great Wall. This area may have intense cold in the wintertime, but not in the spring or fall.

DR. GRAVES: Dr. Caldwell is right about No. 58602 being a mixture. Dr. Gravatt could tell you about that. It is a strain coming from several trees. It's evidently a very fine type, and I think we ought to know for the record just what 58602 is.

DR. GRAVATT: Professor Reisner's 58602 that Dr. Diller has been testing so widely is made up from a collection of seed from a number of isolated valleys of the Nanking area. It is rather southern in its native home, but Dr. Diller's tests and other tests have shown that it's hardy up north and it's hardy down south. As some of you have noticed, the nuts are very variable, with a number of different types mixed in together.

Dr. Diller and I have been discussing the question of hybrid vigor. It may be involved that each of these seedlings is a cross between different local strains. We must remember that the foresters have gone into this question of hardiness in great detail. You will find that you can't plant trees in Germany in a certain area unless the parent trees grew in a certain area, with comparable altitude and latitude. Minimum and maximum temperatures and other factors are also taken into consideration.

Pennsylvania started a program along the same line. They have divided their state into about five areas, and in each of those areas they are locating sources of seed that are going to be suited to those areas. They have evidence that many of these Chinese introductions coming from way down south are going to be hardy way up north, but in this matter of hardiness you sometimes have to wait for 50 or 100 years before you are sure of your conclusions.

DR. CRANE: That's right.

The next question I was going to ask these growers in the areas growing chestnuts is how much trouble they have had with hardiness or cold injury to chestnut trees that they have had. Has there been any?

MR. STOKE: I have had none.

MR. SILVIS: We have had none in Massillon.

DR. CRANE: Wilson, how about Georgia?

MR. WILSON: None.

MR. KEPLINGER: Dr. Meader sent me some stock from seed that he brought from near Seoul, Korea in 1947. They are very productive up there at Durham, New Hampshire. I have two trees from seed from these trees. They have much more narrow leaves, than any Chinese chestnuts I have seen so far.

DR. CRANE: Are you sure they are pure Chinese?

DR. MACDANIEL: I am sure they are not. I have seen pictures and had some correspondence with Dr. Meader on them. They seem to be the Japanese species, C. crenata type, or possibly hybrid, not strictly Japanese.

MR. PEASE: I want to throw in something a little bit aside. I think we kid ourselves and the public in assuming, tacitly, that Chinese chestnuts, no matter how narrow the strain, are going to breed true or anywhere near true. Any one lot of seedlings are likely to show great variation in hardiness, disease resistance and other characters. There is a great difference between resistance and immunity. I speak this way because I have seen plenty of people selling Chinese chestnuts who actually believe they are immune, and

I have seen customers mad enough to shoot them when they have seen half of them die of blight.

MR. MILLER: When considering hardiness, climate is one thing and air drainage is another. In any climatic zone the exact location or site, particularly air drainage is important. I have my orchard on a southwest slope with perfect air drainage. I have 250-some trees that are six or seven years old growing very nicely, and I have not had any loss, even with English walnut, the Carpathian or any of the other trees. I think that many of us are overlooking the fact that air drainage and location of the orchard is one of the main things. I don't think this has anything to do with the particular seed or the varieties, but I think that is one thing that we must consider.

DR. CRANE: No question about that. Chinese chestnuts are like peaches, and they start pretty early in the spring.

MR. GIBBS: Chinese chestnuts are hardy from Maine to Florida. I think they winter kill because of unhealthy condition of the tree. The place that I did live, at McLean, Virginia, was low in a frosty place, and the first spring they killed back three times before they took off. Where I live now in the Blue Ridge Mountains, which is orchard country, the Chinese chestnut killed back in the spring, but there is nothing the matter with their winter hardiness. They stand winter cold as good as a walnut tree.

DR. GRAVES: I want to make the point that it is in part a question of age, to my mind, as to whether these trees get winter killed. I know we had some trees from the Department of Agriculture, Division of Forest Pathology, back in 1925, and in the very cold winter, 1933-34, they killed back almost to the ground. Again in the severe winter of 1943 Chinese chestnuts were killed. But I feel that when a tree is of good size with its roots down in the ground, it's not so liable to winter kill as are the small seedlings.

DR. CRANE: We have spent enough time on this matter. The question of growing seedlings as compared to grafted trees is up for discussion. Mr. Wilson is a big operator growing chestnuts in Georgia. I would like to have him tell what he thinks of this matter of seedlings versus varieties for nut production.

MR. WILSON: Dr. Crane, I am fully convinced if we ever make an industry out of this chestnut business it's going to have to be based on grafted trees of good varieties. I have one block of approximately 200 grafted trees of Meiling and Kuling. Those trees have a nice crop on this year. They have different age tops, but we have a nice crop of nuts on them. I have another block of some 260 seedlings that were planted in 1948. The crop on these trees, with the same fertilization and cultivation ranges from no nuts to a heavy crop of nuts. You can't have an industry on that kind of yield. There are probably only 30 trees out of 260 that have a paying crop of nuts. That won't go as a paying proposition. You have got to have nuts on all the trees, and I am fully convinced if we ever make an industry out of it, the grower has got to produce nuts. Trees are not enough, he can't sell the tree; he wants to keep his tree. He wants nuts to sell, and you can't get them on the seedling trees. I am fully convinced you can't do it.

DR. MACDANIEL: Have any of your grafts gone bad?

MR. WILSON: I have had no incompatibility, except on one tree. My oldest grafts are four and five years old, top grafted in place on two and three year old seedlings.

DR. CRANE: Mr. Stoke, what is your experience?

MR. STOKE: I have two trees in my yard at home. Dr. Reed gave me credit for doing the first grafting of Mollissima in this country. I don't know whether it's true or not. Those were grafted in '31. They made perfect union, and they are perfect today, and they will be perfect when I am dead and gone. I find no incompatibility between Mollissima and Mollissima. One acre of good, select varieties, grafted, will produce as many nuts as three or four acres of seedlings.

DR. CRANE: Mr. Bernath, how about the situation up in the Hudson Valley?

MR. BERNATH: My trees are of small size. We have some in bearing, but as far as having any difficulty with them or freezing back, we have none.

DR. CRANE: Mr. Snyder, how about the situation out in Iowa?

MR. SNYDER: I am not trying to grow Chinese chestnuts anymore. We have had two different lots from U.S.D.A. and both of them have gone out in the winters sooner or later. We have had nice seedling rows, and Dr. Colby sent over a collection of scions, enough to graft each one. Every one grew. This winter they are all gone. We can grow American chestnuts, but we can't grow the Chinese.

DR. CRANE: Joe, you have had a lot of experience, made a lot of observations of this matter of seedlings versus grafted varieties. What do you think of the situation?

DR. MACDANIEL: I will follow Mr. Stokes' opinion on that. I think grafted trees, if you have a compatible graft, are worth several times as much as average seedling trees. At the University of Illinois most of our trees are seedling trees. We are just getting started with grafted Chinese chestnuts.

DR. CRANE: That's the way it is with us. Anybody in the audience that has an opinion that they think seedlings are better than grafted trees?

MR. CALDWELL: I was going to say seedlings are better, but I think this is one thing everybody should realize: The emphasis has been based on early production. In many cases we have found in forest trees that early seed production doesn't necessarily mean heavy late seed production. Some of those that didn't produce early went ahead and 40 or 50 years later produced heavily. So be a little bit careful when you start swinging too heavily on early production.

DR. CRANE: Yes, but, Dr. Caldwell, we in the United States haven't time to wait. We haven't time to wait.

MR. CALDWELL: You are going to have to take it.

DR. CRANE: It's just like Mr. Wilson said. He planted seedlings in 1948, and he is telling me that most of them haven't come into bearing, so he is going to ply the axe or top work them. He hasn't time to wait. He's got to make his bread and butter out of that, and when it comes to growing nuts, we can't wait 40 or 50 years for a tree to come in. That might be all right for posterity, but we have got to be sure of it, or our posterity is not going to be able to pay the national debt.

DR. MACDANIELS: According to the experience I have had, the chestnut is only a little more hardy than the peach, and behaves pretty much the same as regards wood injury. At 30 deg. below zero the trees have been killed outright or to the ground. At about 25 deg. below they will black heart with killing of sapwood and serious injury to the bark. At 20 they will survive. This experience involves perhaps 125 seedling trees from various sources, but mostly from the U.S.D.A. It is quite likely that there may be more hardier strains that will withstand these low temperatures. The other point is the matter of grafted trees. It is my opinion that the failure of the graft is a form of cold injury related to delayed maturity of the tissues at the graft union. Certainly failure of grafts is much more persistent in the north than in the south.

My experience has been that I haven't been able to keep grafted trees. They appear to thrive for three or four years and then die. I have tried it over and over again. It appears that the grafted tree in Georgia and Virginia is one thing. In New York it's another.

MR. WALLICK: I have never bought a grafted chestnut tree that grew. They all die. And seedlings mostly do not have the kind of nuts you want. Also they may be susceptible to disease.

DR. MCKAY: I want to make one observation about our experience at Beltsville on the question of seedlings versus varieties as regards bearing. We topworked scions of some of our good varieties, like Nanking and Meiling, onto large seedlings we have at Beltsville that are poor bearers. These grafted portions in the top of these trees under poor conditions—our soil is poor at Beltsville—set tremendously heavy crops, but the nuts are smaller in size than normal, and therefore the crop is not as desirable as it would be if it were grown under good conditions. The point is that those varieties bear even under poor conditions. Bearing is a variety characteristic, and wherever it grows it will bear though it may not produce a good-sized nut.

MR. PEASE: I believe what's coming out in this discussion on bearing is also true in hardiness, growth, and any characteristic we want. We may select seeds from trees at an elevation of 6,000 feet, and still have some which will be not hardy.

DR. CRANE: That's right.

MR. SILVIS: I'd like to make a point. If in your observation you find a tree seedling in your locality that is producing good crops plant that. Don't get one from Georgia. We can take a little bit of advice from the fruit grower, and not plant too much from the south, even though it came from China.

DR. GRAVATT: I'd like to Comment about conditions in Europe with reference to seedlings and varieties. The general practice there is for each little farmer to graft from the best variety in his section, especially in Italy where you find hundreds of varieties.

In Portugal we were all very much impressed with one area where the government has had an active program in persuading the chestnut owners to topwork all their trees to three varieties. These varieties are very good ones, and they are getting a very greatly increased price on account of the high quality and uniformity of the nuts they export.

It seems to me that in the discussion on the Chinese chestnut in this country we have done a little bit of injustice to the seedlings, so far as the discussion has gone. I am in perfect agreement with what's been said about the low production the first few years, but over on the Eastern Shore Mr. Hemming's trees are producing just about as much in the way of a crop as the tree can bear, and the grafted varieties there don't produce any more than his 17 or 18 seedlings.

DR. MACDANIEL: I believe Hemming has some exceptional seedlings in that lot.

DR. GRAVATT: Yes, they are very valuable, don't misunderstand me. After the first ten years you may find a seedling orchard is going to produce a very good crop, tree by tree. We have had a lot of experience, similar to that reported in New York, with grafted trees dying. We get seedling trees dying, too, but I agree that there is more damage from fall freezes, spring freezes and perhaps from straight low temperature winter injury with the grafted trees than with the seedling trees. Furthermore, I am very critical of the tactics of some of the nurseries. They have grafted on seedlings of absolutely unknown origin or mixed origin. They will take a South Chinese variety and graft it on seedlings that for hundreds of years have been grown in North China. That's just inviting trouble. The nearer you can get to having seedling and scion from the same climatic origin, the better off you are. In fact, we have advised growers to get seedlings of the Nanking and graft Nanking on them.

Dr. McKay is doing a lot of good, basic research work on this problem, and he will have more information for us in times to come. I am firmly convinced that we are going to come some day to the grafted chestnuts, especially in the South, because a lot of the southern producers right now are giving a black eye to Chinese chestnuts, because they are shipping lots of mixed nuts, and by the time they get to the consumer half of them are rotten. This will ruin the market. We have been buying some six or seven thousand pounds of nuts to ship to Italy, and we know something about the conditions of nuts when they reach us. There is no quicker way of killing a market than to be shipping in a whole lot of nuts that are going to spoil or are in the process of spoiling when they reach the consumer. Grafted varieties are one way of getting away from this, especially in the South.

MR. WILSON: I am far enough south so that in peach production we often have winters so warm that the trees don't wake up. This question of rest period is quite important with us. We have a warm winter, and the

Mayflower peach just keeps on sleeping. Eventually bloom will break, and a little peach will sit up there waiting for the leaf to come out. There is apparently a rest period with the Chinese chestnut there also. The time of breaking of the rest period in my seedling trees varies as much as three to four weeks, and that would lead me to believe that, in the long run, we will have to plant locally adapted varieties.

PRESIDENT BEST: I am sorry that we have to stop this very interesting discussion.

At this time is there any item of general interest to the group that anyone would like to bring up?

MR. MILLER: For sometime I have been considering the desirability of changing the name of the Northern Nut Growers. I am inclined to think that maybe some of our southern friends or from the Far West or Southwest would be a little dubious of joining the Northern Nut Growers, because they think we are perhaps exclusive for the north tier of states and we didn't want them.

I thought perhaps the International Nut Growers, or the United States Nut Growers Association were names worth considering. I think that would have a desirable psychological effect on our membership. We are a big organization, and I think a lot of people would think it was a whole lot larger if the name would imply that. I think the "Northern Nut Growers" just looks like we are concerned with the northern tier of states, and I think we would do a whole lot better by changing the name. I would like to have some suggestions. Possibly, it could be American Nut Growers.

MR. KERR: Mr. Chairman, I am a charter member of the American Farm Bureau, and that goes over big. It's a real success as an organization, and I think the American Nut Growers—take in South America and North America—would hit our proposition about right.

PRESIDENT BEST: All right, is there another suggestion? We mustn't take so much time on this, but it is mighty important.

MR. BECKER: My final opinion is that it's best to leave it as it was.

MR. STOKE: It seems to me that this matter was well decided some time ago. We have certain definite problems to work out. I think we had better stay on those problems and work them out before we spread over the whole universe. We will have too many other problems coming in our lap.

MR. DAVIDSON: That matter was taken up some five or six years ago, and for the reason that Mr. Stoke mentioned, the fact that we have special problems and the very difficult problems that don't concern southerners was the reason for voting that proposition down before. I think it would be better, at least, for us to consider the matter rather thoroughly before we vote on it, maybe postpone it until another year.

DR. MACDANIELS: It just occurs to me that the Northern Nut Growers Association was formed to tackle problems that weren't being covered anywhere else. There are other local organizations which are concerned with the Persian walnut and the Northwest Filbert and the Southern Pecan. The Northern Nut Growers Association was organized to save America's nut heritage, as somebody said, in a rather restricted area. Possibly the time has come to get into a larger organization with a greater scope, but I will say with Mr. Davidson that we want to consider very carefully what the gain or loss might be for the change in emphasis.

PRESIDENT BEST: Would someone make the suggestion here that we keep this thing in mind for a year and maybe at our next meeting take a little time to discuss it thoroughly.

DR. CRANE: I'd like to make a few remarks and offer a motion. I believe I am correct as to the history of the organization when I state that the oldest nut growers' organization was the old National Nut Growers Association, and that covered the nut interests of the country of all kinds. Then out of that came the National Pecan Growers Association, and almost at the same time, the Northern Nut Growers Association. The old National Nut Growers Association folded up, as did the National Pecan Growers Association. They were victims of the depression. I think we could discuss this at great length and not get anywhere, and therefore I make the motion that the president appoint a committee of three members to study the possibilities, both advantages and disadvantages of a change in the name of the association and report back to the association their recommendations at the next meeting.

(Motion seconded.)

PRESIDENT BEST: The motion has been made and seconded that we appoint a committee to handle this thing and report back to us. Is there any discussion?

DR. GRAVATT: I'd like to point out that research work is being started in Europe that is going to be very valuable to us. They are now working on the Chinese chestnuts on a very large scale, starting in Yugoslavia, France, Switzerland, and they are already doing quite a bit of breeding work in Spain and Portugal along these lines. The things that they develop, will be Chinese chestnut hybrids so they are going to have the same problems in Europe working with the chestnuts that we have here. In the past they have been working entirely with the European chestnut. I think we are now on a basis whereby the European growers can feel that they can profit by taking our publication, and, that both continents will benefit.

DR. GRAVES: Do you put that as an argument for changing our name to American Nut Growers?

DR. GRAVATT: I don't think "American" would help at all. And, furthermore, when you talk about "American", a lot of people think of South America.

MR. SLATE: It's what the members get out of the proceedings and meetings that brings them in and keeps them, it's not the name of the organization.

DR. MACDANIEL: Mr. President, as former state vice-president in Alabama, Florida and Tennessee, I don't believe the change of name would result in any great immediate increase in membership in the Southeast.

PRESIDENT BEST: Now, are you ready for the question?

(The question was called for, and carried unanimously.)

Development of the Nut Industry in the Midwest

J. F. WILKINSON, *Rockport, Ind.*

The development of the northern nut tree industry in the midwest really began about 1910. Prior to that time W. C. Reed and son of Vincennes, Indiana had done some experimental work with the Indiana and Busseron varieties of pecan, as they had located these two parent trees. E. A. Riehl of Godfrey, Illinois had been experimenting with the walnut and chestnut, and it was at this time that T. P. Littlepage, R. L. McCoy and established our nurseries here in southern Indiana.

We then began the search for the best parent trees for propagation in the midwest.

We located Warrick, Hoosier, Major, Greenriver, Posey, Kentucky, Butterick and several other varieties most of which have since been discarded.

A number of varieties have since been introduced, by Messrs. Gerardi, Whitford, Snyder, Burkhart, Bolten, and others who are either nurserymen or propagators, of pecan, walnut, hickory and chestnut.

The Littlepage and McCoy nurseries were discontinued about thirty years ago though I have continued the search for new and better varieties, and several years ago located, named, and introduced the Giles pecan, in southeast Kansas which is proving very satisfactory. I have recently located, named, and am now introducing a new variety, CHIEF, from Illinois. This is the largest northern pecan that I have ever seen and it promises to be an outstanding variety.

In the territory from southern Indiana to eastern Kansas are countless thousands of native pecan trees in the valleys of the Ohio, Mississippi and Missouri rivers and their tributaries.

On the uplands in this same territory, the black walnut is found almost everywhere. Thousands of pecan and walnut are of suitable size for top-working and could be made valuable by being grafted over to these fine varieties. These may be found in any quantity from a single tree to a native grove (especially pecan) of thousands of trees.

One of the largest pecan groves is in Gallatin county, Illinois along the Wabash river where it has been estimated there are as many as twenty thousand pecan trees of bearing size in one locality.

Other sections where large native groves may be found are in Henderson county, Kentucky near the mouth of Green River, along the Mississippi river in western Kentucky, across the river in southern Illinois, along the Illinois river in central Illinois, along the Missouri river in central Missouri, in eastern Kansas, along the Neosho and Spring rivers, and in Bates county Missouri along the Osage river, in southwestern Missouri.

It has been my pleasure to visit one or more times each of the above places as well as every other section of note where the northern pecan grows naturally.

One of the most interesting places that I have seen is in Bates county, Missouri. I was there in May to top-work trees for Mr. Wesley Heuser, where he has a tract of land along the Osage river on which there is a large native pecan grove making it a profitable possession. Mr. Heuser is increasing its value by planting budded, or grafted trees in the open land and top-working the small native seedlings.

Adjoining this place is one owned by Mr. Fred Marquardt who recently bought it from the estate of the late J. F. Tiedke who had spent years of work there cleaning up the native grove, and top-working the small seedlings to the better varieties. Mr. Marquardt told me there was an estimated four thousand bearing size native trees, and two thousand top-worked trees most of which are of bearing size and many of them top-worked as long as twenty years ago. Mr. Marquardt is taking splendid care of this place making it a profitable as well as a most beautiful nut orchard.

Mr. Tiedke in topworking these small trees, selected those as nearly as possible in rows giving it the appearance in places of a planted orchard.

Along the Illinois river in central Illinois is a great pecan section. It is there that Mr. R. B. Best is located, and he probably has more grafted and top-worked trees than any other person in the midwest. The late Charles Stephens of Columbus, Kansas, had topworked several hundred trees in southeastern Kansas and Stanley Walberts planted a 35 acre pecan orchard there at Columbus that at the last time I visited it was a beautiful and well kept orchard.

Mr. W. F. Thielenhaus of Buffalo, Kansas is doing a lot of work there both in planting and top-working trees.

In western Kentucky, Professors W. W. Magill, and W. D. Armstrong of the University of Kentucky with county agent John B. Watts of Hickman, Kentucky cooperating, interested Mr. Roscoe Stone, who had a large acreage of land in developing the young seedling pecan trees by top-working them to better varieties. Mr. Sly and I went there the first time in the spring of 1948 and each spring since then we have worked trees on this land, and for others around Hickman to the number of possibly 500 trees.

Last year a number of the trees that were worked in the spring of 1948 produced quite a few nuts. I was there in May at which time there was a splendid crop of nuts on these trees. On August 3, I had a letter from Mr. Watts stating "I feel that many of these trees will bear a good crop of nuts this year, and although we are having a drought here, the trees on the Stone farm are not suffering much.

The largest planting of nut trees that I know in the midwest is that planted by the late Harry R. Weber near Rockport which consists of about 70 acres mostly walnuts, with some pecans, hybrids, hickories, and filberts.

Many smaller plantings of nut trees have been made throughout the midwest and thousands of seedling trees having been top-worked.

Most of the native walnut trees through this section have been cut for timber and the native chestnut has been killed by the blight, making a shortage that should be replaced with the better varieties of walnut and the Chinese chestnut.

The earlier plantings of the Persian walnut from France and England were not hardy in the midwest but the Carpathian walnut from Poland seems to be doing well.

Some parts of this territory are suitable for almost any kind of nut trees. There is a vast field in the Midwest awaiting development in nut culture.

Some Aspects of the Problem of Producing Curly-Grained Walnut

L. H. MACDANIELS, *Cornell University, Ithaca, N. Y.*

About 15 years ago a tree of the Lamb Curly Walnut was planted at Ithaca, N. Y. After the tree had grown to a height of about 12 feet, it was topworked about 8 feet from the ground to scions of the Cornell variety of Black Walnut with the idea that it would be possible to grow a trunk of curly walnut and a top of a named variety. The tree grew rapidly and in the fall of 1952 had a trunk 10 inches in diameter at the base. Sometime in 1952 the tree became infected with bunchy-top disease and was cut in an attempt to eliminate this disease from the premises. It was expected that the trunk would show figured curly grain and plans were made to have at least a part of the log cut into veneer. On cutting the tree, however, and examining the wood, there was no evidence of curly grain detectable either by casual personal observation or from samples sent to the Forest Products Laboratory at Madison, Wisconsin. This, of course, was a disappointment because J. F. Wilkinson had shown samples of walnut grown from scions of the Lamb Walnut obtained from the late W. B. Bixby which showed evidence of curly grain. A photograph of the wood secured from Mr. Wilkinson is shown in figure 1. Wood samples from a tree growing at Beltsville, Maryland, which was also secured from Mr. Bixby by C. A. Reed, does not show evidence of curly grain.

The simplest explanation of the failure of the tree in Ithaca to show curly grain would be that somehow the tree was not properly labelled or that scions were mixed in propagation and that the trunk was not derived from the original Lamb Curly Walnut. However, the fact that only a few trees were concerned makes it improbable that trees were mislabelled in the Ithaca planting and there is no good reason to believe that the tree planted at Beltsville was not authentic.

[Illustration: Fig. 1. Radial face of wood of grafted Lamb black walnut grown by J. F. Wilkinson. Wavy or curly grain is apparent on right side which is the outer part of the log (about natural size).]

Another possibility is that the original Lamb Walnut was a chimera. Such a tree would have mixed tissues in its growing points, some having the curly grain character and others not. In such a tree some scions would produce curliness and others straight grain. It may be that these were mixed in the original collection.

A third possibility is that curliness is produced by the interaction of several factors, one a tendency to curliness inherent in the Lamb tree and the others environmental such as growth rate, nutrient supply, the nature of the soil or other such conditions.

Theoretically curly grain in walnut or any other tree is related to the nature of the growth of the cambium layer. In normal growth the cells of this layer are much elongated as seen in tangential section and are relatively straight. The nature of these cambium cells is shown in figure 2.

[Illustration: Fig. 2. The cambium of a straight-grained black walnut tree as seen in tangential section. The nature and regularity of these cells determines the nature and regularity of the cells of adjacent wood and bark (x 150).]

It is well known from studies of cambial growth that irregularities in the growth of the cambium are reflected in the irregularities in the shape and position of the wood fibers and vessels, which it forms. Ordinarily, if the cambium is wounded, the first cells formed are irregular in shape and orientation but after a wound is healed over the cambium cells resume their normal position. In parts of trees in which the grain is irregular or confused such as in the inner angle of crotches the shape of the cambium cells determines the nature of the grain beneath as shown in figure 3 (Ref. 1). This has been established also in the study of the nature of spiral-grained Douglas Fir and in various experimental work where it has been possible to change the direction or extent of the cambium cells through various experimental means. (Ref. 2)

[Illustration: Fig. 3. Section through cambium and underlying wood in a crotch of an apple tree where the grain of the wood is not straight. The shape and direction of the wood fibers or grain of the wood, and bark is determined by the shape and direction of the cambium cells that form them (X 100).]

There seems to be no doubt, therefore, that curly grain in walnut is directly related to the curly condition to be found in the cambium, which produces such curly grain. The basic question to be resolved is what makes the cambium of a curly-grain tree assume the curly or wavy character. As indicated above, one hypothesis is that several factors may be operating. For example, a tree might have the inherent capacity to produce wavy grain but would only do so under special environmental conditions. These environmental conditions might be related to rapidity of growth, water and nutrient supply, or various other habitat characteristics, which affect the nature of growth. The fact that the tree in question at Ithaca was growing rapidly might have been responsible for the failure of the curly grain to develop. There is evidence that trees with figured grain grow slowly. (Ref. 3, 4) On the other hand the specimens from the tree at Beltsville, Maryland, were from a slowly growing plant and did not show curly grain.

Another hypothesis is that development of the curly grain is dependent upon the foliage of the tree. This has been demonstrated to be true in instances where the foliage of fruit plants determines the characteristics of the growth of the trunk and roots and of the fruit itself. (Ref. 5, 6) It might be, therefore, that the failure of this particular trunk to show curly grain is related to the fact that the top of the tree at Ithaca was of another variety than the original Lamb. Possibly the foliage of the original variety producing the curly character is necessary to produce the curly grain. An argument against this interpretation is that the tree at Beltsville, Maryland, is not topworked.

It would be valuable at the present time to survey all the trees of the Lamb walnut, which are growing in various parts of the country, to see under what circumstances they may be showing the curly characteristic of the original tree. Dr. M. Y. Pillow of the Forest Products Laboratory at Madison, Wisconsin, in an unpublished report, has pointed out that it is possible to determine the curly nature of the grain by shaving off the outer bark, exposing the inner bark just outside of the cambium. Inasmuch as the same cambium cells form fiber cells both on the inside to make the wood and towards the outside to make the bark, the direction and nature of the fibers in the bark are a direct indication of the direction of the fibers underneath the cambium in the wood.

The appearance of the normal straight grained wood and bark and wood and bark of a curly grained tree are shown in figures 4 and 5. Shaving off the outer bark in this manner will not harm the trees, if it is done carefully so it would be possible to make this survey without injury to the trees. Examining a number of trees of the Lamb walnut in this way and finding that some were curly, might give evidence as to the conditions under which the Lamb walnut will produce curly grain.

Dr. Pillow of the Forest Products Laboratory, kindly furnished me with his file on curly and birdseye grained wood. In this file is a very interesting group of manuscripts and letters including a report from Mr. Willard G. Bixby reporting a trip to New Hampshire to study the occurrence of birdseye maple and also his early experiments with the Lamb walnut. The Lamb walnut trees at that time were too young to give any indication of curly grain. Other letters of interest on the subject were from Mr. J. F. Wilkinson, A. S. Colby and C. A. Reed. These letters mention the desirability of propagating figured walnut but aside from indicating that trees of the Lamb had been propagated there was no indication that curliness had developed. The first definite indication that curliness would develop in a grafted tree was reported by Mr. Wilkinson (Ref. 7) at the Norris meeting of the Northern Nut Growers Association. At that time the wood photographed in figure 1 was shown.

In the literature somewhat conflicting reports are found as to whether or not curliness will show up early in the growth of a tree or late. Apparently it was possible to trace curly grain into the twigs a few years old in the original Lamb walnut (unpublished letters). Various statements, however, indicate that curliness may not develop until the trees are 20 years old or more. It would seem that with the propagation and introduction of the Lamb walnut in 1926-27 and distribution soon thereafter it ought to be possible to locate and examine these trees which are now more than 20 years old.

[Illustration: Fig. 4. Slightly enlarged photograph of black walnut with straight grain in the wood (light-colored area) and also in the bark (dark-colored area). U. S. Forest Service Forest Products Laboratory Photo.]

In the various literature and other material available on the subject of birdseye and curliness, it appears that the birdseye grain is different in its origin from curliness although both may be related to the functioning of the cambium and definitely seem to be related to slow growth. (Ref. 8)

[Illustration: Fig. 5. Slightly enlarged photograph of black walnut with curly grain in the wood (light-colored area, upper left) and also in the bark (dark-colored area). U. S. Forest Service Forest Products Laboratory Photo.]

Curliness is reported in other kinds of trees. Curly grained white poplar has been propagated from hybrid trees by growing cuttings of shoots from the roots of the curly trees (Ref. 9). In Sweden it has been possible to grow figured birch, much of which has the curly type grain. In birch, seedling strains producing curly grain have been developed and are being grown. It is of interest to note that with these birches, the trees with curly grain grow only about half as fast as the normal trees and have to be staked during their early growth years in order to make straight trunks or to stand erect (Ref. 4).

The original Lamb walnut tree was curly throughout. Other trees, particularly maples and birches may be curly only in part of their trunks and sometimes only in restricted segments. Trees frequently have curly grain at the base where the trunk joins the roots but not elsewhere. Such curliness may be related to the shortening of the curve where the root joins the trunk, thus causing distortion. W. G. Bixby states (Ref. 3) that a birdseye maple tree 170 years old was only about a quarter as large in diameter as normal trees of the same age. I know of no comparison of curly walnut with other types of walnut. The original Lamb walnut tree was apparently a very large one.

In conclusion, it is obvious that our knowledge of the possibility of producing curly grained walnut logs by grafting is as yet incomplete. Much more information is needed and at the present time undoubtedly much can be gained by examining the Lamb walnut trees, which are growing in various parts of the country. This can be done without seriously injuring the trees as described earlier in this paper. Those in the Northern Nut Grower's Association, who have Lamb trees are urged to examine them to find out if we can gain further useful information regarding this rather important subject. Obviously, if it is possible to grow curly walnut through vegetative propagation, we should know under what conditions a grower can expect to successfully produce a curly grained log.

References

1. MacDaniels, L. H. The apple tree crotch, histological studies and practical considerations. Cornell Univ. Agr. Exp. Sta. Bul. 419: 1-22. 1923.

2. —— and Otis F. Curtis. The effect of spiral ringing on solute translocation on the structure of the regenerated tissue of the apple. Cornell Univ. Agr. Exp. Sta. Memoir 133:1-32. 1930.

3. Bixby, W. G. Field work at Warren, New Hampshire. Unpublished Report. 1932. (On file with U. S. Forest Products Laboratory, Madison, Wisconsin.)

4. Heinkinheimo, O. Om odling ar masurbjork (The cultivation of figured birch). Skogen 27:165-167. 1940. (Translation in U. S. Forest Products Laboratory.)

5. Heinicke, A. J. Influence of scion leaves on the quality of apples borne by the stock. Am. Soc. Hort. Sci. Proc. 24:143-146. 1927.

6. Swarbrick, Thomas and R. H. Roberts. The relation of scion variety to character of root growth in apple trees. Wisconsin Univ. Agr. Exp. Sta. Res. Bul. 78:1-24. 1927.

7. Wilkinson, J. F. The grafted curly walnut as a timber tree. Northern Nut Growers Ass'n. Proc. 39:139-142. 1948.

8. Pillow, M. Y. Dormant buds are not the cause of bird's eyes in maple. Wood Working Industries 5:26-27. Sept. 1929.

9. Grober, Samuel. Science shows the way. Chemurgic Digest 5:152. 1946.

DR. CRANE: Dr. MacDaniels, the idea prevails on the part of some I know that this curliness would show up more at the base or crown of the tree than it would be likely to show on the trunk, and at the base of large limbs we tend to have curliness. Of course, the Lamb walnut was supposed to be curly throughout, but in the case of other trees I wonder if that's true. You have emphasized the change in the direction of the grain at the crown between the root and trunk and in the crotches. I wonder just where would be the best place to scrape this bark or pare it down in examination to determine whether it was curly or not. Would that be, in your opinion, more likely to show up on the trunk of the tree or base of some limb or near down to the crown?

DR. MACDANIELS: I'd be inclined to take it where you can work at it most easily; down towards the base. If the grain is curly only in restricted areas the log is not very valuable.

A MEMBER: I have been told by a sawmill man that he could tell by the convolutions of the bark. Instead of being straight, they would be fluted.

DR. MACDANIELS: That might be. I was told during the First World War when they wanted straight-grained spruce for airplanes they found they could tell a straight-grained spruce from a spiral, so they wouldn't waste their time getting logs with spiral grain.

TUESDAY AFTERNOON SESSION

PRESIDENT BEST: The first item on the program is the life story of the Late Reverend Crath and the Crath Carpathian walnut in Ontario. We are going to have Mr. L. K. Devitt of Toronto, Canada, get into this subject for us. Mr. Devitt did know Reverend Crath since 1934. Mr. Devitt supported his expedition to the Ukraine in 1934. He has a few slides for us and then he is going to talk to us about a number of features.

Mr. Devitt is in the school system in Toronto, and he is a graduate of the University of Toronto, and so without further introduction, take over and give us your story.

MR. DEVITT: Thank you. Mr. Chairman, ladies and gentlemen, when I wrote a letter to the secretary of the Association about Reverend Crath, I thought it was also fitting that at the next meeting I should come here and say a little more about the life and work of the Reverend Crath and the Crath Carpathian walnut in Ontario and the progress for the last 20 years.

Late Rev. Paul C. Crath

L. K. DEVITT, *Toronto, Ontario*

Rev. Crath was born near Kiev in Greater Ukraine, Poland, in 1883. He was the son of an Agricultural College Professor. It is assumed that he enjoyed the life of the upper class, being a graduate of two universities; and speaking fluently at least six languages of Central and Western Europe, and having travelled almost everywhere in Europe. He possessed a wide knowledge of the peoples, the history and the culture of all the Central European Countries.

He migrated to Canada in 1908 and settled in Western Canada. He was employed at various, clerical occupations before entering the Theological College of the University of Manitoba from where he graduated as a Presbyterian minister in 1922.

He was the minister of a Ukranian Presbyterian Church in Toronto for two years. From 1924 to 1936 he served as a Presbyterian missionary in Poland, organizing some thirty missions in Galicia and Volynia. For some years before the war, he spent considerable time on a farm near Welcome, Ontario, building up a European Nursery and in the winters he served with the Home Missions mostly in Western Canada. During the last ten years of his life he had to curtail his activities more and more, owing to poor health and a heart condition.

I met Rev. Crath when he was on furlough in 1934.

I went to the National Exhibition and among the various exhibits I came across a rather unique exhibit of nuts, grown by the late Geo. H. Corsan, Echo Valley, Islington.

In the course of our conversation along came the late Prof. Jas. Neilson and we continued to talk about nut-growing. Prof. Neilson was interesting indeed. I could see he was a sincere man and most enthusiastic about the subject. He told me there was a Presbyterian Ukranian missionary in town who had brought out some hardy English walnuts from the Carpathian Mountains—a variety which he was sure would survive in Ontario and the Northern States and that it had great possibilities. The missionary was returning to Europe to bring out a shipment but needed, backing for the expedition. I met Prof. Neilson the following day. The sum required was $400.00 and he agreed to guarantee the sale of $400.00 worth in the U.S. at least. The next day I met Rev. Crath at the Exhibition display. We met off and on for two or three days. I could see no flaw in the project, so I raised the $400.00 by a bank-note. The banker thought I was crazy—and the missionary was on his way by the end of the week.

He arrived by mid-September and having had so many charges in the Ukraine, he knew where to go and just when the crop was being harvested.

The walnuts were selected, dried, boxed and shipped by the middle of October. The shipment arrived in Toronto the first week of November—nearly two tons of them. I received with them, a bill of lading with port charges, export duties and freight. I was out another $100.00.

In two weeks, the Winter Fair opened and Mr. Corsan was invited to put on his nut exhibit as an attraction. In the meantime he was on the radio once a week to talk on health, food and various subjects, always getting around to nuts as a food—and this new discovery, the Carpathian walnut. The radio broadcasts brought interested people right to his exhibit. He gave an hourly talk on nuts and a pamphlet was given out. The Winter Fair sales grossed $300.00 and there was another $100.00 on follow-up sales by Christmas. The situation was at least easier.

Prof. Neilson before Christmas had taken ill and passed away in February. However into the picture came another man, H. J. Rahmlow, secretary of the Wisconsin Horticultural Association with about 600 affiliated societies. He wrote an article in the Country Gentleman and circularized the expedition of Rev. Crath to the Carpathian Mountains. We sent four shipments to Mr. Rahmlow in 25 and 50 lb. lots. Sales came in from all over Canada and the United States until spring. By spring we had cleared all expenses and had about $200.00 on hand, but the next problem was, what to do with the rest of the walnut seeds?

On Mr. Corsan's Echo Valley, there were two fields, one in the valley and one over the road. We broke up out of sod an acre in each field and planted about 40,000 seeds. Rev. Crath took to a farm near Welcome, Ont. about another 20,000. Our plantation required a good deal of attention, work and expense during the growing season. However 90% of the walnuts germinated and grew to trees about 6 inches high. Over 30,000 trees survived the next cold winter.

The following year we could scuffle them between the rows. Our nursery required less care and expense. During the summer they grew about a foot higher (15 in. average) but developed a very thick carrot-like tap root with numerous root hairs. By autumn 1936 it was evident we had to transplant. The seeds were planted originally 8 inches apart. So we divided up the lot by each taking one out of every three trees, thus leaving the trees in Echo Valley now 2 ft. apart. Rev. Crath took his trees to the farm at Welcome, 80 miles east of Toronto. They were planted on a slope below a thick woods from where melting snow and spring rains kept the field cold and wet until mid-summer. Rev. Crath's trees were practically a failure; in fact the area seemed to be unsuitable for walnut seedlings. Mr. Corsan's trees continued to grow, but even here the soil did not seem to be the most suitable.

I took mine to a sandy garden soil that had been under sod for 20 years. The sod was broken and thoroughly disced. The spring was wet and very favourable for transplanting. The trees on this soil grew very well without any fertilizer at all; nor did they require any spraying. The trees continued to grow deep and do better each succeeding year.

In the spring of 1939 I started to sell trees wholesale to the Dominion Nursery, Georgetown, Ont. Mr. Bradley, the president, carried more novelty items in his catalogue than any other nurseryman in Canada. I continued to plant more seeds until 1939.—The war stopped further importations, and I sold out all the trees by the spring of 1943.

So from my nursery probably went out some 10,000 trees; the weaker seedling always perished during the winter. From Mr. Corsan's nursery, another 10,000 trees—about half of these went to his son, Hebden Corsan in Michigan. Rev. Crath's nursery yielded not more than 5000. He imported a number of cherries, plums, grapes and others fruits, all of which did not do too well either.

During the period before the war, orders came in from everywhere—from British Columbia to Nova Scotia, even Newfoundland, besides nurseries in the United States. Orders from the prairie provinces were dissuaded but some customers insisted on a trial basis. Walnut seed, the first two years went mostly to Western Ontario, British Columbia and Nova Scotia. By 1939 the seedling nursery business that I had apparently fallen into, looked good. Rev. Crath and I talked the situation over. We decided to go to the country, lease some land. I would select the land and continue to grow seedlings and besides, import selected grafts to develop in Canada a hardy high quality grafted walnut tree.

In September we prepared to make another expedition. My banker was most agreeable this time. Rev. Crath got as far as New York where, awaiting the S. S. Batory to sail, the war broke out. The S. S. Pilsudski was sunk just out of Gdynia the next day. The S. S. Batory never did sail back to Poland. When he arrived home we went to the bank on a Saturday morning. The travellers' cheques were cancelled.

Rev. Crath in the 1936 expedition brought out a shipment of walnuts selected from the most northerly port of the Ukraine for Mr. Weschcke, St. Paul, Minn. I am not familiar with this part of his work.

Rev. Crath was a cheerful soul, an interesting and pleasant individual to talk to. He loved people and, especially, meeting people. He possessed a great love for humanity; he bore malice toward no one and charity to all except the Bolsheviks. He was a restless man—"always on the go". One could see he preferred to be missionary rather than a resident minister. Although he was away a good part of the time he was dearly loved by his family.

Shortly after his death, as an appreciation of his services as a minister among Ukranian families, special memorial services were held in Toronto, Oshawa and Detroit. I was invited to attend the Toronto service.

On a visit one day last August, 1952, to places where his Carpathian walnut trees were coming into bearing, he examined them and gazed at them with a look of joy and sadness. On the way home he was somewhat upset, he looked at me and said "Mr. Devitt, my good friend, at last our experiment is a success. Promise me two things; continue our work and go to the convention and tell our American friends to continue the work."

* * * * *

This is the story of the introduction of hardy Carpathian walnuts (*Juglans regia*) into Canada and the United States by the late Rev. Crath.

Looking back on the whole adventure (now twenty years ago) it would be only fair that I mention the names of three other men for the work they did to make the expedition a success. The late Professor James Neilson whose research in nut growing in Ontario and the United States was already well known should be mentioned. It was he who really "sparked" the expedition. To the late George H. Corsan whose nut growing experiment at Echo Valley was something unique in Ontario, credit is due for his enthusiasm and support of the late Reverend Crath. The American nut growers who were fortunate to obtain walnut seeds at the time through Wisconsin Horticultural Society can thank Mr. H. J. Rhamlow, then secretary. He took over the task of distributing the walnut seeds through the affiliated societies. He insisted that the seeds be tested for germination, kept in proper storage, and did everything possible to ensure success. However none of these men as I knew them then, and including myself, would want any credit, but we give full recognition to Reverend Crath for his work.

During the years spent in Poland Reverend Crath must have given the idea of growing hardy walnuts in Ontario and the Northern States considerable thought. He examined trees and nuts wherever he went; and continued gathering information each year. When I first met him I could see he had given walnut growing a great deal of study. He had great faith in his idea, and when leaving on his expedition 1934 he felt he was on a great mission. It should be remembered he made this arduous trip without pay and that he made very little money from the sale of walnut seeds or trees. No one did for that matter. It is also significant that in bringing these Carpathian walnuts out of Poland at that time, 1934, he did something that could never be done again. The trees he saw then probably went into rifle butts for use in World War II. The introduction of these walnuts into Ontario, met with varied success. Many bought them on a trial basis and were eventually rewarded; some looked on with skepticism and ridicule and a few thought that the growing of walnuts in Ontario was impossible. The intervening years, however, have brought forth a different picture. These seedling walnut trees are now bearing in Ontario and as late Reverend Crath predicted more than half of them are producing fair to good quality nuts. This is also true in the United States. In Ontario they grow well in the commercial apple districts and with variations mature nuts fully in 90 to 120 days (between Sept. 15 to October 15.) All of the best varieties should now be propagated by grafting to produce hundreds of hardy Crath Carpathian walnut trees. This project should always be one of the foremost with the members of the Northern Nut Growers Association. Twenty years from now and later, the number of hardy walnut trees producing nuts (Crath strain) should make a living monument to this obscure missionary—Rev. Paul C. Crath.

PRESIDENT BEST: Thank you very much, Mr. Devitt, for this very intriguing story. Truth is sometimes stranger than fiction. And we want to keep in touch with you, and we want to keep hearing from you, because you have got a big job to do yet.

MR. DEVITT: There is only one thing, ladies and gentlemen: I don't want to run into 5,000 letters to answer. Keep my name out of this. That is my walking-out request now. That's the story. I am going to continue to keep collecting samples. I hope some day to have a number myself of the best, and I might come back again sometime. I can't say every year; circumstances may be that I can't come. However, it's been a great pleasure for me to be here. I have wanted to come for 20 years, and I thought this year that I should come, because I am on this special mission of Reverend Crath's. Now you know what's going on in Ontario.

MR. SLATE: Mr. Chairman, I think the Association will answer the 5,000 letters, if he will ask.

MR. DEVITT: I didn't ask. Are there any questions?

DR. MCKAY: I'd like to ask a question. Was any scion wood ever brought over?

MR. DEVITT: There was some scion wood brought over by the Reverend Crath in the spring of 1935, and it was brought over on the boat. I remember in those years only one that grew on a tree belonging to Mr. Corsan. I don't think the other scion wood proved any good at all.

MR. STOKE: I got a little of that scion wood, and it had been waxed. The bark was nice and green, but the buds were dead.

MR. CALDWELL: Do you have a plantation of young, producing trees?

MR. DEVITT: No. My place, where I had those trees is now $3 million worth of buildings on 15 acres. You'd be looking down a street. They moved in in 1944, and built up 15 acres where I had one acre in the 15.

The Eastern Black Walnut as a Farm Timber Tree

JOHN DAVIDSON, *Xenia, Ohio*

Most people instinctively love trees. Perhaps this is an inherited result of arboreal ancestry. Even so, very few of us realize what an astonishingly close tie exists between the survival of trees and the well-being of the human race. Probably even fewer realize the very great importance, in the economy of animal life, of trees which bear nuts. Not alone for the sake of their nuts are they important, valuable as nuts are, but also for the sake of the unmatched timber which some of them produce, as well as for the sake of their service as soil conservers and builders, as beautifiers, and as silent, persistent builders of capital values.

In view of these outstanding qualities, it is strange that nut trees are today unfortunately and shamefully neglected in the north. Especially, I claim, is this true of the Eastern Black walnut. Here is a mystery. Why do not northern planters of trees plant more Eastern Black walnuts for their exceedingly valuable timber?

"Backward" Burma could give us lessons in intelligent forestry. It is said that the Burmese are permitted to clear their thickets and tropical woodlands for agricultural use only after they agree to plant a definite amount of that land in teak, perhaps the most valuable of all woods. It is said that, due to the effectiveness of this system, some 35,000 acres have now been stocked with this valuable timber.

There are two or three main reasons why the planting of Eastern Black walnut for timber is thus far not very common in America. (1). The native and favorable area of this tree is limited to a comparatively small section. (2), The tree grows well only in deep, fertile soil where quick-money crops have had the first call. Strip-mine planting is better than none at all, but such soil as is left after a strip-mine operation is hardly the best. (3) We are in too great a hurry. (4) Most farmers must have annual incomes, or they must quit farming.

What, then, are the offsetting reasons why this kind of planting should have an appeal to far-seeing people who are favorably located? In the first place, the Eastern Black walnut yields wood of unique quality. Pattern makers, who must work within tolerances of thousandths of an inch, prefer it. Walter Page, a well known sports writer has this to say: "Few woods come as close as walnut to fulfilling all the demands of a good gunstock: beauty of grain, workableness with cutting tools, resistance to warpage, weight or density in proportion to strength."

Another example of the many-sided versatility of this wood can be found in those timbered regions of America where termites are a problem for home owners. Termites seem to leave black walnut wood very much alone. It probably has a taste which termites cannot stomach. This is one reason why so many of the old rail fences of our ancestors in the walnut area were made of black walnut. The "ground-chunks," in particular, which were laid upon the ground under the corners of the worm-fences were often either of rock, or of walnut.

Just this year I watched the demolition of part of an old log cabin which was being riddled by termites. Many of the ordinary logs were in ruins but the walnut boards which had served as weather-boarding over the ends of some of the termite-infested logs were as sound and as beautifully preserved as they had been when they were placed there.

Is it any wonder that so many of the pioneers who had lived long enough in the termite area to see what could happen to other lumber, chose walnut, whenever they could get it, for structural work and for weatherboard protection?

Safety of operation is still another matter for consideration. If I wish to create an estate for my family or for my last years, how can I go about it with the best chance for success? Shall I go prospecting for precious metals? Thousands have failed at that job where but few have succeeded. Shall it be manufacturing? Count up the failures. For each success, at least ten go broke. Wall Street? The Wall Street journals themselves give the statistics. More than 90 percent of all persistent Wall Street gamblers lose money in the end. Farming? Much safer, but most farmers who have made much money in the past have accomplished it by way of an increase in the value of their land rather than through their farming operations. This is the result of fluctuating prices. Bad years often eat up the savings of good years. Then, too, the good farmer is a busy man. The better the year the busier he is. Very little time remains for side issues, such as the planting of trees.

As a matter of fact, as erosion of the soil progresses, as good, productive land becomes more scarce, and as farm labor becomes more and more difficult to employ, the attention of informed farm owners and operators has been turning more and more to soil-building, perennial, permanent and labor-saving crops. Of these, grass and tree crops are, far and away, the most promising today.

In view of what I have found out during the last 20 years, I am quite sure that, if I were starting now, I should expect to make farming a major element in my estate building, but it would be mostly tree and grass farming, not grain farming. I should need livestock, of course, to make use of the grass. And I like livestock.

This is what I ask of life: First of all, I must enjoy my work. I do not care to spend all my days in getting ready to live. My job must lie along the road I like to travel. I do not care to work at a task so burdensome, so time-consuming that I have no heart for the enjoyment of living. At the same time, a big part of the plan must be to find a good, safe way to build an estate. It must be feasible, practical, enjoyable.

I believe, in the light of my own recent experience, that if one is properly situated, there is much to be said for the idea of undertaking the practice of forestry upon a rather liberal scale using Eastern Black walnut trees as a foundation.

In the first place, I ask, what living thing upon one's farm will cause less labor than a forest tree? I know of none. This fulfills the first requirement. A forest tree calls for a minimum of attention as compared with other crops. This is especially true if one permits livestock to keep down weeds and brush. And here I am likely to be called a heretic. The authorities say, "No grazing in a forest". However, in this field of forestry there are some traditional maxims which, to say the least, are not capable of universal application. The authorities, too, have been known to rely upon what other authorities tell them—without investigating the facts for themselves. It is not well to rely too implicitly or trustfully upon the "authorities", either ecclesiastical or scientific. "No grazing" is a valid enough rule to follow in the ordinary forest, but I have found that after the trees are well grown we can graze the land under a deep-rooted walnut tree which is planted in deep, rich soil as we would graze any meadow land—in reason and in moderation. The practice is profitable for annual income and it keeps down the fire hazard. One bad fire in an ungrazed or unmown piece of brush-covered undergrowth can destroy in an hour 50 years of timber growth. If we plant deep-rooted nut trees in deep, rich soil, and if we fertilize that soil as any valuable permanent pasture land is fertilized, we can graze that land without injury to the trees or the land.

One other reason which is given for the prohibition of grazing is the desire to save young tree growth. This is justified in ordinary forestry practice by the need to get annual income through successive cuttings. The young growth must be encouraged to come on. Even so, it must be thinned as it comes. However, a forest of black walnut trees yields its annual income in another way—through its nuts and its livestock.

Trees in such a forest should be planted close enough together to cause them to reach straight up as they grow. They will not all reach straight up, of course, but enough will do so to produce as many saw-logs as will normally grow in a forest; that is, if they have been properly planted in the first place.

In my own modified forest-type planting, the black walnut trees stand 8 feet apart in rows 20 feet apart. The 20-foot spacing between rows was planned to provide more sunlight for nut production during the early years. No one ever planted a forest in that way, so far as I know. The trees are now 17 years old, about 3250 of them in all. In the best soil of this 20 acres I can count about 1000 forest-type, straight, well-grown trees. There are about 1500 lesser trees, low-limbed trees which will eventually be used, perhaps, for posts or some such purpose. There are, I regret to say, about 750 trees that will never be worth anything. An eroded slope and a hidden clay bed explain these misbegotten dwarfs.

The variable growth of these trees proves that the first care in making a planting of walnut for timber should be to plant in good soil, deep and well drained. Bottom land, even some that is occasionally overflown with flood-water, and therefore not the best wheat land, should be excellent for Eastern Black walnuts if the drainage is good. Rule two:—Select your seed or seedlings from large, straight-growing, healthy parents. This rule needs explanation.

In last October's NUTSHELL, an organ of the Northern Nut Growers' Association, Spencer Chase, its editor, called attention to a showing of Carpathian Persian walnuts by Mr. H. F. Stoke which illustrated what was called "the variability of seedling trees." The progenitor of these seedlings was a Lancaster Carpathian Persian walnut tree. Differences in size, appearance and quality of nuts from these seedlings were said to have been remarkable. Such differences, we know, are greater with some species than with others. A variable ancestry often results in a variable progeny. On the other hand, I know that my Eastern American black walnuts do tend to reproduce the characteristics of their parents. I have long rows of seedling trees, all from one parent tree, standing alongside long rows of seedlings from another parent. The similarity of the tree growth and nut production of the trees in their own rows, and their contrast in growth of trees and nuts to those in adjoining rows is striking and to me conclusive. A photograph taken by Dr. O. D. Diller, of Ohio State University, in 1946, shows trees in a right-hand row grown from seed of a tree on the Kinsey farm, while on the left are seedlings of a tree on the McCoy farm. The circumference of trunks of Kinsey seedlings averages more than twice that of McCoy trees. Same soil, same age (11 years), same treatment.

Those same trees, now 17 years old, still show these striking characteristics. It is true that each tree in a row of seedlings is an individual in its own right. No others are exactly like it. Nevertheless, the family resemblances in that row are very like those in human families. They are especially noticeable in the nuts—with, for example, rough shells in one row and smooth shells in another; mainly large nuts in one and mainly small nuts in an adjoining family. Also, some rows have mostly straight-growing trees, others are predominantly branchy, like the Thomas.

It should be said in this connection that practically all of the parent trees of these seedlings stood in isolated positions and little subject to pollination from other trees.

So much for the Eastern Black walnut's evidence of hereditary influence. So, let us take inventory.

Today, I figure that the thousand well-grown trees in this planting are each adding a dollar per year to the value of the 20 acres upon which they stand. $1000 per year in all. This estimate, which of course seems optimistic, is based upon the statement of a walnut tree buyer—a sawmill man—who tells me that a well grown, deep-soil, 50-year-old Eastern Black walnut tree should average about $50 in value. Thus far, my 17-year-old youngsters, some of them nearly 3 feet in girth (9-1/2 to 11 or more inches in diameter at breast height) look promising.

In addition to the potential added value of $1000 per year, this 20 acres has produced about two tons of in-hull nuts from selected trees only, in each of the past two years, (with more than that in prospect this year), while the land beneath the trees grows good pasture and helps to support a small herd of cattle and calves.

Once the trees were thoroughly established, the labor investment has been very small. Nature, for the most part, has done her own pruning, and has done it better than I deserve. Since the first half-dozen years, there has been no cultivation. The trees have been practically trouble-free. Winds have damaged a few and one wet spot has killed three trees. There are a few black locust trees among the walnuts. I can see no evidence that the walnuts have made either better or poorer growth because of the proximity of these nitrogen storers. Perhaps the evidence will show up later. We shall see.

The last item in this inventory, added value to the estate, is still potential, but the potential is surprising. If my walnut timber buyer's estimate is trustworthy, in 17 years the best 1000 trees have added 17,000 potential dollars to the value of that 20 acres. And they have done it with safety, with little labor on my part and, lately, with annual dividends of excellent nuts and good pasture. No other kind of forestry that I know of can do that.

It would, of course, be foolish to claim that the kind of management here described would be wise or workable with other forest species. Wise forest management requires, first of all, that the choice of species shall be adapted to the soil and climate favored by that species. It requires a proper density of stand. Finally, good management demands that a choice be made of the most valuable type of timber that can be produced upon your land. If you can grow walnut successfully, it would be foolish to grow Willow or Box Elder.

One necessary thing I must do, a thing that I should advise others similarly situated to do, namely, place a tight legal fence around this twenty acres in order to assure the trees' survival until 50 years have proved or disproved my faith. For, after all, these trees are guinea pigs—pioneering. They break some traditional rules. The land they stand on is grazed. They are not set the traditional 80 feet apart. Their nut crops may dwindle away. One never sees walnut trees growing in pure stands—always with other species which scatter their seeds and push in. They are not monopolists—like the pines.

Very well, we shall see. My own small experiment in unorthodox ways has the temerity to suggest a new treatment for a species of timber tree which I personally regard as America's very best gift of its kind to the world. For 17 years my modified forest-type planting of black walnut trees has not disappointed me. That is why I now believe that the farmer in the

Eastern black walnut's native habitat who fails to set out these nut trees wherever he can is losing a good opportunity.

The McKinster Persian Walnut

P. E. MACHOVINA, *Columbus, Ohio*

The McKinster Persian walnut first attracted public attention when it received first place in the preliminary Persian walnut contest conducted by the Northern Nut Growers Association in 1949. In the follow-up contest of 1950, the variety was granted third place. The McKinster tree resulted from Crath Carpathian seed secured through the Wisconsin Horticultural Society by Mr. Ray McKinster of Columbus, Ohio. The seed was obtained and planted in the spring of 1938, hence the tree is now 15 years of age. Probably this seed was secured by Rev. Crath during his last trip when, presumably, he made some of his most careful selections.

Altogether, Mr. McKinster planted eleven Crath nuts in the back yard of his small city lot, nine of which germinated. All but two of the resulting seedlings were distributed to friends and relatives living in the countryside. Many of these trees have disappeared due to accidents and lack of care; a few, however, have produced nuts which apparently are not exceptional. One such nut examined was of medium size with a fairly thick shell; the kernel was of good flavor but somewhat bitter. Of the two trees retained by Mr. McKinster, both were permitted to grow where the seed was planted, however one died of an unknown cause when five years of age. Nuts produced by this tree were inferior to those produced by the survivor which later became known as the McKinster variety.

[Illustration]

[Illustration]

The McKinster tree may be viewed in the accompanying illustrations which show it without foliage and with foliage. The pictures were taken in March and August, respectively, 1953. Since it is a very beautiful and relatively clean tree, the McKinster would be desirable in any yard. From the pictures, it will be noted that the site is unfortunate being restricted by two garages, an alley, and with numerous overhead utility wires. Some effort was made two years ago to keep the tree out of the wires by cutting back top growth. The trimming stimulated the usual vigorous, annual growth to produce terminals as great as 10 feet in one year. Ordinarily, annual growths of 6 feet of husky wood are not unusual. New wood and buds are hardy in appearance and assume a rich brown color upon maturing. With such growth, cutting 1000 feet of scion wood annually would be no problem. The tree is now about 35 feet in height with a like spread.

The bearing record of the McKinster Persian has been excellent. Its first crop of five or six nuts was borne at five years of age and large crops have been consistently set each year since with but one exception. Crop records have been impossible to maintain since the tree is located in a section of the city where squirrels abound. Any nuts saved must be protected by screen-wire cages. The hunger of the squirrels for the nuts is amazing. For example, in 1951, they descended upon the tree during the first week of July and destroyed all nuts of the large crop within two weeks. These nuts could not possibly have been filled and, consequently, could have been of little nutrient value. In their voracity, the squirrels frequently work on the cages and sometimes manage to break through. To facilitate this endeavor, limbs up to one inch in diameter carrying cages are sometimes cut off so the squirrels can attack more conveniently from the ground.

[Illustration]

It could be that nuts saved by caging are sometimes inferior. The cages used are made by folding window screen into a doubled, 4 to 6 inch square, producing an "envelope" with wire sewn edges. Crowding from one to three nuts into a cage may result in inhibited development, especially since considerable leaf surface must be removed when installing a cage. Because Mr. McKinster has been ill for several years, it has been difficult to accomplish the caging; consequently, but few nuts are saved. For example, in 1950, there were insufficient nuts to meet the 25 nut sample required by the contest judges. All available nuts, some probably inferior, were entered, and it is a matter of conjecture whether the nuts might have been judged higher under different circumstances. Also conjectural are the questions of crop size and regularity of bearing in the event the tree was permitted to mature its nuts.

The McKinster nuts, which were the principal consideration in the contests rather than the tree itself, are excellent in nearly all aspects. They are of medium size, averaging around 35 to the pound, with about 52 per cent kernel. The shell is moderately thin, light in color, well sealed, of a satisfactory shape (see illustration), and with excellent cracking qualities. The kernel is light, plump, of excellent flavor, and in the words of one authority, "probably rank with the best in freedom from bitterness." The nuts are matured by the middle of September and, later, drop, free of the husk.

Blooming of the parent tree usually occurs during the first week of May. In 1951, the staminate flowers were first observed April 29 and the pistillate flowers May 2. The narrator visited the tree on May 4 at which time some catkins had fallen; it was estimated that one-half to two-thirds of the pollen had been shed. The pistillate flowers appeared to be either receptive or slightly past at this stage. Mr. McKinster commented that the blooming period of 1951 was from a few days to a week earlier than usual. In 1952, the shedding of pollen started on April 29. From the foregoing, it may be noted that the McKinster Persian is entirely or largely self-pollinating. No other Persian walnut trees which might assist in pollination occur in the vicinity and all known seedlings raised from nuts of the parent McKinster tree have appeared to be pure Persian. Leafing out starts about a week before the bloom appears. In the fall, leaves are colored a beautiful bronze and are brought down in a great shower by the first frost.

A sample of the soil in which the McKinster tree is growing, taken at a depth of 6 inches, was tested in July 1950. The results specify that the soil is mostly silt with an average amount of organic matter and that evidence indicates it to contain ashes. The acidity is specified as "neutral", potash "high", and phosphate "low". No mention is made of available nitrogen; however, the dark green color of the leaves and vigorousness of growth would indicate a satisfactory supply. Fertilizer in small amount was applied once or twice during the early life of the tree; also, during this period, Mr. McKinster "spaded in" garbage, etc., to increase the humus content of the soil. In 1951, the narrator checked the pH of the soil near the surface and obtained a value of 6.5.

Only one instance of damage due to climatic conditions and none whatsoever from insects and diseases has ever been observed with the parent McKinster tree. Undoubtedly, the city location offers some protection from frost, but may also be detrimental, on occasion, through heat reflected from the many surrounding white-painted buildings. For example, an unseasonable warm spell occurred in Columbus during the latter part of the first week in April of the current year. The heat, lasting for several days, reached a high of 80.4 degrees and, as a result, the McKinster tree started vegetating. Leaf growths of from one-half to one inch had been reached when normal conditions returned. Two weeks later, a cold spell with snow and temperatures of 22 degrees killed the new growth but did not injure the wood. Following this, leafing re-occurred, but at a slower rate and somewhat later than normal. The size ultimately attained by the leaves is about one-half their usual size, and, consequently, the accompanying illustration, taken this summer, does not exhibit the usual luxuriant appearance of the tree. A large part of the bloom was damaged by the cold, hence the tree set a lighter crop of nuts than usual.

In connection with early vegetating, it may be remarked that Mr. McKinster, several years ago, presented two small grafted trees of his variety to a relative living in eastern Kentucky. These trees were planted on low ground and were killed the first year by late spring frosts after leafing out twice. Thus it seems evident that the McKinster tree has the fault, common in Carpathians, of leafing out too early and being injured by late spring frosts, especially when planted too far south. Three other trees, grafted by Mr. McKinster and now about four years from the graft, are situated in the countryside several miles south of Columbus, Ohio, where they are doing excellently, having never been damaged.

The writer has several three year old McKinster grafts at his property in southeastern Ohio which were deliberately set on stocks located in a bad frost pocket. The grafts, which are adjacent to a woods, have made fair growth each spring but are injured during the summer by an insect laying eggs in the succulent growth. The portion of terminal above the point of sting invariably dies the following winter and has the appearance produced by winter killing. This damage has not been unique with the McKinster, having also occurred with the McDermid, Watt, Burtner, and other Persian varieties growing nearby; some of the latter were killed outright the first winter after grafting.

A one-year McKinster grafted tree with three feet of growth above the graft was cut back and transplanted by the writer to the yard of his Columbus, Ohio, home during the winter of 1952. Growth the following spring was about two feet and obtained in rather poor soil. After a long absence during the summer which was attended by a prolonged drouth, the tree was found in a dying condition, having lost all its leaves. Hurried watering resulted in a complete new coat of leaves and a small amount of additional terminal growth. The tree matured its growth and withstood the winter nicely, but suffered, similar to the parent, from the April, 1953, unseasonable weather. Growth this summer from adventitious buds has been poor.

Unfortunately, the McKinster variety saw but little testing in other parts of the country prior to its recognition in 1949. So that this report might be as complete as possible, requests were sent to several dozen experimenters who are known to have grafted the McKinster, asking for their experiences and opinions of the variety. The requests went to people scattered generally throughout the northeastern portion of the country, a very few of which had received scion wood in 1950, a larger portion in 1951, and the bulk in 1952. For the most part, replies indicate satisfaction and even enthusiasm; very few report failure. Definite conclusions cannot be drawn because of the short time of trial; however, a general description of experiences will provide indications.

Few experimenters report failure in grafting, most stating the variety to be "easy to graft." Any who mention the characteristic, state that "grafts are vigorous," or that "it is a fairly rapid grower." For the experimenters, the McKinster seems to be about "average" in its time of leafing out. Many report a set of nuts the second year after grafting. As to time of maturing new growth, the reply of Mr. Stephen Bernath of New York, "New growth matures about the end of September," is fairly typical, as is the reply of Dr.

R. T. Dunstan of North Carolina, "It appears to harden wood well ahead of frost." Most reports indicate no winter injury but are tempered by cautious observations that temperatures had not been low. Mr. H. F. Stoke of Virginia, who grafted the McKinster in the spring of 1950, reports: "Pistillate buds developed during the summer of 1951 were killed by a frost catching new growth in the spring of 1952." Mr. John Howe of Missouri was the sole reporter of catastrophe when he stated: "My McKinster graft was killed by the November, 1951, cold while the Lake and McDermid varieties close by were not hurt." Mr. Sylvester Shessler of northern Ohio reports: "The McKinster withstood, without injury, the 1951 winter which killed 4 hybrids and a Crath, and injured several others." Mr. Harry P. Burgart of Michigan reports the variety as doing extra well for him. The only reply mentioning disease came from Dr. Dunstan who says: "It has been fairly clean in foliage so far, less susceptible to leaf spot than some." Mr. John Gerstenmaier of Massillon, Ohio, grafted the McKinster in 1951 and reports excellent growth with a diameter of 2 inches at the graft after two years. He reports temperatures of 16 degrees very early in November which caused no harm, and pistillate bloom from May 8 to 16, 1952, which materialized into a crop of two nuts; pollen was supplied by adjacent Carpathians. Leafing out ordinarily starts about a week prior to the bloom for Mr. Gerstenmaier; but, in April, 1953, the unseasonable weather conditions also occurring in his vicinity caused early vegetating and killing, while at nearby Orrville, the variety was undamaged.

Mr. Gilbert Becker of Michigan, who is enthusiastic about the McKinster variety, believes the qualities of the nut to be superb and the characteristics of the tree satisfactory. He is of the opinion that the too-short dormancy of the variety is not a serious objection, particularly with climatic conditions such as those experienced in Michigan. Even in central Ohio, where peach and apple crops are frequently lost due to spring frosts, the McKinster has not been injured when located in the countryside and injured but once during its 15 years, with a resultant smaller than usual crop, when located in the city.

In closing, it might be well to comment on the fact that nuts of the McKinster, Hansen and Jacobs varieties alone placed high in both the 1949 and 1950 N.N.G.A. contests and that different panels of judges served in the two events. Certainly the nuts of these varieties are of a superior quality, and it would seem important to determine those parts of the country where these varieties are sufficiently hardy to be of commercial value. Certainly these varieties should be given every opportunity to prove themselves.

Carpathian Walnuts in the Columbia River Basin

LYNN TUTTLE, *Clarkston, Wash.*

Mr. Chairman and friends of the nut culture, I regret that I cannot meet with you at this time, but fate seems to have decreed otherwise. The pleasant memory of the meeting at Guelph is still with me and I must admit a feeling of humility as I prepare this paper for a group of sincere and devoted people united in a common interest.

The Pacific Northwest extends from the Rocky Mountains to the Pacific Ocean. This area is divided by the Cascade Mountains which run north and south. Between the Cascades and the Pacific we have a coastal area wherein winters are generally mild, summers cool, and rainfall abundant. Under these, conditions many plants do not attain a high degree of dormancy. Zero weather in Seattle will damage walnuts as much as will twenty-five degrees below zero in the more continental climate east of the Cascades. Carpathian walnuts have proved their value under both coastal and interior conditions.

This hardiness is at least partially due to their tendency to mature their buds and harden their growth earlier in the fall than do other types of English walnuts.

Between the Cascades and the Rockies is a vast area part plateau and part mountains. It is scarred with deep canyons and crossed by swift streams fed from springs and mountain snows. Roughly the elevation of farm lands varies from five hundred to over forty-five hundred feet. Depending largely on slope and elevation, rainfall varies from about eight to twenty-five inches. In general, summer days are bright, dry, and fairly hot. Nights are clear and cool. Winters are unpredictable but always vary much according to location and elevation. Infrequently temperatures may drop to more than twenty below zero at Clarkston. Other areas of similar elevation may be five to ten degrees colder.

For the sake of clarity and to reduce the territory covered, we will confine ourselves largely to that part of the Columbia Basin irrigated and to be irrigated in Central Washington. The application is general, however.

Grand Coulee Dam has made feasible the irrigation of about 1-1/4 million acres of sage brush, bunch grass, and marginal wheat lands. Irrigation is already practised over other vast acreages. This land is level to rolling, and is of sandy loam nature. It is deeply under-laid by layers of lava rock—in places thousands of feet thick. As in most arid climates the soil is rich in minerals but low in nitrogen and organic matter. Under irrigation production is amazing. The growing season is sufficiently long for Carpathian walnuts anywhere in the irrigated area.

Walnuts originally from Southern Europe have proved unsatisfactory because they killed at 20 to 25 below zero. It was discouraging to have a ten or fifteen year old tree killed outright by an unusual winter. But it was just these conditions that led to the discovery of the Schafer walnut. This tree survived the winters of 1936 and 1937 in a part of the Yakima valley where all other varieties similarly located were killed. So far as I know, none of these were Carpathians.

Many Carpathians are now being planted, mostly for yard trees, but promise to eventually become one of the big commercial crops of the area. However, skepticism on the part of the public and scarcity of nursery stock has delayed commercial planting. A fair portion of good growers are now convinced that commercial growing is profitable and stock, our own and others, is becoming more plentiful.

Our experience has been confined largely to the Schafer walnut and it is, aside from some promising seedlings, so far as we know, the only proven Carpathian in this area. We do not wish to discredit possibilities of any other variety, but must speak out of our own observations. There are numerous small, commercial plantings now producing, the nuts being sold locally. Accurate production figures are not available and if available would vary greatly due to the care given the trees. The Schafer, and this will undoubtedly hold true of some other Carpathians, bears more at five years than a Franquette does at ten. I have seen apple boxes (about one bushel) of nuts harvested from five and six years old trees. Production increases rapidly with age.

As with fruit trees good air drainage and good soil drainage are desirable for the walnut orchard. The Schafer starts fairly early in the spring and new leaves are easily nipped by late frosts. A severe late freeze might also injure new growth although I do not recall a crop having been lost due to this cause. Although pollinizers have not been used, we think that on young trees and in some years they might insure a better crop. We are now propagating two pollinizing varieties the catkins of which come out later than the Schafer.

Trees planted sixty feet apart permit inter-planting to row and other crops for several years. Columbia Basin lands under irrigation produce enormous crops of potatoes, beans, sugar beets, rutabagas, green peas, clover or alfalfa seed, peppermint oil, and fruit. Average potato—20 tons, alfalfa hay—7 tons (three cuttings), alfalfa seed—800 pounds, dry beans—2,500 pounds, wheat—70 to 100 bushels. In some areas peach or apricot trees make good fillers.

Carpathians also fit into the picture as yard trees, for border plantings,—either to utilize run-off water or to use water wasted along ditches and pipe lines and for wind breaks. This open country is naturally windy and trees greatly reduce the ground velocity of wind.

Nut production in this area appears to be much heavier than on the coast or in California with varieties now being grown there. So far we are pest-free. The potentials of good Carpathian walnuts in this area are unlimited.

Walnuts and Filberts in Southern Wisconsin

C. F. LADWIG, *Beloit, Wisc.*

My farm is located a few blocks north of the Illinois-Wisconsin line on a rise overlooking the city of Beloit, whose western limits are almost adjacent to my land. Temperature in this section ranges from 100 degrees above to 30 degrees below zero; rarely reaching either extreme—with an average frost free period of 173 days. Rainfall averages approximately 35 inches. Walnut, butternut, bitternut, hazel and hickory are native, but just about non-existent in my vicinity except on my place in the young state.

The land on my place has been tobaccoed and corned out for over 100 years and its once rich clay loam with sandy streaks was unable to grow ragweed over 2 inches high when I bought it. Trying to grow nut trees in this soil presents problems as you well know. My problem was not to get them to grow vigorously but to get them to grow at all. However, by using fertile spots, formerly barnyard and around the house, I got several walnuts and filberts started.

I have an eight year old Crath #1, two Myers black walnuts, about the same age, Cochrane and Thomas, 6 years, all obtained from Mr. Berhow, and a fine assortment of Jones hybrid filberts from Mrs. Langdoc, a Rush filbert from Mr. Burgart, two European filberts from the New York State Fruit Testing Association, some hybrid seedlings, some native hazels from seed, some bitternut seedlings from Mr. Weschcke, a few native hickory seedlings, an American chestnut seedling from Scarff, 2 butternut seedlings, 2 nice Chinese tree hazels from Mr. Shessler, several Jacobs walnut seedlings, and *regia* & *hindsii* hybrids from seed of Mr. Pozzi, some Crath seedlings and a number of Thomas black walnut seedlings—also native walnut seedlings.

Mr. Shessler and Prof. J. C. McDaniel have been a source of help, advice, and inspiration to me and I am deeply indebted to them, as well as to many other members of the N.N.G.A. who have shared their experiences with me.

How have the trees done? The Crath #1 is bearing a few nuts this year. It had no catkins, but the Cochrane was loaded with staminate bloom at the right time. I got busy with the Cochrane pollen and a brush and went to work on the Crath pistillate bloom. Very pleased with this cross I looked the Crath over a few days later to check on progress. I picked the little nutlets off the ground and inspected them carefully, then threw them into the chickens to see if they would eat them. Back in my mind was the feeling that Mother Nature thought I was getting too big for my britches and decided to teach me a lesson. However she generously allowed a few air pollinated nutlets to grow, and so there will be a small crop of the round and plump smooth green balls.

The Crath #1 is not perfectly hardy as it freezes back an inch or two in the cold winters. Two years ago the warm wet fall left it unprepared for the sudden onslaught of winter and several whole branches died and the trunk split open, the split sounding like a rifle shot one cold, crisp evening. I happened to be standing by it at the time.

The Myers black walnuts are splendid trees and just about hardy. They bore a few nuts and second and third year from planting, which sapped their vitality. They then bore nothing for about three years, which happened to be unfavorable years for walnuts anyway, and began to bear again this year with a moderate crop. It looks like the plum curculio, my arch insect enemy, is trying the nuts for size. I saved some Cochrane pollen and went to work on the Myers, with you know what results. However three of the nutlets stayed on the tree; so that I may have effected a cross between Myers and Cochrane.

Thomas has acted peculiarly for me. It went thru the devastating winters of 1950 and 1951 in fine shape, then froze back last winter when the temperatures never went below 5 degrees below zero. The very dry fall should have ripened all branches to perfection. My mule, Zombie, took a liking to the branches and leaves of this tree, so it is now trimmed up like an umbrella. The small nut crop must have also gone down Zombie's gullet. He is more destructive to walnut and plum than the curculio. (Tie him up. Ed.) Thomas does not seem to have a great future up here.

Now Cochrane is different. If that little tree has as many nuts on it as it had catkins this year, I'm going to have to move the corn out of the crib and put the walnuts in there. It is not a fast growing tree, but this may be the fault of the spot it is in, judging by the color of the leaves. I never got around to fertilizing it.

Now that I told you about the Cochrane, I'll have to tell you about the "Wayne" black walnut. It is eight years old, stands about eight feet high and is hardy. My Black Walnut seedlings stand from six inches to six feet high. They go back to six inches every other year when I cut them down to graft them. Nobody in the nut tree field can call me a grafter. I'll make him prove it!

The hickory, butternut, bitternut, and chestnut are step children and fend for themselves on less desirable soil. All are small. The regia-hindsii hybrids are small and young and are being given special care, but may not be perfectly hardy. They grow well.

The Jones hybrid filberts stand from six to eight feet high, except those planted recently. This year they have a fair crop. The catkins came thru the winter in good shape for the most part. My two European filberts, which have lost their identity, but are either Italian Red, Cosford, or Medium Long, (one of the three perished) usually suffer the loss of their catkins and occasionally lose a branch or two to winter's icy fingers.

To me, the filberts are fascinating at all times of the year. When the snow is deep and the cold bites deep, their tight little catkins always hold forth the comfortable promise of spring. When spring does come the thrill of the tiny red blossoms and lumbering catkins is as real and enduring as the promise of a crop of the shiny nuts is fickle. Then, of course, after the last tiny blossom has faded and the last catkin has withered, the leaves push forth. To me, these tiny leaves are a sight comparable to the opening and unfurling of the various varieties of the grape. Then enters the element of suspense, between the time of leafing out and the time when the little nut clusters appear.

My bushes are all growing together on a rise of ground near an old barn foundation. The ground is rich and they love it. Each bush is individual and distinctive as are their nuts—some tucked far in the husk, some bulging out in a precarious fashion, some fat and round, others long and narrow. They're interesting. I can let the butternuts, bitternuts and hickories pass, the heartnuts, chestnuts, and pecans can wait until I am sure they will bear here. The walnut will grow up along with the other trees—blending into the landscape, but the filberts, like Zombie, call attention to themselves every day of the year.

Somebody said recently that the emphasis in England is in being, and in our country in becoming. I imagine our land stopped being with the disappearance of the Indian and the primeval forest and is now in the process of becoming something else. What that something else is we don't know, and each generation carries a new set of values, but we all know that to become something better, trees must and will figure in the plans of all generations—better and more useful and more disease resistant trees. It is significant that nut trees lead in these requirements.

Biology, Distribution and Control of the Walnut Husk Maggot

DR. F. L. GAMBRELL, *New York State Agricultural Experiment Station, Geneva, N. Y.*

DR. GAMBRELL: Mr. Chairman, ladies and gentlemen, some 22 years ago, I believe it was, I attended one of your meetings at the Experiment

Station in Geneva, and at that time I gave a little talk on the walnut husk maggot. Perhaps some of you are old enough to have been there and remembered something about it, or maybe you are old enough so that you have forgotten as much as I have, so it would be worth talking over again. At any rate, when the chairman of your program committee wrote Dr. Chapman, asking him if he might talk, he came to me and said, "Would you be willing to do this?" I said I'd be willing but I didn't know whether I'd be able. But finally, the pressure was so great that I said yes, and I am here.

After I accepted the invitation, I made up my mind that I would like to bring myself up to date as much as possible on recent developments on walnuts, so I took the liberty of writing to a lot of our entomological colleagues and talking to one of your members, Mr. Slate, in the hope that I might get some more recent information on the maggots, or, particularly, the control of this walnut husk maggot. I wrote to some 10 or 15 entomologists in 15 states, as well as the United States Department of Agriculture in Washington and to our neighbors on the north in Ottawa. I must say that I have had a very fine response from everybody. They were all very willing to help, but practically all of them had the same answer: while they knew there was such a bug, they didn't know too much about it as an economic pest. So that left us all right in the same boat, with about two exceptions, as when we began. Our friends to the north in Canada sent some very nice information. We also had some information from the U. S. Bureau of Entomology and Plant Quarantine, Washington, D. C., together with some illustrated material. Also our good friend, Dr. Boyce, at the Citrus Experiment Station, in Riverside, California, with whom I have discussed the walnut husk maggot problem quite a few years ago, had a very nice bit of information and illustrative material which he provided. Incidentally, he is the man who has been mainly responsible for the development of the walnut husk fly control program for the nut industry in

California. I would certainly like to take this opportunity to acknowledge any contributions he or the other people have made towards this discussion.

In New York State we have in our official list of insects about 30 species of fruit flies that are catalogued, but only about five of these can be classified as of economic importance. Two of these occur on the cherries, both sweets and sours, and are called the cherry maggots. Another one on apples, known as apple maggot, and a related form on blueberry. And then, of course, the walnut husk maggot, and one other which occasionally occurs on currants, but this one, of course, is of less importance than the others.

The fruit industry, of course, in New York is quite large, both apples and cherries, so that there is a considerable problem there as far as control is concerned. The growers spend thousands of dollars every year in combatting the various species of fruit flies. The interesting thing in this connection is that throughout the last 25 years with which I am familiar with the cherry fruit flies—in fact, that was one of the first projects I worked on in cooperation with Dr. Hugh Glasgow when I came to the Experiment Station in 1925—the control measures which we developed in 1925 to 1927 are essentially the ones which we are still using today; that is, for the most part. There have been various attempts to change the control program through the introduction of these newer insecticides, and some progress has been made, but in every case they have been wrought with some difficulties. At the present time the official state recommendations for the control of apple maggot and cherry maggot still include the use of arsenate of lead under some conditions. I mention that at this point because it is of some significance in the overall control. I am going to discuss that later on.

As far as the host plants and distribution of the walnut husk maggot is concerned, according to the original description which was published almost a hundred years ago, it was listed as occurring in and to the Middle States. That is a little bit indefinite, but at least it occurs all over the Eastern United States and as far west as Kansas. Then the one which occurs in California, which has since been called *Rhagoletis completus* (?) looks very similar to the one that we have here, but there are slight taxonomic differences, so at least it is considered a different species. At any rate, it is very similar to the one we have here, and this whole group of fruit flies that we have been talking about have a lot of similarity in their wing patterns and things of that sort.

And the fact that I mentioned the control as generally as I did is of significance in that all of these flies of the various species are apparently susceptible to the same type of control measure.

As far as the host plants are concerned, I have personally observed injury on all of our common *Juglans* species that I have run across in New York State and in some of the states to the south of us, including butternut, Japanese walnut, English walnut and black walnut. I have seen reports of infestations which were recorded in hickory, but I personally have not seen them.

I'd just like to have a show of hands. How many in the audience here have had experience with the walnut husk maggot or had injury on the fruit? (Showing of hands.) I see the majority of you certainly know what it is, but just as a brief reminder, the type of injury, of course, varies somewhat depending possibly on the variety and time of year at which the fruits first become infested. We know, of course, that the flies do not begin to puncture the husk until they attain a certain degree of softness. Early in the season they are not able, apparently, to penetrate the husk with the ovipositor, and that, of course, varies not only with hardness but with varieties. The flies, of course, may be seen on the fruit even though they are not able to penetrate the husk and deposit their eggs. These husks, of course, many of them, become dry and hard after they have been tunnelled out, and it is almost impossible to clean the shells. Occasionally you have nuts in which you have a separation of the suture, and in those cases you very frequently get the exudate from the husk penetrating through the suture in the shell onto the kernels themselves, and in those cases molds may grow on the kernels so that those fruits are no good.

In connection with this injury I am going to show you some slides in a few minutes, but the preceding speaker made reference to a type of injury which occurred on the terminal growth of a walnut tree and that is one that we have had a lot of inquiries at the Experiment Station about, injury to the new terminal growth fairly early in the season. That probably, in most cases, is caused by the butternut or the walnut curculio. Early in the season these adults begin feeding on the new terminal growth, and they even puncture the new growth and lay their eggs there before the nuts are large enough for them to attack and very often considerable killing back of the terminal growth occurs. I have seen it on English walnut seedlings in nursery rows where there would be very large kill-back from the walnut curculio. Superficially the injury on the fruit is quite similar to that of the husk maggot.

(First slide.) This first slide is just to give you some idea of the general areas of fruit growing and distribution in New York State. The eastern section, right-hand side, Champlain Valley and Hudson Valley, are primarily apple maggot regions. Some walnut husk fly probably occurs there, but they are predominantly apple-growing areas. In the central part of the state, northern, particularly, we have fruit, and as far as I know, there are no plantings of walnuts there, though you people may know of some. In The Ontario Plains section south of Lake Ontario is one of our big fruit belts in the State. Some walnuts are also grown here. Consequently this area has in it apple, walnut and cherry maggot flies, and, of course, they will be lapping over in all those areas into surrounding territories. But this gives you an idea, in a general way, of the distribution of the host plants and the flies about which I have been speaking.

(Next slide.) Those flies get pretty big when you get them up there. They are not that easy to see in the field. The ones on the top are the species found on cherries. The one on the lower left is the apple maggot, the one on the lower right is walnut husk maggot. The only difference you can see here is in the wing pattern but in nature they differ in color. They all have a little different wing pattern. Also, there is a little difference in size, the walnut husk maggot being the biggest of the four species shown here.

(Next slide.) I have shown here the emergence date of the various species, including the cherry fruit flies, the apple maggot and the walnut husk fly. And you notice that beginning over about the first week in June you have emergence of the cherry fruit flies, and you have a continuance of emergence of some of these species up until at least the first or second week in August. These points going up and down just show the number of flies that were taken on given dates, and there is a very definite correlation between the proportion of flies that emerge on any given day with the temperature or moisture condition. Some years, when you have very hot, dry weather, there is considerable mortality of these flies as they just do not seem to be able to emerge from the soil, which is a good thing.

(Next slide.) This photograph is one that I wasn't sure I was going to get back in time for the meeting, but it is a Kodachrome of a pair of flies mating on an English walnut. This happened to occur on some of our own trees at the station, so that we are not immune from attack by this bug.

(Next slide.) That is a close-up of an egg puncture, just a very tiny little hole in the husk, and once in a while they lay an egg even on the surface.

Those eggs are quite small, about a millimeter in length and about two-tenths of a millimeter in width, but the next slide will show you that what they normally do is to put them inside that puncture in groups. They vary quite a bit, but the average number of eggs is about 20 in each puncture. But that doesn't mean you won't have maybe four or five different punctures on a given nut, so you may end up with at least a hundred or more maggots in a shuck.

(Next slide.) And the next picture is a photograph of the same English walnut taken about six or seven days later, showing the young maggots that have just hatched out. What they will do, they will begin boring in, and they will just radiate out in all directions into the shuck. When they have gotten that far along, of course, there is no hope for control.

(Next slide.) This slide is one taken when the maggots were almost mature, showing the type of damage that you get.

(Next slide.) This is the resting stage, or the pupa, the one which spends the winter in the soil and from which the flies emerge in New York, at least in our section, beginning about July 15th and going through up until August 15th.

(Next slide.) The one at the top is normal fruit. I mentioned a while ago that this butternut curculio causes quite a bit of concern and also spoke about its being in terminals. If you look carefully you see a very definite hole here in the husk. That is where the adult punctured the husk. It may have been a feeding puncture first and later an egg was laid inside, and then you get the maggot or the grub of the curculio developing in there, so that superficially that discoloration looks very much like the walnut husk maggot. But in this case you may not find over one or two maggots in a nut. And the other difference is that these fruits which are attacked usually fall during July and August, whereas the ones that have maggots in, many of them stick right on the trees and don't come off at all.

(Next slide.) I have two or three slides just showing the variations in the degree of injury on English walnuts from the point where you'd have an egg puncture. The puncture was made on the other side of the nut, on top here, and this is just the exudate running down around the nut which dries and becomes black. But these walnuts up above show just a lot of dark spots where the maggots are beginning to find their way through the husk. I have with me some injured nuts similar to those shown on the screen if you'd like to see them when I have finished my talk. They will give you a little idea what maggot injury looks like.

(Next slide.) This is the same type of injury on butternut. Maybe you'd have one egg puncture and as many as a hundred or 120 maggots inside the shuck.

(Next slide.) This is a picture of maggot injury on black walnut. They don't seem to like the black walnuts as well as they do the Persian walnut and butternut.

(Next slide.) This is one of the hybrid English walnuts that is located on the grounds at the Geneva Experiment Station. It's quite a large tree. I don't know the name of it. Maybe you do, George.

MR. SLATE: It has no name.

DR. GAMBRELL: It's not very fruitful, anyway, is it? But it is also susceptible to injury.

(Next slide.) This photograph was made quite a few years ago, and that explains some of the lines around it, but at any rate, this pile of nuts shows the damaged ones that came from one tree, and also the ones that were not infested. In other words, about two-thirds of the nuts on that particular tree had been infested with maggots.

(Next slide.) That's a close-up view and is the type of thing I was trying to describe to you earlier where the shucks dry up and stick to the nut so that you cannot remove them. Those on the left, of course, would be absolutely no good for commercial purposes.

(Next slide.) Now, I suppose you are all interested in this matter of control. Unfortunately, I must admit that I have not worked on the walnut husk maggots very much in the last 15 or 20 years. You may recall that we had a severe freeze back in 1933 or 1934, which took out quite a lot of our Persian walnuts in Western New York, and only the hardier trees remained. But prior to that time we had been getting numerous complaints, from growers about injury from walnut husk maggots, and we did some work at that time and also worked with the Farm Bureau people in the counties where walnuts were grown fairly commonly. In many cases these Persian walnuts were grown on fruit farms where they also have apples and other fruits. So that in those cases it was not a difficult problem to obtain control. We worked out a program whereby, say, beginning about July the 20th to the 25th, at which time quite a few of the flies would have emerged, if the orchardist, when he was going through with his regular spray operation on his fruit trees, would give his walnuts at least two applications at about two weeks intervals, he'd cease to have a maggot problem. That pretty well solved it, as far as they were concerned. But there were also these other plantings where you'd have just a few trees, or possibly one tree in a back yard, something of that sort, which is a little bit more difficult to control.

Dr. Glasgow and I found that on cherry maggot in the city, while a material like lead arsenate is very effective in a commercial orchard, it's very ineffective for just one little tree in your own back yard, providing your neighbors have some trees and they don't spray them. The reason is very obvious: the flies don't necessarily stay on the same tree. They visit around from tree to tree, they feed on the surface of the leaves or fruit. Therefore, it's possible for them to be over on someone else's unsprayed tree and still come over and lay eggs in the nuts of a sprayed walnut tree before being killed. So you can see that such activity may create somewhat of a problem.

At any rate, the lead arsenate spray of three pounds to a hundred gallons, with or without fungicides, has given good control in the past. That No. 3 combination of lime sulphur and lead arsenate was used west of Rochester here around Hilton where this grower had a commercial fruit planting, but he also had a number of English walnuts. The year prior to the time these trees were sprayed he had about 40 per cent of the nuts infested, and the year these were sprayed the infestation dropped they came down to about one percent. Notice the comment at the foot of the table which states that the trees that were not treated the following year went back up to 20 per cent of the nuts infested. There were about 20 per cent of the trees that had infestation. Of course, the flies moved around enough that the trees became reinfested. It simply brings out the point that unless you have a pretty good-sized planting, you are going to have to spray pretty thoroughly in order to get control, and also, if you only have one or two trees and you have a lot of surrounding shrubbery and a lot of trees, it would be very wise to also spray those, unless they are plums or peaches, which are quite susceptible to arsenical injury. But most things would stand the arsenate of lead, and it would be very desirable, wherever you can, to spray surrounding trees and shrubs close to the walnuts themselves, and in so doing you would get pretty effective control. It is quite possible to use this control method and obtain over 80 per cent reduction in infestation.

I am sorry to say I don't have any information on these newer materials, like DDT, methoxychlor and parathion. You have probably read about all of those in the magazines. Some of the men in our department have done quite a bit of work with these insecticides on the apple maggot in the Hudson Valley and in Western New York and they find, as I mentioned earlier, while it's possible to obtain control of apple maggot, say, with DDT, it requires much more frequent application. In that case, if any of you are orchardists or follow the apple-growing insect problems at all, the first application of the walnut maggot spray should go on at about the time the last cover spray for the coddling moth goes on for the first brood. That sounds a little involved, but from the calendar point of view it would be about July 25th in Central or Western New York. Normally, with us here the cherries are being harvested by about July 15th, sometimes a little earlier, but at any rate, that's the time the flies usually begin to emerge.

We have what we call a pre-oviposition period of about two weeks, during which time the flies are not laying any eggs in the shucks and are moving around feeding. Of course, that is the time you have to get this spray material on, before they have punctured the nuts and deposited eggs inside.

I think, unless there are questions, that's all I have to say.

A MEMBER: You recommend No. 3 to be used?

DR. GAMBRELL: Lead arsenate at 3 lbs./100 gallons and 2 gal. of lime sulphur would be an effective insecticide-fungicide mixture. I have used both the wettable sulphur and lime sulphur, as shown here, without any injury to foliage. Sometimes, as you know, if it's real hot, like today, sulphur could cause you a lot of foliage injury. Dr. MacDaniels will certainly bear me out on that.

PRESIDENT BEST: Now I think Joe McDaniel has a little idea here he wants to introduce at this time.

DR. MACDANIEL: I have been talking with Mr. Devitt. He is interested in following up these Carpathian trees in Ontario and is willing to act as our agent in securing seed nuts from some of the better selected trees. As I understand it, this Association couldn't properly act as a sales agency for them, but I believe there are some of the members who would like to get these superior seed nuts of Ontario, and I would be willing to take the names of persons who are interested in them, either for their personal planting or for resale. Mr. Devitt thinks he can secure the nuts at about 60 cents a pound from the owners who have these good trees and deliver them to the United States at around a dollar a pound. Anyone who is interested in that, see me or Spencer Chase during the remainder of the meeting.

Panel Discussion: The Persian Walnut Situation

Moderator: S. B. CHASE; *Panel Members*: H. L. CRANE, GILBERT BECKER, J. C. MCDANIEL, H. F. STOKE.

MR. CHASE: To introduce the subject, Lynn Tuttle sent a paper, and in addition he sent a few slides. We won't give the paper, but we are going to run through a few slides very hurriedly, because he took the trouble to send them. I am going to read the captions off very quickly. (A series of slides of Persian Walnut were shown).

The moderator isn't going to do anything other than ask for any questions that you folks have on Carpathians at this time. I am going to ask Dr. Crane to comment on this question: Are we going overboard building up our varieties as we know them now? In other words, we have selected four or five varieties that won a contest and our judges selected them as best, and these are the only ones we are hearing about.

DR. CRANE: Mr. Moderator, I don't believe we are, provided that we maintain high standards in the varieties distributed and tested. I feel that when we select a variety we should select it because it is a good nut, not that it's a world beater, for big size and thick, rough, rugged shell that is not sealed, and which is of no value for human use or consumption, excepting for firewood or fuel. Those big nuts won't fill. The best nuts are of reasonably large size, well filled with a well sealed shell and with a kernel that is sweet. Don't figure on selling nuts that have bitter kernels to anybody else. We have nut varieties of the Carpathians that are not going to go over because of the faults that I have mentioned. I should say, too, that we do not know how widely a variety is going to be adapted to different climates. If we select rigidly for good, outstanding varieties that bear good nuts and good, vigorous trees, we won't get too many.

MR. CHASE: That was one point I wanted brought out, that we are now just in the preliminary stage of this Carpathian variety selection business. Of the selections made some have been made by default, because there weren't enough of other samples to compete with. On the other hand, the several we have we all consider outstanding in some respect, or other, and are of value as a beginning provided we bear in mind that we haven't scratched the surface on Carpathian walnuts yet.

MR. STOKE: And let's not confine ourselves to Carpathian walnuts, because Hanson is not Carpathian walnut, and that's an excellent nut.

MR. CHASE: Mr. Stoke, what is going to be NNGA's policy in trying to give recommendations for the planting of Carpathian or Persian walnuts? In other words, does it make any real difference whether it's a Carpathian or whether it is not, as long as it has proved hardy and of good quality?

MR. STOKE: We are dealing with Persian walnuts, and Carpathian happens to be one class of Persian, and Broadview happens to be a Persian that came from Russia, and Lancaster is one that came from somewhere in Europe and landed up in Lancaster, Pennsylvania. I would emphasize the name Persian as the over all name. The Carpathian is merely a Persian walnut which has been brought from the Carpathian mountains of Poland.

MR. BECKER: Last summer a group of nut growers went to Lee Sommers', which is in the central part of Michigan. In the invitation to our nut growers I said, "This is the only pure Carpathian orchard we know in Michigan." That didn't set well with some of them and they took issue with me. In answering this issue, I said that Mr. Sommers had planted Carpathian Persian walnut seed that came from Poland direct. Many of us have a mixture. Even Mr. Shessler has the Hanson and Jacobs and a number of others. If he sells you seed, you are going to get it mixed. In a few years we will have a job keeping pure Carpathian.

DR. MACDANIELS: Isn't it a matter of straight terminology? *Juglans Regia* is the Persian walnut. Carpathians are a regional strain of *Juglans Regia*.

MR. CHASE: I think we all understand that.

MR. MACHOVINA: Can we speak of a Carpathian strain. Crath himself said there were many. He even found walnuts growing in clusters like grapes.

DR. MACDANIELS: It would be a regional group of clones with a certain origin not a strain in the genetic sense.

MR. STOKE: They are just Persian walnuts that happened to come from the
Carpathian region.

DR. CRANE: There is a little difference. I believe that in the northern countries we have had more or less inbreeding and we could consider them more nearly a line, not a strain, because of that. When the original seed was introduced by Reverend Crath, probably each one of those lots of nuts come from different trees, as a line, but, now this second generation stuff that's coming along, it's just *Juglans Regia*. It's a hardy Persian walnut.

MR. STOKE: I think I can offer a word of explanation of those growing in clusters. I have no doubt that when the barbarians swept over the wall centuries ago they brought Asiatic walnuts with them from as far as Manchuria. They grew in clusters there like butternuts and heartnuts. No doubt some of them reached Europe, and some of them may have hybridized with the Persian, and I think really that's the answer.

DR. MACDANIELS: The same situation existed with peaches 20 years ago. We had five geographical races of peaches that were more or less distinct. With the exception of one, the Peento, they have all lost their identity now because there has been no attempt to keep them distinct.

DR. CRANE: That's right.

MR. CHASE: Then we end up, there is no such thing as a Carpathian, it's just a name for a hardy walnut that came from a certain region, that distinguishes it from others.

MR. KEPLINGER: In my parents' old home in Eastern Germany in the Bohemian mountains there is an English walnut tree that's 300 years old and bears a hundred bushels of walnuts a year. They stand 40 below zero there, too, and the nut cracks and hulls well. It has a record on standing the cold, but there hasn't been any of them brought out here and planted in this country, but they are there. I know they are there, because they are on our estate.

DR. CRANE: Mr. Moderator, there is one remark that I want to make. Here we are, the Northern Nut Growers Association, and yet we still use the term, "English walnut," when we are talking about Carpathian walnut and Persian walnut. This "English walnut" is the worst form of terminology that can be used. England doesn't have any walnuts; they have never grown any Persian walnuts or English walnuts, they haven't in the past and they aren't today. They have a few trees but are in the same fix that we are in the Northern Nut Growers Association; they are trying to find a variety of Persian walnut that they can grow in England, and yet here we call them English walnuts. They should be Persian walnuts, or Chinese walnuts. We don't know where they came from. The best authorities seem to think that they originated in Persia; others think they originated in China, but the abundance of evidence is on Persia.

We want to get this thing kind of straight. They are all the same thing, *Juglans Regia*.

MR. CLARKE: I'd like to make a suggestion. I don't know as you have any authority or power to change, but the term *Juglans Regia* means "royal walnut." Why not work for the adoption of a name like that, and it will include all of them.

DR. MACDANIEL: That's what they call them in France. This country has a little complication; there is another Royal walnut, one of the hybrids between the California black and the Eastern black.

DR. GRAVATT:-While we are talking about bringing English walnuts, Persian walnuts, whatever you want to call them, from Europe, I want to give a warning about a disease that is killing thousands of trees in Southern France. Just recently I saw quite a few of them in France and the edge of Italy. I don't know whether it's virus or what it is, but it is certainly killing out the English walnuts there at a very rapid rate, and I advise very strongly against introducing walnut seed, scions and such, from those areas in France and Switzerland or other areas in southern Europe where this disease is prevalent. We will know more later about it, because quite a team of pathologists is working on it in Europe.

MR. CHASE: Has anybody else got any comments about *Juglans Regia*? I am afraid to say anything else.

DR. MACDANIEL: I will say that this Carpathian strain, of *Juglans Regia* is the first walnut of the Persian type that we have had for Illinois. The Pomeroy, other Eastern strains and California varieties have not survived very long in the climate of the state of Illinois. We do know now that some of the Carpathian seedlings have been fruiting for 10 or 12 years and do show considerable promise there. I don't know whether it will ever develop into a commercial industry but they are worth growing.

MR. CHASE: Thank you. I'd like to ask George Slate what he knows about the Northern Star Persian walnut. Very hardy, and so forth? I think maybe the members might be interested in that.

MR. SLATE: Spencer asked me to find out about the North Star *Juglans Regia*, which was advertised in the Flower Grower. I called up the local nursery that was selling them, and they said they got their seeds from some

Pomeroy trees in the western part of the state. I guess they are just *Juglans Regia*.

MR. STOKE: Down in Virginia we have Virginia Thin Shell purchased sometimes one place and sometimes another.

MR. CHASE: The secretary's office had an inquiry from the executive secretary of the American Nurserymen's Association wanting to know if those claims could be substantiated. I couldn't say on the basis of what information I had, and I so told him. Apparently they, through their organization, have stopped further advertising of that strain under the claims that they made for it.

MR. KORN: We find our public at large, not only our members, seem to be fascinated by the fact that the Persian walnut can be grown in this latitude. So in speaking to them about it, when I am speaking to our members, I try to say Persian walnut, but when speaking to the public at large, they don't know what I am talking about so I come out flatly and say English walnut. I tell them that we can't expect to grow the California type, but we have a hardier type coming from the Carpathian mountains or Germany or Russia or Holland, that can be grown successfully in this part of the country.

MR. CHASE: I think that's the only approach you can use.

MR. KORN: That's the one I use, and I think it quickly helps people to understand what you are talking about, and doesn't get them confused. If you talked to them about Persian walnuts, they wouldn't know what you were talking about, but if you say English walnuts, immediately they understand, or should, at least.

MR. CHASE: I believe Dr. Crane meant that in our inner sanctum he would prefer *Juglans Regia*.

DR. CRANE: I would like to ask if there are any growers here who have propagated the Persian walnut on Eastern black walnut, that is, experienced any trouble with graft union failure on them.

MR. STOKE: I haven't.

DR. MACDANIEL: Mr. Oakes?

MR. OAKES: I haven't.

MR. CHASE: No graft union failure on *Regia* and *Nigra*.

MR. STOKE: And my experience is they come in much quicker than on their own roots as seedlings.

DR. CRANE: How old are your oldest grafts?

MR. OAKES: Put on in 1938?

DR. CRANE: That's 15 years.

MR. STOKE: I have them at least 20 years.

MR. BECKER: Mine are twenty.

DR. MACDANIEL: Mr. Moderator, I have in my brief case a translation of the French book on walnut culture, and there is a section on root stocks. This was a publication issued about 1941, and according to that book, *Juglans Nigra* is the best stock they have for general use in France. They have reported no difficulty on this. A second one they were trying of the

American walnuts, with some promise, was *Juglans major*, the Arizona black walnut.

DR. CRANE: The reason why I asked, as I reported in previous meetings—they are having very serious difficulty in Oregon and in parts of Washington with graft union trouble which is known as "black line." All or practically all of the walnuts in both Oregon and California and also what few are grown in Washington have been propagated on the Northern California black walnut, *Juglans Hindsii*. No graft union trouble evidence shows up before the tree has been grafted about 8 years, and then such cases are very rare. But after the trees or the grafts attain an age of 15 years or more, graft union failures are numerous. For three years now we have been making surveys in the State of Oregon, and we have surveyed tree by tree, year after year, the same orchards, the same trees, and our observations now go into the thousands, and we find that this black line is a terrifically serious thing. In some orchards 22 per cent of the trees will develop black line in one year's time. So, you see, at that rate it would only take you five or six years with a good bearing orchard until you wouldn't have any.

DR. MACDANIEL: Is that always with the Franquettes?

DR. CRANE: That is not only true with Franquettes but also with other varieties in California, even in Contra Costa County.

MR. STOKE: Where those trees are so grafted, does it tend to overgrow, or just the opposite?

DR. CRANE: No, it appears much like our *Crenata-mollissima* chestnut graft union failure.

MR. STOKE: Is there a tendency for the top to be more vigorous, to have more growth, or vice versa or is growth uniform?

DR. CRANE: It may be uniform. Depends somewhat on the varieties and the seedlings. There may be some overgrowth or some outgrowth, but there is only one test for it, and that is at the graft union. With an axe or knife and you cut out a strip of bark across the union. It may look absolutely perfect, but if there is a black line developed there that is just like a lead pencil line between the stock and the scion, the tree is on the way out. It's just a matter of time. Ultimately the bark between the stock and scion will split, and you get infolding, just like on the chestnut.

One of the reasons that they have propagated their trees on Northern California black walnut was that they had the idea that the Northern California produced a stronger, more vigorous seedling and that they grew much faster than seedlings of the Persian walnut. And, furthermore, somebody at some time circulated the idea that Northern California walnuts were immune to infection by the mushroom root rot fungus. We have surveyed thousands of trees of Persian on Persian roots, and we have never found a single case of black line developing or graft union failure as long as it's a Persian on Persian, and we find the same percentage of infection from mushroom root rot fungus on Persian as on Northern California black.

MR. CHASE: In other words, we should watch our stocks and perhaps try out some *Regia* on *Regia*?

DR. CRANE: That's right.

MR. CHASE: Now, folks, we could talk for a long time, but let me make one request before we close our panel: I would be interested in receiving from any member pictures, good, glossy photographs of the newer Carpathian varieties so that we can perhaps publish them in the newsletter and give some folks an opportunity to see what these nuts look like. Some of the folks who never come to a meeting never see a sample and just read about it. It's much better if we can show them a picture now and then. So if you have some good pictures, or plan to take some good pictures, remember, I'd like to have a copy.

TUESDAY EVENING BANQUET SESSION

PRESIDENT BEST: We will now hear from the Resolutions Committee, Mr.
Davidson.

MR. DAVIDSON: Before reading any resolutions, I have been asked to read a letter that came to Mr. Chase dated August 16th of this year, from Dr. W. C. Deming:

"Mr. Spencer B. Chase, Secretary, NNGA.

"My Dear Child and Grandchildren:" What a beautiful greeting, that.

> "This is to let you know that your father and grandfather still holds a house at this hospital and rejoices in your vitality and in your coming convention but especially in the energy and ability of your secretary who gets out those wonderful Nutshell letters which are so stimulating to all nut growers.
>
> "More than 20 years ago I planted an Italian chestnut tree on the grounds of this hospital. The main trunk was killed by blight, but many shoots have come and now it appears to be flourishing because there are no other chestnut trees near. About that time I grafted nut trees commercially in Westchester County, New York

at the Westchester Country Club, asking and getting $50 a day for my services and material and never a kick. But I have forgotten the results and the name of the beneficiaries. From my home in Litchfield, Connecticut, my sister, aged 85, saved for me—that is, saved from the squirrels—a double handful of nice chestnuts—no other chestnut tree nearby—and three green walnuts, Carpathians. Both were from my grafts.

"I shall never forget the NNGA and your splendid services. Ever faithfully devoted, Dr. W. C. Deming."

A beautiful letter.

Now, then, the Resolutions Committee recommends that we send this letter;

"Dear Dr. Deming:

"Once more we are happy to greet you and to wish you well. Today the representatives of more than a thousand members of your thought child, the Northern Nut Growers Association, are here gathered in Rochester, New York, to carry on the work in which you had so large a part in starting. It must be a source of great satisfaction to you to be able to see so important a project which you helped to start continuing and expanding fruitfully. We envy you.

"May your tribe increase. Affectionately, the Northern Nut Growers Association."

Now, shall I go on with the rest of the resolutions, and perhaps you can act on them all at once.

"*Be it Resolved*: That we hereby acknowledge our longstanding indebtedness to the men of the United States Department of Agriculture and of similar departments of the various states who have so faithfully and efficiently upheld the work of this Association. Without their loyal help, doubtless our efforts would languish or suffer severely. It is such a spirit as theirs that continues to make America the great pioneer it has always been."

"*Be it Resolved*: That the members of this Association acknowledge with deep appreciation the outstanding' hospitality of the City of Rochester at the hands of its representative Park Commissioner Wilbur Wright, Dr. Roy B. Anthony, Mr. Harkness, and their helpers who have done so much to make the visit of this organization not only welcome but extremely enjoyable and informative. We shall always remember Rochester's exceptional hospitality and its generously free provision of so beautiful a meeting place. This is sincerely appreciated."

"*Be it Resolved*: That this Association extends to Mr. George Salzer, Mr. Victor Brook its thanks for their work which has resulted in so pleasant and profitable a meeting here in Rochester; also to many others due our thanks, to Dr. McKay for organizing a splendid program, to Mrs. Negus for organizing the registration, to Mrs. Gibbs and finally to our outstandingly efficient officers who have so skillfully organized our work and the Association's expansion."

"In order to correct a tendency toward increasing confusion arising from the too great multiplicity of names and nut varieties, the Resolutions Committee offers the following motion: We move that the President be authorized to appoint a self-perpetuating

> Northern Nut Growers Association Committee on Variety Nomenclature, and we recommend to our members that they refer to this committee for its official approval any new nut discoveries they may wish to name and to propagate." That is in the form of a motion, which, I believe, requires a second and some action.

PRESIDENT BEST: I think we had better act first on this motion of Mr. Davidson's about this committee for naming of nuts, and then we can have another motion to accept the resolutions. Is there a second to that motion that we have a committee on nomenclature of nuts?

DR. MACDANIEL: Mr. President, I second that motion.

PRESIDENT BEST: Is there any discussion?

DR. CRANE: I wanted to suggest that the motion should provide that the committee use the rules of nomenclature approved by the American Pomological Society.

DR. MACDANIEL: I will accept that amendment, Mr. President.

(A vote was taken on the amendment, and was passed.)

MR. MACHOVINA: Is that a proposal to amend the by-laws of this organization? It would, if it's a self-perpetuating committee.

MR. DAVIDSON: May I suggest you withdraw the word "self-perpetuating."
The idea, Mr. Best, was to make this a permanent committee, if possible.
That was the reason for putting that word in there, but if it is an
abridgment of the constitution, we don't want to do it, of course.

MR. KINTZEL: I'd like to know what the rules of nomenclature of the American Pomological Society are.

DR. MACDANIEL: The rules cover about two pages. I can give you the gist of it, I think. One provision is that the discoverer or introducer of a new variety has the privilege of selecting a name for it. Another rule is that it shall not duplicate a name given previously for a variety of the same class of fruit or nut. The name should preferably be one word or, at most, two words, without hyphens, without possessives. That a nut not be named for a person without his permission during his lifetime. That covers the meat of it.

MR. CHASE: Such a committee would give official status and recognition to your discovery. I believe it would prevent, on a large scale, such things as this Morning Star hardy English walnut. In other words, we'd have a committee to examine a nut sample from your tree, anybody's tree, pass on it and see that the name that you select meets the requirements of this American Pomological Society's rules of nomenclature, which are quite reasonable. I think it is an excellent step that we should take at this time.

MR. CALDWELL: Mr. President. The variety we are using is not a variety, it's a clone. Maybe we had better get together with taxonomists and botanists. That's all they are, selections, they are not varieties, in the botanical sense, even though the term has been badly misused by the nut growers. I don't see why we should continue with mis-application of a term just because somebody set up rules for application of names.

DR. CRANE: Mr. President, I want to get this straight. This Association is talking about horticultural varieties, not botanical varieties. A correct term for a horticultural variety is a clone.

The American Pomological Society is over a hundred years old, and they have followed all types of experiences and usages, and they are up to date, and we can't follow any better pattern than what the American Pomological Society has done all down through the years. The Northern Nut Growers Association would be the laughing stock of the world who are in the know if they don't adopt the rules of nomenclature as set forth by the American Pomological Society.

MR. SLATE: Mr. President, we already have a committee on Varieties and Standards. I don't see why that committee can't be revived. If we set up another committee by resolution, we are duplicating the work of that committee, or overlapping. I'd like to see this matter referred to the Committee on Varieties or Judging Standards and possibly report another year. I am not in favor of setting up this committee at the present time.

I would like to amend that motion to refer this matter to the present Committee on Varieties and Judging Standards.

DR. MCKAY: Second that amendment.

DR. MACDANIELS: There is one other angle to this. The International Committee on Nomenclature of Cultivated Plants, has during the past year published a report. I think it would be only wise for us to delay our action on this matter until our committee at least gets the opportunity to study the suitability of this international code for nomenclature of cultivated plants and see how it applies to our situation. In other words, I am complete in accord with Mr. Slate's motion.

MR. DAVIDSON: I would suggest that we refer it to that committee and think about it for another year.

(A vote was taken on the amendment, and it was passed.)

PRESIDENT BEST: The motion of the Resolutions Committee is now referred to the Committee on Varieties and Judging Standards.

MR. STOKE: I move we proceed with the adoption of the resolutions that were presented before the motion.

(Motion seconded and passed.)

PRESIDENT BEST: We'd like to hear from the Nominating Committee.

MR. SLATE: For president the Nominating Committee proposed R. B. Best. I think he is about one of the best presidents we have ever had. For vice-president Gilbert Becker of Michigan. For treasurer W. S. Clarke of Pennsylvania, and for secretary, our very efficient and very effective Spencer Chase of Tennessee.

PRESIDENT BEST: Are there further nominations?

A MEMBER: I move nominations be closed.

A MEMBER: Second the motion. (Motion passed.)

DR. CRANE: Mr. President, I move that the Secretary of the Association be instructed to cast a unanimous ballot for the nominations made by the Nominating Committee. Seconded and passed.

PRESIDENT BEST: Is there any further business to come before the group?
Spencer Chase here has an item.

MR. CHASE: Now, for the feature of the evening, that honor which is bestowed upon the most deserving member of the organization, I will call on Mr. Wilkinson for the comments about the Big Nut. Mr. Wilkinson.

MR. WILKINSON: Four years ago at Beltsville, Maryland, Dr. Crane made a suggestion that someone ought to be the King Nut of the Association. If I remember, Mr. Stoke immediately took the floor and nominated Dr. Crane, and he was unanimously elected the Big Nut. One year later he bestowed that honor on Spencer B. Chase. The next year Mr. Chase passed it on to Dr. Colby. One year ago Dr. Colby passed it on to me. Now it's my duty to pass this on to someone else tonight.

Well, I didn't know just exactly how to do it, so I fell back on my friend, Mrs. Negus for her suggestion. She suggested that the King Nut should wear a crown, so I said, "Now, that's your suggestion; I will leave it up to you." So here is the crown she made, with an ornament from the Chief pecan, which in my opinion today is the king nut of the Northern Nut Growers Association.

I don't know whose head she measured to make the crown, she didn't tell me, but it looks to me like it would just about fit George Salzer. (Applause.) George, it's a pleasure that I pass that on to you as I received it, and I hope you will wear it for a year or longer (putting crown on Mr. Salzer's head).

MR. SALZER: Well, they can't say I am big-headed. Why, honestly, folks, my very good friends of the Association, honestly, I don't know what to say. This is the greatest honor that I ever thought would come to me. I always refer to myself as one of the buck privates in the rear rank, and here I am the King Nut. I will assure you, every one of you that I really appreciate this, I honestly do, right from the bottom of my heart.

Ever since I have been a member of this organization and attended the meetings, I have had the finest times, most pleasant associations and the closest friends I ever had in my entire life right here among you people. Thanks a million.

MR. CHASE: Now I think we are entitled to a few words from our new and best president, Mr. Best.

PRESIDENT BEST: Ladies and gentlemen, it is quite a responsibility to take this job on again. It's the first time that I have ever questioned your judgment about anything, but I think there are other people here that could have done the job better than I could.

When I was asked if I would accept if I were elected, I turned to my wife, and I said, "Are you willing to do the work again for another year?" and she said, "Yes, I suppose I'll have to." And I said, "Well, then, I will accept." There is a lot more truth in that than there is poetry. Honestly, we just don't give these officers that work for us enough recognition. There is a whole page of them, as you know, about 11 committees, and all those folks have all done a fine job, at the expense of their work at home. I am not talking about myself, because I don't do any of it, I have it done, as I explained. But Carl Prell made a great sacrifice when he handled the Northern Nut Growers business in a very, very fine, thorough, business-like way.

I ought to give you a good example of what salesmanship really means and how it operates. This morning Carl was going down to the museum in a taxi. The taxi man professed an interest in nuts. Well, what did Carl do? Did he say, "Well, that's all right, but I can't get into that?" No, he said, "Man, you ought to belong to the Nut Growers Association. The fact that you don't know anything about it, that's nothing. Come right into the museum here, and I will show you the exhibits," and he took the taxi man in, and I don't know whether he sold him a membership, but he passed him on to the next man. He's got him going out to see Irondequoit, and we, are going to get a sale there. That's the spirit that it's going to take to get this job done.

I am reminded of a little story in Kipling. You know the story about the sergeant in India. He was a sergeant in the cavalry. They had been out in the hills, and the weather was hot, and they had an awful, awful time. Well, when the men came in and lined up, this sergeant got off his horse and he said, "Well, boys, I realize it's been hot, I know you sweat. But," he said, "from here on in this campaign we are not going to sweat, we are going to lather." That's what it's going to take to get this 2,000 members that we have set for our goal. It's going to take a lot of hard work, and our job is not to peer into the dim future, but to attack those problems which are right with us every day and ask some of our friends to join the Nut Growers Association. We are all widely separated in different walks of life, and each in his own world is just apt to see things a whole lot like the goldfish in a bowl. That is, he will see it twisted and distorted. So when all is said and done, it's up to us to support these committee heads and help get this job done.

A preacher had in his congregation an old lady who was ill. On one of his visits to her she appeared to be growing weaker all the time, and fearing the worst he said as he left her, "Well, sister, I suppose that we will meet Up There." And she looked at him and she said,

"Well, Parson, it's up to you." So from here on out now it's up to you.
Thank you. (Applause.)

MR. CHASE: Now I think we ought to have just a brief word from Gilbert
Becker, our new vice-president. Mr. Becker.

MR. BECKER: This was really a great surprise to me. It is humbly I tell you this, because I was on the Nominating Committee myself, and that is a very embarrassing position to be in, to find that I, as a member of Nominating Committee, appear as an officer. But it was a pleasant surprise, and in doing vice-president work I shall try my best, and I shall surely spend much time and much thought to it. Thank you. (Applause.)

MR. CHASE: Now, Bill Clark, will you come up here for a few words? Bill will succeed Carl Prell as Treasurer and handle your finances during the coming year.

MR. CLARK: Friends, I thank you for this honor and that you should have enough confidence in me to trust me with your funds for the coming year. I will do the best I can, and thank you very much.

MR. CHASE: There will be a joint meeting of the new officers and old officers immediately after we adjourn.

George Salzer says the last time we met in Rochester was 1922, and we figure the next time we will be here is 1984.

(Whereupon, the meeting was adjourned.)

Walnuts in Lubec, Maine

RADCLIFFE B. PIKE, *University of New Hampshire, Durham, N. H.*

In the 1930's, when the Wisconsin Horticultural Society distributed seeds of the Carpathian Walnut from Poland, my brother secured some and we planted them. From these nuts we now have four trees, one of which has been bearing for the last four years.

Lubec, Maine is located at the extreme eastern tip of the United States just a few miles south of the 45th parallel. The site where the trees are planted is on a peninsula extending out into the Bay of Fundy which gives very low summer temperatures and moderate winter temperatures. The night temperature in the summer is usually in the 50 deg.s F. with day temperatures rarely reaching 80 deg. F. Winter temperatures seldom go to -10 deg. F. and only lower than this about once in ten years. During the early summer, fogs are usually heavy and continuous. Day length is, of course, longer in the summer than in most of the United States but it is similar to that of the Northern tier of States from the Great Lakes West.

The trees have grown well being about 18 feet tall and even more in spread. They are multiple trunked having never been pruned. The foliage is remarkably clean and glossy and has not been bothered by insects or disease and it ripens and turns yellow in the late fall before killing frosts at the end of October. Excessive late terminal growth is usually winter-killed but this sort of growth has not been as great since bearing started. The staminate flower buds are more likely to be winter killed than the pistillate but the whole of them have never been entirely killed.

The trees do not leaf out until mid-June, after the danger of killing frosts is over. They have not been frost injured at any time in the spring. The nuts ripen and are shed from the husks in late September and early October while the tree is in full foliage. The nuts are shed perfectly clean with husk either falling separately or remaining on the tree. The nuts will germinate and seedlings have been raised. In 1953, one tree bore 315 nuts. This number represents just a fraction of the pistillate bloom, for while this tree is self fertile, the catkins bloom for a much shorter period than the pistillate blossoms, the latter extending over nearly a month.

In the same year that the above Carpathian variety of *Juglans regia* was planted, my brother and I also planted some *Juglans mandshurica* secured from F. L. Skinner of Dropmore, Manitoba which had originated in Harbin, Manchuria. The resulting trees agree well with many of the specimens in the Herbarium of the Arnold Arboretum at Boston labelled *Juglans mandshurica*. These trees have done remarkably well in Lubec, the trunks being 8-9 inches in diameter, while the height is 15 feet or more and the spread 20 feet. They have borne annually since 1939. They are planted less than 1000 feet from the ocean exposed to the summer storms, winter gales and salt spray. These trees leaf-out a month earlier than the Carpathians yet the foliage has only been partially frost injured once. Wind whipping sometimes injures the leaves in early summer while they are still tender but this sort of injury has never been serious.

The nuts are borne in clusters of up to six and the shells are hard and thick. The flavor of the kernels is excellent having more character than the butternut yet not as strong as the black walnut. Cracking is easy with the Hershey nut cracker. The kernels resemble our American butternut in shape which may account for the fact that *J. mandshurica* is sometimes called the Oriental butternut.

The nuts germinate well and make trees quickly. In one case, I had mature nuts five years after planting the seed. This particular tree was unusually vigorous having leaves 36 inches long and 23 inches wide.

In my experience, *J. regia* and *J. mandshurica* do not hybridize easily if at all, at least with the individuals and under the conditions with which I have been working. After several attempts I now have two progenies of reciprocal crosses of which a few seedlings seem to show hybridity in the vegetative parts. However, there is such a range of characters in the herbarium specimens labelled *J. mandshurica* that there will be a doubt in my mind until I see the mature trees, or it may be possible that some of the herbarium specimens may have been collected from naturally occurring hybrids, as the two species overlap in their distribution in Manchuria. If the best vegetative and fruiting characters from these two species can be combined the result should be good for our northern sections.

My Thirty Years Experience With Nut Trees

CARL WESCHCKE, *St. Paul, Minn.*

From time to time I have submitted articles for our annual report, as well as other publications, which had to do more or less specifically with certain species of nut trees, but since there are so many species, and since most nut growers are interested in at least two or more, it might be well to bring the story up to date of how a nut orchard might be viewed or evaluated after twenty years.

Thirty years ago we did not have knowledge which has been gained by the experimenters in the nut growing industry in the interim. Therefore no one could foresee what the future would be. We hopeful ones of that era planted trees and experimented with seeds from all over the world because we thought nut trees deserved a place not only in the orchard but in the dietary needs of the human being as well. Many of the wisest and most respected experimenters of this era have passed beyond this life; however, their lives were made much more interesting because of their horticultural activities.

Here in the midwest in the 45th parallel we have established probably what would be considered the practical northern limits of nut tree cultivation. When I purchased trees it was by the hundreds, and sometimes thousands, because I knew from reading Luther Burbank's works that this work had to be done on a rather large scale in order to make any kind of an adequate test.

Let us start by taking the most obvious species, the black walnut, which, because of its native hardiness, and public popularity might have succeeded the best in a commercial way if everything had gone right. I have planted at least five hundred black walnut trees altogether; these included the Thomas, Ohio, Ten Eyck and Stabler, and later on the Patterson, Rohwer, Pearl, the Throp, Adams and others were added. The Ohio probably produced the first nuts, with the Thomas a close second. For a few years I was able to make good reports on the Stabler and its behavior but since that, our severe test winters of recent years have wiped them out and have substantially proved that the only one of the older varieties which can be trusted in this territory is the Ohio, and although derogatory things have been said of the Ohio because of its hull, I am inclined to put it high on the list because of its fine cracking quality and excellent flavor, also it has been a prolific variety, bearing good crops most of the years.

The Thomas grew faster, and the nuts were considered a better commercial product when hulled, but, alas, it could not take our winters nearly so well, and today the Thomas has a poor physical appearance although it shows tremendous power of recovery and seldom a tree will die entirely.

The Ten Eyck was a negligible experiment, and the Stabler as mentioned before, is much too tender for this climate. The Rohwer and the Patterson from Iowa did much better and even in an off year, like this one, some of these trees had fairly good crops. I like the Patterson the best of these—it is a roundish nut that cracks quite well and the kernels are on the sweet side.

The Throp was a curiosity and we did not have any of our grafted Throp trees bear.

Pearl has borne several crops of good nuts; they are large but are inclined not to ripen in time.

Vandersloot was considered the largest nut of any variety at one time. It has a very rough appearance but aside from its size it is of no particular interest as compared to others.

Adams, a long narrow type of nut similar to the Ohio, but still more elongated, was one of the best crackers I have ever seen, but did not seem to be prolific although it has lived and demonstrated its hardiness.

I am patenting a new walnut at this time which I consider the best for our locality. Some day it may produce well in orchard form if trees become available. One thing is certain about it—it is very hardy and is reasonably easy to propagate.

And so we can conclude the walnut chapter by saying that at least we have some giants in the orchard to show for our trouble and expense, which bear nice edible walnuts in favorable seasons. When comparing this with the wild butternut crop from butternuts in the adjacent woods, which has consistently failed each year for the last ten years, it is quite encouraging.

It was my hard luck to have an uncongenial soil for my experiments in chestnuts, and the knowledge of this came so late that I thought the chestnut was not meant to succeed in our territory. So I put my efforts on hickory nuts and filberts. Both of these succeeded to a degree and with my present knowledge and experience on hickory nuts I would not be a bit afraid to start an orchard on good deep clay or other satisfactory soil which hickories like, using grafted trees of Bridgewater and Weschcke.

A few Kirtland and Deveaux No. 2 would be planted for extra pollination and the extra variety in nuts. There are of course many other varieties of hickories that have succeeded in this territory but those above mentioned, have possibilities of commercial success in orchard formation.

The hickory is a difficult tree to transplant and I would advise that grafted trees be dug with a ball of dirt for shipping, similar to an evergreen, as I have found that, with the greatest of care and experience, the hickory is very slow to re-establish itself unless handled that way.

The hybrid hazels are perhaps the hardiest and certainly bear the earliest of any of the nut trees. My own hybrids show great possibilities for commercial enterprise, but as yet no nurserymen are carrying these varieties and I have not found help enough to promote them myself.

I am convinced that had I spent as much time with the chestnuts on favorable soil as I did with hickories that they would probably head the list of successful nut trees growing. Recently I have purchased an adjoining piece of property which has the necessary well-drained, sandy or gravelly soil, which chestnuts seem to like, and I have started my chestnut orchard there along with a sprinkling of hickory and walnut trees, merely as a matter of test.

This year the chestnuts are again putting on a fair crop for the number and the size of the trees involved. As yet, in order to get a reasonable number of nuts for planting, I have to cross-pollinize them by hand, and I was surprised and pleased this year to find one Chinese chestnut tree with staminate bloom, allowing me to make a cross pollinization with an American sweet chestnut and a Chinquapin type chestnut, which grows to be a tall tree. These crosses ought to insure trees with a great degree of hardiness, and should the blight ever strike this territory in the future they should be highly resistant as well. A few of my chestnut trees produce nuts that may be the size of the best Chinese chestnuts, but I am just as fond of the smaller and sweeter chestnuts of the several Chinquapin type trees which seem to be consistent bearers and certainly are prolific. There are three trees in a close group which are strains of the European chestnut combined with American chestnut. These bear rather large nuts and usually every year have a few and of high quality. It is conceivable that by crossing this hybrid with Chinese pollen that something unusual could be produced.

The pure Chinese strain has not proved hardy in this territory and I have never matured a pure variety. However, there are dozens of seedlings that are not old enough to prove whether there might be a hardy specimen among them that may at some time in the future be relied upon for this species of chestnut.

One other species of nut requires a little space here since it has shown that it can bear crops and is hardy enough to be included among the hardy nuts. It is the Gellatly heartnut. It is very subject to the butternut curculio, but in spite of that it continues to grow quite well when grafted on black walnut,—a difficult piece of propagation, however. A tree in St. Paul, on the boulevard, thrives next to a large butternut, and bears nuts practically every year which the squirrels delight in cutting down while still green. This tree is not bothered by the curculio since the curculio does not infest the large butternut near it.

In summing up the whole situation, I would say that my experiments over thirty years quite adequately prove that the walnuts, hickories, hybrid hazels and chestnuts can most certainly be set out in orchard form and in favorable locations. However, pecan, hiccan, English walnuts and almonds have not proved hardy enough to indicate that they can be relied upon for steady crops of nuts although in some instances varieties show a great hardiness such as the Rockville hiccan. Of course the native butternut is perfectly hardy and prolific but until such time as the butternut curculio ceases to be a major pest we cannot expect to have good crops of them.

Growing American Chestnuts and Their Hybrids Under Blight Conditions

ALFRED SZEGO, *Jackson Heights, N. Y.*

An interesting group of young American chestnut trees growing on my land near Pine Plains, N. Y. has been under observation since 1946. As they are growing closely together which suggests a common parental origin, we have named this group the "Dutchess Clone" for reference purposes. This name was chosen merely because Pine Plains is situated in Dutchess County.

Their reaction to the deadly chestnut blight was studied at great length and at different seasons. Sometimes branches were inoculated with the fungus to test resistance more precisely. It was learned that blight resistance, in this group of trees, was at an apparently low ebb from March until May. After this period the fungus seemed to make almost no progress at all. This might suggest that the resistant substance was manufactured by the leaves. Of course, such conclusions cannot be accepted in a scientific sense without an involved system of checks and measurements.

Pollination problems are exactly the same as with our Chinese Chestnuts that we are more familiar with today. Unlike the latter, in the American, species the bloom is concentrated near the top of the tree.

The burs are so high up as to create difficulties if we intend to anticipate nature and harvest our crop prematurely. The burs open during the month of October with or without frost. High temperatures in 1953 did not interfere with the harvest. The best method of harvesting is to use a long slender pole with a metal hook at the extreme end, and by gently pulling and twisting, remove the burs from the tree.

Unless this is done promptly before the nuts fall, the rodents will get almost every nut.

Tree growth is about 2 to 3 feet per year in height. At present some are nearly 40 feet tall. Bearing starts at about 12 years of age. The nuts, three in a bur are somewhat wedge shaped and average 5/8 of an inch in diameter. One tree has nuts almost an inch in diameter. This is definitely worth propagating and I will gladly furnish scions in the spring free to anyone who is interested. These are probably incompatible with Chinese understocks, but may be grafted on European and some Japanese seedlings.

As we are listed as cooperators with the U.S.D.A., Division of Forest Pathology, Beltsville, Md., we prepare semi-annual reports for Dr. Frederick H. Berry and also send a portion of our American chestnut seed to him. In this way we insure the continuation of the "Dutchess" clone after our lifetimes.

The American chestnut is not as sweet as Chinese chestnut but is much finer in texture and richer in subtle pleasing flavor. We would say that the quality is higher. *Castanea dentata* has the most uniformly delicious nuts. It is excelled, however, by many individuals of *C. pumila*. In our opinion these possess the highest quality nuts in the entire genus.

Our American chestnuts hybrids, especially those with *C. Sequinii*, are very interesting. The latter make a dwarf tree that bears incredible amounts of small chestnuts. They have pollination problems to be solved and the nuts are seldom filled. Pollen sterility is a common feature with them. They are also everbearing.

Some Northern strains of Chinese chestnut seem barely hardy but promise to survive. Of the grafted varieties we have, Abundance is the most vigorous. "Nanking" has winter-killed here and it has been replanted this year. These are very blight resistant, and rarely lose a branch to this disease after winter injury. The Japanese behave in much the same way.

We have many obscure chestnut species and hybrids growing here. They are grown for study, hybridizing purposes, and as a source of supply to interested members. When mature, we hope to obtain some cash crops from our Chinese and Japanese Chestnut trees. Blight in Europe will no doubt, in about 5 years more, reduce imports of chestnuts thus creating higher prices and a more favorable market.

Chinese chestnuts do not keep well when stored using standard commercial practices. European chestnuts are shipped in barrels and kept in open fruit boxes for weeks at a time in front of fruit and vegetable stores in New York City. Storekeepers never moisten these believing that rot would result. These are viable even in January and sometimes as late as March. Will our present Chinese chestnuts keep as well under these conditions? We think not. American Chestnuts can be kept in bulk only.

We are continually striving to obtain by selection and subsequent hybridization, the best chestnuts that can be grown in our severe climate. The Chinese chestnut has performed miracles in the Southeast, but we regret that it is not the answer to our problems. Only a long period of seed selection will turn up better trees of this species.

Prolonged heat and drought caused us much concern this year. Some one year old seedlings died outright but older trees only suffered varying degrees of defoliation. In some areas, the subsoil was reported powder dry to a depth of six feet. Even the native forest trees dropped much foliage and went into premature dormancy. Oddly enough, the American and Japanese chestnuts suffered much less defoliation than the common Allegheny chinkapin, *C. pumila*. *C. henryi*, a rare species, a native of China, and the several chinkapins native to the Gulf Coast seemed inherently adjusted to drought and heat, and thrived without apparent damage. The Ozark tree chinkapins did well also.

Hybrid hazels and choice native seedlings have been set out here in the last few years. We are adding a few every year and planting them between chestnuts to prevent the latter from forming extensive root grafts. This is done in anticipation of oak wilt, which has not yet made its appearance here.

Experiences and Observations on Nut Growing in Central Texas

KAUFMAN FLORIDA, *Rotan, Texas*

In view of my membership in the Association for some twelve or fourteen years it would be quite reasonable to expect of me more observations in connection with nut growing in my area than I'm able to make. Though I've followed the proceedings of NNGA with great interest, the difficulty of earning a living (from farming) and putting a little something aside has caused me to neglect and put off from year to year the planting of the kind of experimental orchard I've long hoped for. I have lately acquired a reasonably well situated plot of land and, barring a continuation of the drouth of the past two or three years, plan to put out a few young trees next year.

My original interest in nut trees sprang from the hope that a tree combining beauty, utility and long-life might be found to replace the Chinese elm—a "weed tree" if there ever was one. In spite of many shortcomings the Chinese elm (along with two or three other equally undesirable trees) is to be found in most homestead plantings in my area.

Here, in my locality of north-west central Texas, the total rainfall ranges from a low of about twelve inches in some years to a high of about forty-two inches in others, and the annual average is about twenty-one inches. Our principal limiting conditions in nut tree growing is want of sufficient rainfall, though late spring frosts following a period of balmy weather would be a hazard in some instances. It appears to me that if a nut tree planting in this part of the country is to live, every drop of water that falls must be conserved; if it is to thrive, additional water falling on adjacent uplands and carried down in flash floods must be diverted to it. Terraces and retainer dams are usually essential. Cultivation and weed control are necessary. The addition of a mulch helps.

I have tried the Chinese chestnut here. The plants arrived in good condition and had excellent care with what I believe was adequate water and fertile soil. They put out in April and grew off most encouragingly until about July, and then, in an interval of about a week, every tree withered and died as though from heat and drouth. But until other evidence to the contrary comes in, I shall strongly suspect that the real trouble was that the Chinese chestnut demands an acid soil and is highly allergic to even a slight alkalinity. My impression is that the soil here has a reading of about pH 7-7-1/2.

Experience and observation here on the western fringe of the native pecan belt lead me to believe the pecan, black and Persian walnuts do well when they can be irrigated, or when they are planted on a site where a first class water conservation system can be devised and properly constructed.

The black walnut has not been damaged by any insect, disease or mineral deficiency of the soil that I know of. A very limited and inconclusive experience with Clark, Thomas, Myers, Mintle, Sifford, Snyder and

Sparrow varieties led to the suggestion that the Thomas might be a slightly more thrifty tree.

The pecan (both nut and tree) seems more subject to insect damage than the walnut. It is also sensitive to a zinc deficiency in some soils. But a proper mineral and insecticide spray usually serves to control these problems when they occur.

I have observed only one named variety of Persian walnut—a Mayette. The tree was a vigorous grower and precocious in putting on nutlets, but to my knowledge never bore staminate blooms and over a period of several years matured only one nut. No other Persian walnuts grew in the locality and I assumed the matured nut must have been pollinized by a black walnut. The tree never seemed damaged by late spring frosts or other cause.

A few members of NNGA have manifested an interest in the honey locust and the Chinese jujube. Both of these trees grow well in this region with a minimum of care. The Oriental persimmon, like the nut trees, requires more than casual attention and ordinary growing conditions.

The Chinese jujube, a little known but hardy and attractive tree may deserve more attention in the southwest. I have trees of the Li and Lang varieties which bear annually and have never been bothered by insects or disease. I am not overly enthusiastic about the fruit but understand it "compares favorably with the fig and date in food value. Dried jujubes carry more protein than dried figs or dates and more (50%) sugar than figs."—T.A.E.S. Bulletin no. 41. But the jujube has the disagreeable habit of sending up root sprouts which are a nuisance to destroy and, because the tree is grafted, the sprouts are worthless seedlings. It has occurred to me that this bad feature of the jujube might be partly offset if cuttings of the improved varieties could be made to grow by means of some of the root inducing chemicals.

Propagation of the Hickories[1]

F. L. O'ROURKE, *Department of Horticulture, Michigan State College*

The genus *Carya* comprises all the hickories and pecans found in the United States. The eighth edition of Gray's Manual of Botany lists the following species as being native to the United States:

[1] The survey of literature pertaining to this review was completed in August, 1952.

Carya aquatica—Water hickory, Bitter Pecan *Carya cordiformis*—Bitternut, Swamp Hickory *Carya glabra*—Pignut *Carya illinoensis*—Pecan *Carya laciniosa*—Shellbark, Kingnut *Carya ovalis*—Sweet Pignut, False Shagbark, Red Hickory *Carya ovata*—Shagbark *Carya pallida*—Pale Hickory *Carya texana*—Black Hickory *Carya tormentosa* (C. alba)—Mockernut

Nut growers are interested primarily in the pecan and the shagbark, although a few selections have been made of the shellbark species. The bitternut is quite often used for rootstocks for the shagbark and shagbark hybrids.

Hickories, like other nut and tree species, do not come true from seed, so superior selected clones are propagated by budding and grafting on other trees known as rootstocks. These rootstocks are produced from seed.

Seed Propagation

Investigations by Barton(1) showed that some seedlings were produced when the nuts were planted immediately in a warm greenhouse without pretreatment, but that germination was markedly increased when the nuts were held in a cool moist environment from one to four months before bringing into the greenhouse. She also found that fall planting of hickory nuts resulted in a good stand of seedlings the following spring if the soil was mulched, but that the freezing and thawing of unprotected ground resulted in an exceedingly poor stand of seedlings.

Burkett(6) advocated stratifying pecan seed over winter in moist sand and planting in moist soil in the very early spring. He observed that thin-shelled nuts germinate more quickly than thick-shelled ones, and warned against "damping-off" fungi which often killed young seedlings.

Brison(5) stated that some nurserymen prefer seed of certain pecan varieties as Riverside and Burkett for rootstock purposes as these produce strong vigorous seedlings. He reported that while the pecan seed does not have a rest period, germination is increased by stratifying in moist sand for 2 to 3 weeks or soaking in water, changed daily, for 4 to 5 days previous to planting.

Propagation by Layering

No records are available in regard to any hickory species or variety other than pecan having been propagated by any method of either soil or air layering. The writer(14) while experimenting with aerial layering in 1945 found one instance of root production on a hickory where the branch was girdled at the base of the one-year wood. This method offers possibilities, especially now that polythene plastic is available for retaining moisture in the moss about the girdle or wound on the layered branch.

Gossard(9) reported success in producing roots from the tops of small grafted and budded pecan trees by trench layering and from older trees by aerial layering with marcot boxes. He indicated that a favorable combination of etiolation, moisture, rooting medium, and a root-inducing chemical was desirable for successful rooting.

Propagation by Cuttings

Hardwood cuttings of pecan were rooted by Stoutemyer and O'Rourke(23) in 1938 by first callusing the bases of the cuttings in warm moist peat moss, and then treating with an aqueous solution of indole butyric acid before planting. Both roots and shoots grew well for three to four weeks and then the shoots wilted and died. It was observed that the roots were thickened and presented an abnormal appearance. Trials during succeeding years gave no better results and the experiments were discontinued. Cuttings taken from native hickories during these same years failed to produce roots.

Romberg(17) reported a small measure of success in rooting hardwood stem cuttings to which young seedlings had been grafted by the inarch method. The influence of the seedling on the nourishment of the cutting was gradually diminished by girdling caused by a copper wire which was tied about the seedling stem.

Apparently root cuttings of pecans and other hickories have never been tried. In 1896 Corsa(7) observed that "when the lateral roots of the pecan are broken by the plow, the ends of these roots frequently send up thrifty shoots." Such a response would indicate that adventitious shoots may arise from roots and that root cuttings may be successful.

Propagation by Grafting

A search of the literature failed to reveal a discussion of any method of bench grafting with hickories, although presumably it must have been tried. Propagators may have been discouraged in using bench graft methods by the sparse roots usually found on two-year seedlings. It is suggested that undercutting and root pruning the seedlings several times while in the nursery row should produce a more adequate root system which would transplant well after grafting. Brison(5) remarked that bench grafting is not used in the propagation of pecans in Texas on account of transplanting difficulties.

Commercial nurserymen now prefer to bud hickories and pecans rather than to graft, but formerly Reed(15) reported the whip-and-tongue method was used on thrifty one-year seedlings in the nursery row. It is conceivable that the cleft graft could be used at this stage when the diameter of stock and scion are quite similar but no record of its use is available.

Top-working or grafting in the branches is commonly practiced on seedling trees and sometimes used to change varieties in the orchard. Reed(15), Sitton(19), Rosborough et al(18), MacDaniels(11), and Stoke(22) have described various methods that have proven successful. Practically all agree that the bark graft or a modification thereof is best. Morris(12), Benton(3), MacDaniels(11), Wilkinson(25), and others have shown that a greater per cent of survival is secured when the stocks are cut 10 days to 2 weeks before grafting. During this time the stubs heal somewhat and excess bleeding is decreased. It has been reported by Becker(2) that the success of walnut grafting is greater when the grafts are set just after the leaves are full grown but no such data is available for hickories. The use of paper bags or other shading device over the scion is advocated by Morris(13), MacDaniels(11), Shelton(20) and others.

Propagation by Budding

Patch budding is now almost universally used by commercial nurserymen in the propagation of hickories and pecans. Patches are usually cut with a double-bladed knife although some use the rectangular Jones patch-budding tool. The "plate" or "skin" bud is also used to some extent. The thick bark of hickories and pecans discourages the use of the shield or "T" bud.

Budding is usually done in late summer with mature buds of the season growth which remain dormant until the following spring. Occasionally dormant budwood taken in winter is held in cold storage until the bark of the stock slips in the spring. These spring-set buds are forced the same season by cutting the stocks back shortly after setting. Patch-budding is described by Reed(15) and by Rosborough et al(18). Reed(15) mentioned that it may be advisable to make the cuts in the stock from one to three weeks before the bark is removed so that the healing process may be under way at the time the bud patch is inserted.

Storage and Handling of Scions and Budwood.

Shelton(20) reported an easy and unique method of keeping scions moist by storing in a closed container with a small amount of sodium sulphate (Glauber's salt). Slightly moist peat moss is an excellent packing material. Brison(4) reported that a temperature of 32 deg. F to 38 deg. F in storage is satisfactory for keeping the buds dormant, and that a few days from 80 deg. F to 85 deg. F will stimulate cambial activity so that the patches will "slip" easily when cut. Scionwood is sometimes dipped in wax, paraffin, or plastic resin before storing in order to prevent loss of moisture and guard against pathogenic organisms.

Waxes and Wound Dressings

Sitton(19) used a large number of variously formulated waxes on pecan and found that the most successful from the standpoint of graft survival was one composed of 10 parts rosin, 2 parts beeswax, and 1 part filler such as kieselguhr, talc, or aluminum powder. Under Louisiana conditions a light-colored wax was preferable to dark colored one. Asphalt emulsions were not satisfactory.

Rootstocks and Interstocks for Hickories

Reed(16) summarized the rootstock studies at Beltsville, Maryland, by stating that pecans were best on pecan seedlings and that shagbarks were successful on either shagbark or pecan rootstocks. He reported a lack of congeniality between shagbark and bitternut hickory. Smith(21), however, found that pecan stocks were unsuccessful for shagbarks as few scions lived and growth of those which survived was poor. He also reported that bitternut was practically as good as shagbark for shagbark varieties. He stated that pignut was absolutely useless as a stock for shagbark. Weschcke(24) reported that shagbark varieties grew well on bitternut but also indicated that a slow growing variety would be stimulated in growth by working on pecan stocks which are more vigorous in growth than the other hickories. Dunstan(8) reported that pecan provides a perfectly satisfactory rootstock for shagbark, shagbark hybrid, and hican varieties. A number of varieties have been tested over a period of several years with favorable results as shown by lasting unions and better than average yields.

The Fairbanks hybrid has often been used as an intermediate stock between bitternut and some shagbark varieties and Last(10) has stated that the variety Rockville is useful for interstock purposes on account of its exceptional vigor.

Nursery Problems

Hickories and pecans have long tap-roots with few branches and hence do not transplant well. Some few have grown the seedlings for one year in beds underlain with wire screen netting or have undercut the seedlings to promote branching of the roots. The stocks must grow two years from seed to attain a diameter permitting of patch budding and must remain one or two years more to allow the scion to form a tree. The resulting plant is large in both root and stem and requires careful handling in digging, shipping, and planting in the permanent location. The vicissitudes which befall the production of the northern hickories are often so great as to discourage nurserymen who otherwise would grow them. This is an unfortunate fact but a real one, as the would-be purchaser often learns when he attempts to buy named varieties of hickories. The situation with the pecan is much better, due perhaps to the greater demand for such trees but also to the greater ease of propagation in general nursery practice.

Conclusions

Good varieties of hickories bear good nuts and more people should plant good trees which should be produced by nurseries with well-branched fibrous root systems so that they will transplant easily. Research is needed to determine practical methods of propagation which will permit of inexpensive quantity production of superior named varieties of shagbark and shagbark hybrid hickories.

Literature Cited

1. Barton, Lela V. Seedling production in *Carya ovata*, *Juglans cinerea*, and *Juglans nigra*. Cont. Boyce Thompson Institute 8:1-5.

1936.

2. Becker, Gilbert. Notes from Southwestern Michigan. Rpt. North. Nut Grow. Assoc. 28:135-136. 1937.

3. Benton, Wm. A. Report on propagation of nut trees. Rpt. North. Nut Grow. Assoc. 29:90-92. 1938.

4. Brison, Fred R. The storage and seasoning of pecan budwood. Texas Agric. Expt. Sta. Bul. 478. 1933.

5. ——. Personal correspondence. 1952.

6. Burkett, J. H. The pecan in Texas. Texas Dept. of Agric. Bul. 111. 1932.

7. Corsa, W. P. Nut culture in the United States. U. S. Dept. Agric. Div. of Pomology. 1896.

8. Dunstan, R. T. Personal correspondence. 1952.

9. Gossard, A. C. Rooting pecan stem tissue by layering. Proc. Amer. Soc. Hort. Sci. 38:213-214. 1941.

10. Last, Herman. Personal correspondence. 1952.

11. MacDaniels, L. H. Some experiences in nut tree grafting at Ithaca, N. Y. Rpt. North. Nut Grow. Assoc. 28:52-55. 1937.

12. Morris, R. T. Top working hickories. Rpt. North. Nut Grow. Assoc. 11:105. 1920.

13. —— Nut growing. 1931. Macmillan, New York.

14. O'Rourke, F. L. Unpublished data. 1945.

15. Reed, C. A. Nut-tree propagation. U. S. Dept. Agric. For. Bul. 1501. 1926.

16. ———. Hickory species and stock studies at the Plant Industry Station, Beltsville, Md. Rpt. North. Nut Grow. Assoc. 35:88-121. 1944.

17. Romberg, L. D. Use of nurse seedlings in propagating the pecan from stem cuttings. Proc. Amer. Soc. Hort. Sci. 40:298-300. 1942.

18. Rosborough, J. F., F. R. Brison, C. L. Smith, and L. D. Romberg. Propagation of pecans by budding and grafting. Texas Ext. Ser. Bul. B-166. 1949.

19. Sitton, B. G. Pecan grafting methods and waxes. U. S. Dept. Agric. Circ. 545. 1940.

20. Shelton, E. M. Glauber's salt for humidity control in scion storage. Rpt. North. Nut Grow. Assoc. 28:70. 1937.

21. Smith, Gilbert L. Our experience with root stocks. Rpt. North. Nut Grow. Assoc. 40:62-64. 1949.

22. Stoke, H. F. Grafting methods adapted to nut trees. Rpt. North. Nut Grow. Assoc. 37:99-102. 1946.

23. Stoutemyer, V. T. and F. L. O'Rourke. Unpublished data. 1938-1940.

24. Weschcke, Carl. The importance of stock and scion relationship in hickory and walnut. Rpt. North. Nut Grow. Assoc. 39:190-195. 1948.

25. Wilkinson, J. Ford. Preparation of stocks for propagation. Rpt. North. Nut Grow. Assoc. 28:65-66. 1937.

A Root Disease of Persian Walnut

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On three recent trips to southern Europe I noted large numbers of dying Persian (English) walnuts, *Juglans regia*, in France and Switzerland and scattered trees in other countries. Dying of Persian walnuts from a root disease of undetermined cause has been reported from various European countries for many years. The extensive dying of mature Persian walnut in a number of areas in southern France is very serious. Farmers and orchardists are discouraged from planting the Persian walnut even though it is a very profitable tree when not attacked by the root disease. In area after area I noted that the farmers had scattered their Persian walnut trees, separating them as much as possible or planting them along the boundary of fields instead of in orchard plantings. They had found too frequently that solid plantings of walnut die from the root disease. The total number of Persian walnuts in southern France has decreased alarmingly in the last sixty years. In Tessin Province in Switzerland many unhealthy Persian walnuts were noted this past summer showing the same symptoms as in southern France.

Studies By French and Italian pathologists have indicated that the fungus *Phytophthora cinnamomi* is the most likely cause of this dying of walnuts. I was informed that it is worse on soils inclined to be wet or poorly drained at certain times of the year, conditions favorable for attack of many hosts of this Phytophthora. The work reported by B. S. Crandall and me in Phytopathology, March 1945, showed there was a rather direct relation between soil conditions and *Phytophthora cinnamomi* damage to black and Persian walnut seedlings. Long periods of heavy rainfall were very favorable for an epidemic outbreak of this fungus on walnut and other nursery stock. Another species of *Phytophthora*, *P. cactorum*, has also attacked black walnuts in nurseries in eastern United States; this fungus has also been reported on Persian walnuts.

We are interested in receiving reports of the dying of Persian or black walnuts in orchards or rows of trees in the Eastern half of the United States. Persian walnuts suffer from winter injury in many areas and sometimes this injury is confused with the root disease. However, where there are indications of continuing dying of walnuts year after year with a progression from one part to another of the planting, we would like to receive a report.

Some root diseases are difficult to diagnose, especially when the small roots are the parts affected.

The symptoms of the root disease of the Persian walnut in Europe are in many ways very similar to those of the Phytophthora root disease of chestnut and chinkapin in this country as described in the report by Gravatt and Crandall in the Northern Nut Growers Association Proceedings for 1944. In some cases Persian walnuts die slowly and in others death is rapid, with the entire tree browning in summer. Some trees will show less green color than normal during the summer and gradually die over a year or two. Trees in different stages of dying can be seen in the same planting.

Persian walnuts in the Western States in recent years have been dying from a disease of undetermined cause. Dr. Paul W. Miller and others have reported on the black line graft union failure, *Armillaria mellea* and dying of roots from undetermined causes. As *Phytophthora cinnamomi*, an imported fungus, is a comparative recent invader of many parts of the west, Dr. Miller is giving the fungus some attention as a possible parasite. On some hosts, this fungus attacks primarily the very smallest roots at certain favorable times of the year, which makes determination of its role as a parasite rather difficult.

Factors That Influence Nut Production

W. B. WARD, *Extension Horticulturist, Purdue University, Lafayette, Ind.*

The profitable production of fruit on nut trees under cultivation has no doubt been influenced by several factors. Assuming that the present-day seedlings and propagated varieties are winter hardy and the tree of bearing age, 10 to 15 years old, one may expect a reasonable harvest. It is somewhat disappointing to the owner of a single nut tree or for the grower on a semi or commercial basis to find that the tree or several trees have failed to set fruit.

The commercial fruit grower of apples learned, many years ago, that certain varieties when planted in solid blocks failed to set satisfactory crops. Rather than lose several years of growth and expense, the better growers top-worked the trees with a good pollinizer. The result was a profitable harvest of red and yellow apples, or varieties of different maturity. The peach grower liked the looks of a very fine peach and after a short trial found that the variety was not hardy enough to withstand the winter and early spring temperatures. The experiences of the commercial fruit growers could be well used by the nut grower. Only trees adapted to local conditions should be planted regardless of the recommendations of the nurseryman. Hardiness of wood and bud, ample production of pollen, reasonable climate during the growing season, and the control of insects or diseases determine, for the most part, the success of the harvest.

Soils and Fertility

The nut trees require good drainage and a good supply of moisture. A reasonably fertile soil should be selected for a planting site but through mulches, manures, and commercial mineral fertilizers any soil may be built up to a high state of fertility. A weak tree has little chance and may come into bearing too late to be of value for the present owner. The annual growth should be checked each year and, unless 10 to 12 inches of growth has been made the previous year, some means of stimulating more growth should be employed. The hickory, pecan and black walnut, as a rule, make little new annual growth while the Persian or Carpathian walnut, heartnut and chestnut ordinarily produce good annual growth and an abundance of good foliage grown where it counts the most, on the new wood.

Those who have observed the growth habits of nut trees know that the catkins are found on last year's growth, or two year old wood, and the fruiting flowers at the end of the present season's growth. There are times when the new growth developed in a matter of a few days to sometimes as long as two weeks. During the period of prolongation of the new growth and the formation and receptiveness of the pistillate flower much can happen. The catkins shed pollen when the temperature and atmospheric conditions are normal. Many times the pollen is dispersed before the pistillate flowers are formed.

Cross vs Self-Pollination

There is no assurance that a nut tree which fruits on the average of eight out of ten years will continue to do so in the future. Occasionally trees take on an alternate year bearing habit that could be caused from injury, insect or disease damage, or the relocation of plant food. The nut trees on their own roots should do better than when grafted or budded. The compatability of stock and scion is yet to be worked out and any constriction at the union may alter the fruiting habits.

The possible sources of pollen for hickory and pecan are from self-fertile trees, seedlings, and various natural crosses that may or may not produce edible fruit. The walnut family, which includes the black and Persian or Carpathian walnut; butternut and Asiatic nut (heartnut) have been used as pollinizers on the Persian walnut with some success. The butternut is the first to shed pollen in Indiana with the catkins dropping, in some years, by late April and the first week in May. Some years the black walnut has produced the peak pollen from May 5th to 12th but during the spring of 1953 the Thomas, Stabler, Rowher, Ohio and Stambaugh dropped the pollen from May 18th to 22nd. The Asiatic walnuts were in full bloom on May 14th and 15th. The above dates do not correspond to observations of other years, except for the butternut. The pistillate flowers on the Persian walnuts were fully opened by May 16th. The catkins of the Persian trees had dried by May 12th. Catkins from the Asiatic walnuts were kept fresh and distributed throughout two Persian walnut trees and by mid-afternoon a heavy rain came. On May 18th a few catkins were again removed from the Asiatic walnut and only enough for one Persian walnut tree were found and hung in the tree. The first tree has no fruit while the second tree has a fair crop in the making.

A letter from H. F. Stoke, Chairman of the Survey Committee on the blossoming dates of the Persian walnut said: "Payne, Lancaster and Broadview staminate flowers were out on April 9, 10 and 11. The pistillate flowers of McKinster, Caesar and Crath #1 were receptive on April 11, 10 and 10." The above dates were over a month before spring came to Indiana. Whether or not the Stoke varieties in Virginia would do the same in Indiana or elsewhere is still the problem.

The black walnut varieties mentioned previously set very few fruits at Lafayette this year while a promising new variety, Sol, from Ferd Bolten, Linton, Indiana, has a full crop, and has been a consistent producer for the past several years.

How Many Pounds per Tree

Throughout the Middle West the elm, native chestnut and some of the oaks are dying from disease troubles. The homemaker wants to plant a tree that will provide shade, fit well in the landscaping of the home, be a clean tree and yet be fruitful and bear early.

The age of the tree and the growth has much to do with production. Some pecan varieties have produced several hundred pounds per tree and the same for black walnuts with hickory, butternut and chestnut in smaller quantities. There are four Persian walnut trees growing in Franklin, Indiana, that are 20 years old and have fruited continuously for the past 10 years. The trees were seedlings, two of which are very promising for distribution. Tree #1 produces an average of 10 pounds; tree #2, 15 pounds; #3, about 40 pounds and #4, 100 pounds. Good pollination under common growing conditions of the Midwest and a good variety acclimated for general planting will no doubt make a host of good friends and a wonderful contribution from the members of the N.N.G.A.

Rootstocks for the Walnut in France[2]

J. C. MCDANIEL

[2]This is a translation, by Dr. R. T. Dunstan, of the section on "Rootstocks" in Chapter XI of *Les Noyers*, by two Doctors of Pharmacy, P. Peyre and E. Lancosme. This 447 page book with 140 figures was published in 1942 by Jouve et Cie, 15, rue Racine, Paris, and is a very complete treatise on the subject of walnuts.

The French experience with the eastern black walnut and the related Arizona walnut as rootstocks is interesting, as is the discussion of one method of propagation, where dormant whole-root grafts are started in pots under glass. This differs somewhat from the indoor grafting procedures described in our recent Reports by Mr. Stephen Bernath and Dr. Philip Brierley. (Incidentally, Dr. Brierley tells me that he got uniformly good grafts—96 to 100% growing—in his 1953 experiment. The use of growth substance powder did not significantly increase the "take". The controlling factors seem to be the use of healthy scions and rootstocks, followed by high enough temperature and humidity to promote rapid callusing of the grafts.)

The "old Royal Walnut" of the French is, of course, what we call Persian (or English) walnut, and not Luther Burbank's "Royal Hybrid", the unfortunately named cross of two black walnuts, *J. nigra* x *J. hindsii*. *J. torreyi* is a synonym for *J. major*, the Arizona walnut.

Rootstocks fulfilling two essential conditions should be chosen, those capable of adapting themselves to soil and climate where they are to be planted and of resisting diseases that may attack them under unsanitary conditions or under too intense cultivation. Among the numerous varieties tested, two deserve attention as choice rootstocks, one native, the other American = *J. regia* and *J. nigra*.

J. regia, our old Royal Walnut, so common in France, is excellent when planted in new, light and fertile soils, preferably clay-lime or clay-silicon.

But as the roots are very spreading it is important to stir the soil well but slightly and avoid deep plowing, for it is well known that through accidental injury to the roots the various "armillaria" enter the trees to develop the "pourridie" or "pus disease", or "circle disease". It is better, then, to use a rootstock immune to this malady so wide-spread among our native walnuts.

J. nigra enjoys this happy advantage of offering no foothold to this parasite, so harmful to its sister species. It accommodates itself well in many soils in which *J. regia* will grow, even dry and gravelly, but prefers soils which are fresh, open, rich, and especially, deep. Its roots are long and vertical and their development stops in contact with an impermeable layer of soil.

It produces specimens magnificent in height and rapidity of growth. Color of bark differs, though diameter of tree is more or less the same. This slight objection may be easily avoided by grafting regia on nigra at ground-level when wood is well matured and in mild weather.

Proof that this species of walnut is resistant to "pourridie" was given in a report to members of the Congress of Grenoble in 1936 by Mr. Bourne of Saint Marcelin. "At Blache de Vinay, we are told, some black walnuts, planted more than thirty years ago in an infested field, have shown full resistance. One tree, grafted at ground-level and planted too deep, was infected many years ago by the "pus" above the graft on the *J. regia* part. The diseased part was treated as was the custom then, with sulphuric acid, etc. The wound healed and the rootstock remained absolutely clean. A photo by Mr. Roy, Director of Agricultural Services at Isere, establishes this absolute proof.

Other varieties of walnut have been tested as rootstocks—*cinerea*, *cordiformis*, and *Siebodiana*, but only the first seems to have given any satisfactory results.

Reporter Bourne concludes, "The primary purpose of our research on rootstocks will be to obtain a hybrid of *regia* x *nigra* that will combine the resistance of nigra to the "pourridie" and regia's habit of vegetating late in spring.

By virtue of the ability of the female element to transmit its rusticity and vegetative form it seems, *a priori*, that we shall get a good rootstock by crossing *nigra* as mother by Franquette (sic) and then if need be, by backcrossing to Franquette in the second generation.

There exists a 4th type of walnut graft, dating from 1880, which if done intelligently, permits the rapid multiplication of the walnut—the root graft.

In a short but very interesting report to the Nut Congress of Grenoble in October, 1936 by Mr. Leon Treyves, and very kindly sent on to us, the author says, "This procedure, devised by my family around 1880, consists of grafting on one year old roots, branches from selected, vigorous trees, either by cleft or English grafts, whichever gives best fit of scion (which is generally smaller than root) and stock. Graft is then tied with raffia and waxed to avoid all contact with air and placed in a moderately heated frame. After a month of this treatment the graft has taken. Then it is gradually accustomed to open air and the frame is removed. In the fall or the following spring the graft may be planted in its permanent location or in nursery row.

This system presents numerous advantages:

1. Rapidity, since the plants can be grafted after one year, instead of three or four.

2. Economy of time and expense, since considerable numbers of grafts can be made rapidly and in limited space.

3. More rapid development of growth and fruiting. Saplings of 1 to 2 meters planted in winter of '28-'29 measured in October '36 25-27 cm. in circumference at one meter from ground. Trees two and three years old, still in nursery, are bearing one to two normally developed fruits.

The author indicates that he uses *nigra* for stocks, "since that is the only one that has proved its adaptation to grafting and its resistance to the "pus disease"."

At the time he gave his paper in '36 Mr. Treyves announced that he was continuing his grafting experiments on *J. Sieboldiana*, *cordiformis* and *torreyi. [J. torreyi* = *J. major*—J. C. McDaniel.]

Mr. Treyves, whom we cannot thank too much for his favor, was kind enough to set forth the preliminary techniques of his method of root-grafting. We give a resume of them here.

1. Preparation. Plant nuts well-spaced in rows in good soil, convenient to irrigation, if needed. Clean nuts of good quality, previously stratified, should be planted in winter. Plants are lifted before the following spring and heeled in. For scions wood of 7-8 cm. is cut from young, healthy and vigorous trees and passed to the grafter at the same time as the roots, which have been previously lifted, washed and cut off at the crown or a little

below. Scion, bevelled, is set either in English or cleft graft, tied with raffia or with a numbered wool strip, waxed and potted in rich but light soil, moderately firmed around roots. Pots are then set in some homogeneous material (waste tan-bark or sawdust) and left in a moderately heated bed.

2. Care. Watering. Temperature of beds should be kept constant around grafts and they should be watered every other day. Of course, grass and mold should be prevented.

As soon as grafts begin to grow (usually around 15 days) the pots are gradually removed from sawdust, and when plants have made 15-20 cm. of growth (after 30-40 days) they are slowly hardened to air and sun, replanted in well-shaded beds, properly watered and cared for until they are set in nursery row.

3. Planting in nursery. The following spring they are set 60 cm. apart in nursery rows 1 m. apart in well-manured and well-prepared plots. Usual care during growth. With the 2nd year plants attain 1-1/4-1-1/2 m. and it is not uncommon to discover a nut. The 3rd year they make 2-1/2 m. at least with 8-12 cm. of girth and are ready for transplanting to permanent site.

4. Soils and situations. Mr. Treyves tells us that the walnut plantings in "lower Gresivuaudan" are on old alluvium of the Isere Valley and in limy marl soils of the upper slopes. A little farther away in Savoy, the walnut is vigorous in Jurassic or clay limestone soils. The same is true in Dordogne, in Correze, and in the Lot, where soils are of similar origin.

Walnuts are found at an average altitude of 600 m. but grow up to 1200 m. in Savoy, and particularly in Switzerland.

The best exposures are SW, W, and SE, sunny slopes, well protected from the north wind and late frosts.

Mr. Treyves has personally some plantings of walnut in Sologne, (where calcareous soils are lacking) and in Champagne, where the soils lack lime. He has noted that these trees grow and fruit normally.

Cultivation. It is important to keep soil around isolated trees well stirred and to increase the area of cultivation as the rootspread increases.

5. Rootstocks. For the present the plants chosen for stocks have come from nigra, the only one that has proved itself in the matter of "take". It does well in moist soils.

Mr. Treyves has personally tried to graft Mayette and Franquette on *Torreyi*. He has found the "take" and the union perfect. But even though vegetation is promising we must wait 22 years for a full test.

He proposes to lest all the "rootstocks placed at his disposal in order to acclimatize the good French varieties to all the soils which suit these stocks. Grafting on *J. Torreyi* will be useful to a [*sera utile a un*] stock that grows in dry soil, like *nigra*.

"But that is a matter we shall have to examine again in ten years, first as regards vegetation and then in 20 years as concerns fruit production."

Since these experiments date from '36 at the time of the Grenoble Congress we have only 13 years to wait to learn what sort of fruit these trees will bear and only 5 to see how they behave vegetatively.

It remains, then, only to wish "good luck" to our kind and devoted correspondent and to thank him for his valuable documentation.

Pictorial Record of Grafting at Climax, Michigan

W. M. BECKERT, *Jackson, Mich.*

Top-working black walnuts to Persian Walnuts has long been practiced by various members of this organization. It is hoped by this series of Kodachrome slides that a record of such top-working by one of our members would be of interest and also show the details of just how the work is done under actual field conditions.

Mr. Gilbert Becker, of Climax, Michigan, has been quite successful in top-working black walnuts. Needless to say, these pictures were taken to show how an expert goes about grafting black walnuts. Mr. Becker was contacted as to when he would do his grafting and he mentioned that on May 80, 1953, he would be top-working his stock. Plans were made to be present on that date and we were fortunate in having bright weather for taking the pictures.

The first two slides show Mr. Becker removing the scionwood from his storage pit, selecting the scions and preparing to go out to do the grafting. On the way to the trees that were to be grafted, the pictures for the next two slides were taken to show the stage of leaf development and the length of the catkins of the Thomas Black Walnut, so members in other sections of the country can see how far out in leaf the Thomas variety was when the grafting was done.

The following series of slides show how Mr. Becker top-works the black walnuts.

First, removing that portion of the stock, note he used a pruning saw, makes a cut at the point where he wants to graft. He uses the bark slit method. The scion is shaped by one stroke of the grafting knife; a long slanting cut is made and the scion inserted in the stock. Just prior to placing the scion, the bark of the stock is slit, two cuts with the point of the knife, approximate width of the scion and down along the bark to the length the scion is to be inserted, then the scion is placed. The next step is to cut off the little sliver of bark which is pushed out, at the point where it does not contact the scion. In this tree, two scions were placed, the scions being wrapped tightly with waxed muslin which was prepared beforehand, using strips about one-half inch wide. Enough was used to firmly bind the scions to the stock. Please note that a small piece of wax muslin was placed on the inside of the scion to prevent the wax from going down between the scion and stock. The final step is waxing the scion and brushing with hot wax, in order to prevent the scion from drying out; to provide shade, Mr. Becker, in this case, used grass and made a hood over the scion, tying it with string.

The following slides show the same procedure of grafting other trees. You will note in one case he has climbed up into the branches of the tree. To shade the scions, he used aluminum foil, folded around the scion and tied with a rubber grafting strip. In all these pictures the scions used were Colby.

Mr. Becker is very adept, quick, and does the grafting so that it actually seems effortless. His technique is so fast, there is very little chance of the scion drying out before it is placed.

On July 26th, I returned to Climax and the grafts were successful, as you can see by the following series. The one failure was the first tree that was grafted, and which had the grass for shading of the scion.

Rock Phosphate for Nut Trees

HARRY P. BURGART, *Union City, Mich.*

My soil is of the sandy type and I have to watch the mineral content rather closely for nitrogen and phosphate deficiencies. Winter-killing of one year black walnut and pecan seedlings is serious during seasons when our winters are less severe than usual and during winters when we had had plenty of snow cover for protection. This worried me a great deal and I decided there must be a deficiency. Soil tests repeatedly showed a lack of phosphate.

I applied ground rock phosphate to my larger bearing English walnut trees and there has not been the least sign of winter injury since.

Many of my smaller nut trees have been bearing earlier for me since I have been using the phosphate. Customers who come here often remark at the way some of my little grafted trees are bearing crops and I tell them that I believe in keeping plenty of phosphate in the soil for root growth and nut production.

I am writing this brief article thinking that it might help solve the problems of other nut growers who have repeatedly been having trouble with winter-killing of their Carpathian, or English walnut trees. Phosphate seems to prevent a late sappy-condition from causing winter injury.

I prefer to apply the phosphate and nitrogen early in April or early May. Fall applications of any kind of fertilizer are apt to cause winter injury. I usually scatter the rock phosphate around the trees using about four handfuls around a first year tree. Then I turn over the sod bottom with a shovel, which puts the phosphate down where the roots can get it. I use the phosphate around all the young trees we set out and seldom lose a tree as the phosphate encourages the starting of new feeder roots on the nut trees.

A Report From Southern Minnesota

R. E. HODGSON, *University of Minnesota, Southern School and Experiment Station, Waseca, Minn.*

We have 20 odd Carpathian walnut trees growing from nuts planted about 1931. So far, I have never seen a flower on any of them. They grow up 6 or 8 feet in a year and that seems to be their difficulty. They do not stop growing in time to harden off before cold weather comes. I think a lot of the winter killing is also due to sun scald which would indicate an inability to retain dormancy during a January thaw. Some of the trees have lived through two winters with only minor damage and then when the right conditions come along, they are killed to the ground. Wrapping the trunks with aluminum foil has not solved the problem. I have purchased one or two grafted trees which were recommended as more hardy but so far they have had the same experience as the one I grew from nuts.

Black walnut and hickory do well here and I have a hiccan perhaps 20 feet tall but it has never borne any nuts. Chinese chestnuts are not entirely hardy and grow very slowly. This year I set out about 20 American chestnuts from Minnesota grown seed and I hope that we are far enough from other trees of this variety to escape the blight. Tree growing is just a hobby and lately there has been very little time for hobbies.

Chestnut Breeding

Report for 1953

ARTHUR HARMOUNT GRAVES and HANS NIENSTAEDT, *The Connecticut*
Agricultural Experiment Station, New Haven, Connecticut

The chief aim of this breeding work, which has been carried on now without interruption since 1930, is to develop a tall timber type of chestnut by breeding the American species with the blight resistant but comparatively low-growing Japanese and Chinese chestnuts, *Castanea crenata* and *C. mollissima*, respectively. Practically all trees of our valuable American chestnut of any appreciable size have now been killed to the ground by the blight fungus, *Endothia parasitica*. Shoots arising from the base of the old stumps often live long enough to bear pollen, and this we have lately been forced to use in our breeding work with the disadvantage that we can not know definitely the nature of the genotype of the pollen parent. American pollen from a good phenotype near Washington, D. C., was kindly furnished us in the early 30's by the then Office of Forest Pathology of the U.S.D.A., and this stock is now incorporated with our older Japanese-American and Chinese-American hybrids.

As indicated in the following pages, we are not neglecting the nut-bearing potentialities of the chestnut tree.

Weather Conditions in 1953

The disastrous ice storm of the 9th and 10th of January caused slight damage to some of the Chinese trees. Their numerous, more or less horizontal branches and characteristically brittle wood make them prone to damage of this sort; nevertheless, only a few branches were lost. After a comparatively warm February, the warmest since 1925, March brought us more rain than for any March in the 81 years records have been kept[3]—a total of 10.78 inches. This was all to the good, as later events proved. Because of the preceding warm February the ground was for the most part unfrozen, so that, instead of running off, the water was largely absorbed in the soil, and thus added to the water table. The precipitation of April was again heavy—5.6 inches—the normal per month for this area being about 3-1/2 inches. After an unusually good growing season in May, June and July, about the middle of August a long drought of nearly 10 weeks duration commenced. The conditions were similar to those in 1952, except that in that year the drought began later, in October. However, the large amount of water in the soil from the spring rains prevented serious consequences, just as in 1952 the heavy rainfall in August and the normal one in September mitigated any serious results from the later drought of that year.

[3] Weather records are taken from the monthly reports of the municipal airport at New Haven, Conn., and are compared with the New York City records for the same period, kept by the U. S. Weather Bureau at 17 Battery Place, New York City.

Hybrids of 1953

As in former years we continued the formation of hybrids of the combination CxJxA[4] which has to date given the most resistant individuals and the best timber form. 277 hybrid nuts of this combination were obtained by crossing JA with C, and C with JA. JAxJ crossed with C yielded 25 nuts. CJA crossed with pollen from the Roxbury Americans gave 20 nuts. The Chinese-American hybrids are also promising both in form and in blight resistance. By crossing these with American pollen from Thomaston, Conn., and from Clinton Corners, N. Y., we secured 48 nuts. CA crossed with a good native American in Thomaston, Conn., resulted in 30 nuts, and the same combination using an American in Newfoundland, N.

J., produced 9 nuts. The total number of nuts derived from all crosses was 504, a much smaller figure than that for the two preceding years. The reason for this is that considerable time was consumed in experiments designed to determine the length of the receptive period in the pistils.

[4] C = Chinese, J = Japanese, A = American chestnut.

The 1953 nut production (Table I) compared well with last year's. The total yields were as good as, or better than, the 1952 crop and the average weight per nut was not significantly lower than in 1952. Apparently the late summer and fall drought had no effect on nut yields and average weights. Obviously we have the warm winter and abundant early rains to thank for this situation.

Table I. Natural Yield of Nuts (open pollinations) From Sample Trees.

Age in years	Species	Location	Total yield in lbs. 1952	Total yield in lbs. 1953	Av. weight per nut in grams 1952	Av. weight per nut in grams 1953	Approximate no. of nuts per lb.[A]
23	*C.crenta*	13-2	27.5	43.3	14.5	14.0	32
27	*C.mollissima*	1-3	22.2	20.8	10.6	10.5	43
27	*C.mollissima*	1-9	28.2	26.2	9.9	9.7	46
27	*C.mollissima*	1-15	6.8[B]	20.6	12.9	11.7	39

A. based on the 1953 weights

B. a considerable part of crop lost before it was collected

Fig 1. gives a fair idea of the extremes in size of the Japanese chestnut. Since the smaller size is probably close to that of the wild chestnut in Japan, the figure illustrates what has been done by the centuries of selection and cultivation that the chestnut has undergone in Japan.

[Illustration: Fig 1. Nuts of *C. crenata*, Japanese chestnut, showing approximately the limits of size in the species. Left: from a tree on Long Island, N. Y, owned by Mr. John Vertichio. Right: from one of our forest type Japanese trees given to us by the Office of Forest Pathology in 1930 and now growing at the Sleeping Giant Plantation, Hamden, Conn. The tree is probably representative of the wild type of nuts in Japan—a little larger than the native American chestnut. However, it is probable that smaller nuts of the Japanese species exist. About 1/2 natural size. Photo by B. W. McFarland, Conn. Agric. Expt. Sta., Nov. 27, 1953.]

Anent the large nuts in the photograph, which weigh about an ounce apiece or about 28 g. (compare figures in table I), Mr. Ferguson, Instructor at the Long Island Agricultural and Technical Institute, through whom we received the nuts, states that "the nuts of the seedlings from the tree do not average better than half the size of those of the parent tree." This illustrates the fact, now well known, that the chestnut tree is self sterile. Nuts are always (with exceptions) a result of fertilization of the flowers with the pollen from *another* tree.

We should like to reproduce this tree in our plantations, but the only way it can be done is by grafting scions of it on to some other, preferably Japanese, stock, or by rooting cuttings from it—a method which we still have not been able to accomplish readily.

Moll-Seg, or Chinese Prolific

In the report of the senior writer for 1934 (*Brooklyn Botanic Garden Record 24:62*) it was stated: "In our form of the Chinese chestnut only one, if any, bur appears at the base of the flowering branch. The dwarf species, *C. seguini*, from eastern and central China, on the other hand, is most prolific; and in addition, blooms from June to October. It will be seen that crosses of these two species may produce valuable breeding stock." As a result of this cross, made in 1934, we obtained seven nuts, and from these nuts we have today, nineteen years later, four trees, three of which have shown marked blight resistance. One of these three is much larger, partaking more in its stature and form of the character of its Chinese parent, and in contrast to the latter, bears burs and nuts in profusion, usually clustered at the ends of the branches. (Fig. 2). The nuts are small but of good flavor. It is a good nut tree, not suitable for timber. However, as we stated in our 1951-2 report, it is subject to considerable twig blight, caused by the attacks of the weak parasite, *Cryptodiaporthe castanea* (Tul.) Wehmeyer, and this is due apparently to the influence of its tender parent, the Seguin chestnut, which habitually dies back in the winter. The parasite easily enters the dying ends of the twigs. We should like to see this tree tried out in a warmer climate—Georgia, Florida, Alabama, etc. Possibly it might prove adaptable to a southern European environment.[5]

[5] Systematic descriptions of this and other valuable chestnut hybrids are being prepared for publication.

Pollen Receptivity of Female Flowers

Chestnut is monecious. The flowers are borne on the present year's growth in long catkins. These are of two distinct types; near the base of the flowering branch they consist of male flowers only. The catkins near the apex, on the other hand, are bisexual; pistillate flowers are found, solitary or in clusters of two or three, near the base of this type of catkin. The remainder of the catkin bears male flowers similar to those on the all-male catkins.

The unisexual male catkins are the first to start flowering and not until two to three weeks later will the male flowers of the bisexual catkins be in full bloom. Normally, the pistillate flowers will reach full development sometime between these two periods of pollen shedding.

The Length of the Receptivity Period.—During the summer of 1953 an experiment was conducted to determine more definitely when the pistils became receptive and how long they remain in this condition. Two Chinese, two Japanese and two hybrid chestnuts of the combination (Jap. x C. pumila) x Jap., the so-called S8xJ, were used as the females in the study. Emasculation and bagging was done at the beginning of anthesis, that is, when the first unisexual male catkins began to shed pollen. Three different pollen sources were used on each female parent; they were of the same species or hybrid combination as the female. The following diagram shows the pollination schedule used.

Pollen source Time of pollination in days No. after beginning of anthesis 5 9 13 17 22
————————————————————————————- 1 X X X 2
X X X 3 X X X

For example, the Chinese female trees were pollinated with pollen from three other Chinese trees (in the diagram "Pollen source No. 1, 2 and 3), which open their anthers on successively later dates. This pollination schedule was used to avoid prolonged storing of the pollen.

[Illustration: Fig. 2. Showing, above, *C. mollissima*, Chinese chestnut, left, and *C. seguini*, seguin chestnut, right, parents of *mollissima* x *seguini* hybrid below. Note clustered burs in hybrid—more than twice the number appearing in the *mollissima* parent. Leaves and habit of tree resemble more the *mollissima* parent. About 1/6 natural size. Photo by B. W. McFarland, Conn. Agric. Expt. Sta., Sept. 25, 1953.]

To carry on pollination the bags are removed and the pollen-bearing catkins are brushed lightly over the stigmas several times, one or more fresh catkins being used in each bag. These catkins are left in the bag. The bags are then replaced and permanently removed when danger of outcrossing is eliminated, in this case 10-14 days after the last pollination.

The number of nuts collected at the time of harvesting compared with the number of female flowers pollinated was taken as a measure of how successful the pollinations were.

The results showed that five days after the commencement of anthesis a high proportion of the female flowers is receptive. The Japanese and hybrid trees have a definite peak of the period of receptivity between 9 and 17 days after anthesis begins; thereafter, receptivity drops off sharply. The data from the Chinese trees indicated that the period of maximum receptivity is longer than in the Japanese and hybrid chestnuts tested. They maintained full receptivity on the 22nd day after the beginning of anthesis.

It is commonly believed that bagging as well as emasculation may seriously affect the yield from controlled pollination. This is not always the case. One of the Japanese trees and one hybrid tree (S8 x J) yielded fully as many nuts from controlled (under best conditions) as from open pollination. On all other trees the effect of bagging was more or less adverse.

The Effect of Emasculation on Nut Yield

Emasculation involves the removal of the unisexual male catkins and the male part of the bisexual catkins. In the course of the controlled pollination work it has often been found that the female flowers drop off in the bag before the burs start to develop. This has especially been encountered in Japanese x American hybrids and back-crosses. It was thought that this perhaps was due to injuries resulting from emasculation. The following small experiment was carried out in order to determine if this was actually true.

A Japanese x American and a Japanese x (Japanese x American) hybrid were used as the female parents. On these trees some flowering branches were bagged which had been emasculated normally, on other branches only the unisexual catkins were removed, while the bisexual catkins were left intact. Some branches were bagged without any emasculation, and some flowering branches were just tagged. The number of female flowers was counted in all cases. Pollinations were performed 3 times, that is, were repeated on the third and fifth day after the first pollination. This is the procedure ordinarily used for our controlled pollinations. Chinese pollen was used on both trees. Nut set expressed as per cent of the number of pollinated flowers, times three, (because ordinarily there are 3 nuts in every bur) was taken as a measure of how successful the pollinations had been. The results are shown in Table 2.

Table II. The Effect of Emasculation on Nut Yield

Type of Treatment	No. of [Symbol: female]'s at time of first pollination		Nut set as expressed in % of [Symbol: female]'s at time of 1st pollination x3	
	[Symbol: female] parent		[Symbol: female] parent	
	4-4 (JxA)	7-4 (JxJA)	4-4 (JxA)	7-4 (JxJA)
Normal emasculation	39	17	29.1	13.7
Only unisexual catkins removed	23	19	14.5	0.0
Not emasculated, but pollinated	28	18	25.0	3.7
Not emasculated, not pollinated } Control	28	25	1.2	0.0
Not bagged, branches tagged, open pollinated	26	23	44.9	17.4

Bagging apparently was partly injurious on these two trees and caused some decrease in nut yield. There is, however, no evidence that emasculation in itself causes a decreased nut yield, rather it appears to be somewhat beneficial if we are to judge from the results of this experiment. At least, one would be justified in concluding that any harmful effect is negligible. Completely emasculated flowers yielded 29.1 and 13.7 per cent as compared to 14.5 and 0.0 per cent where only unisexual male catkins were removed, and 25.0 and 3.7 per cent where no emasculation was done. The controls which were not emasculated and not hand pollinated, show that the trees are practically completely self sterile; only an occasional nut is set from self-fertilization.

Vegetative Propagation

In vegetative propagation the tree breeder has a very important tool. For instance, if a number of desirable phenotypes have been selected in the forest, they can be propagated vegetatively and planted under uniform conditions where it will be possible to "estimate" their genotype, without the time-consuming progeny testing. Trees propagated vegetatively from old mature trees usually will start flowering very soon after they become established; thus the necessity of doing pollination work in very large trees can be eliminated. Furthermore, it enables the tree breeders to maintain trees of a desirable genotype unaltered for an indefinite length of time without first establishing pure lines through inbreeding. Finally, it may be possible to make valuable individuals available to the forest owners for field planting if they can be propagated vegetatively in large enough numbers at low cost.

Compared to propagation by grafting, the rooting of cuttings is both simpler and cheaper, if it can be done. Chestnut cuttings are, unfortunately, very difficult to root. In the past six years numerous experiments have been conducted in order to find a way to root the various chestnut species. We have tried to root dormant, as well as greenwood, cuttings, the conventional twig cuttings as well as leaf-bud cuttings; numerous hormone treatments using several different hormones in solution and as powders, over a wide range of concentrations, have been tried; a special chamber in which an automatic atomizer nozzle sprays the cuttings intermittently has been used.

Results have always been poor. Dormant cuttings have broken dormancy, sent out new leaves, formed an abundance of callus on the basal end, but failed to develop any roots, and finally after several months have died. Greenwood cuttings also have failed to develop roots in almost all cases. The best results have been obtained with leaf-bud cuttings. In some cases 10 to 20 per cent have rooted; here, however, the difficulty has been the failure of the bud to break dormancy and start growth, and all the rooted cuttings have eventually died.

The rooting by airlayering has been tried in a few cases. Airlayering is the rooting of twigs while they are still attached to the tree. Some distance from the terminal end of the twig an oblique cut is made, or the bark is removed around the twig for about 3/4". The cut or ringed area is treated with a hormone powder, wrapped in sphagnum moss and covered with a wrapping of polyethelene. Attempts to root twigs on older trees by this method have so far failed. Recently successful rooting of twigs on young seedlings by airlayering has been reported from Spain, and from France comes the report that stooling of young seedlings is highly successful. In the stooling method the young plants are cut off a short distance above the ground level. As new shoots grow out, their basal ends are gradually covered with soil until a 5-6" mound has been formed. Left in this manner they may develop their own root system and can eventually be detached from the mother root. That the rooting of young seedling material should be possible, while that from older trees will not root, is not unusual. It is generally accepted that the younger the tree from which the cuttings are taken, the easier the cuttings are to root. Experiments along these lines are planned for 1954.

Cooperative Hybrid Chestnut Plantations

In last year's report we described our hybrid test plots established in cooperation with the U.S.D.A., Bureau of Plant Industry, Division of Forest Pathology.

On March 18, 1953 a new plantation was started at Grafton, West Virginia. As usual, we furnished 50 plants for this test plot.

From the reports from Dr. J. D. Diller, of the then Division of Forest Pathology, our hybrids so far have shown a promising performance, although their average growth rate so far is slightly slower than that of the U.S.D.A. hybrids and straight Chinese chestnut. From the standpoint of blight resistance and growth habit they are at least equal to the two other sources and may be slightly better; however, it is still too early to make any definite evaluation of the results.

Insect Injuries

The most damaging insect pests in the Sleeping Giant Plantation are the spring canker worms, the mites (*Paratetranychus bicolor*), Japanese beetles and the chestnut weevils.

A spraying schedule has now been worked out which keeps these pests under control:

1. Dormant spray with "Scalecide" in middle April against the canker worms and mites.

2. Two applications of "Aramite" (6-7 lbs. per acre) in the middle and toward the end of June. These sprayings have given good control against mites. Where the outbreak is very severe it may be necessary to spray with "Aramite" also in July and again in August.

3. Spraying with DDT in middle August and the first week in September has controlled the weevils successfully.

In concluding this report we desire to express our thanks to the many interested persons who have contributed pollen, nuts and/or scions to further this project.

Dr. W. C. Deming

On November 17th, 1910, twelve dreamers met in the Botanical Museum, Bronx Park, New York City, to form an organization of nut growers in the north. It was largely an organizational meeting. No papers were read, but some solid foundations were laid. Dr. W. C. Deming served as temporary chairman of the meeting and, fortunately for the cause, was then elected as the new body's Secretary-Treasurer, an office which has always called for executive ability and untiring industry.

This election paid off. At the second meeting, held at the New York State College of Agriculture, in Ithaca, it appeared that the new Secretary had communicated with a large number of leading nurserymen, with national and State horticulturists and with others. It was reported at this meeting that only two nurserymen had accepted the invitation to attend. "So", reported Secretary Deming, "evidently the others do not think the northern nut grower is one whose acquaintance is worth cultivating. We hope to convince them to the contrary."

This was done. At the second meeting, the Association could count sixty members. Professor John Craig, of Cornell, in noting this growth, said, "Dr. Deming has not merely performed the routine duties of the secretary, but he has studied the case and has presented a good many facts not apparent on the surface. It seems to me that this augurs well."

The augury proved prophetic. The Association continued to grow. But without this first intelligent, persistent effort upon the part of. Dr. Deming, it could hardly have survived.

This small bit of history is illustrative of the whole life of Dr. Deming. His deep interest in the purposes and hopes of our Association has never ceased. Upon his own ground he planted, and budded and grafted many nut trees, and has given away the fruits of his labors with a prodigal good will. Deming's Burnham pecan and the Deming Purple black walnut are the only introductions, so far as this writer knows, which bear his name.

Again, some thirty years after the first meeting mentioned above, Dr. Deming thought up and carried through another project which makes the Association repeatedly his debtor, an Index of the first thirty volumes of the Association's Annual Reports. It is a work which saves the conscientious worker in northern nut culture hours and hours of labor.

And now our Dean, the last of the founding fathers, has left us for the Elysian Fields. His gentle, kindly face will be sadly missed by those who knew him, but he lives on in every tree whose planting his labors inspired and in every mind which has been, even unconsciously, his heir.

A letter from Miss Charlotte Deming, a sister, assures us, somewhat touchingly, but happily, of this fact:

"My brother's heart was with and in the work of the Association. He was happy to know of its expansion into such a wide-spread organization, and very proud of having been made its Dean."

Dr. Deming lived a full life. He was a physician of distinction, a graduate of Columbia University's College of Physicians and Surgeons and was retired from the army after World War I with the rank of Major. After graduation from Columbia, he served his internship in a New York hospital, then on the medical staff of the State Immigrant Hospital, Ward's Island. He began private practice in Westchester County, New York and, later, for many years, served as examining physician with the Veterans' Administration in Hartford, Connecticut.

It is interesting to know, as told by his son, Hawthorne, that Mrs. Deming, formerly Imogene Hawthorne, was the youngest granddaughter of the immortal Nathaniel. It is evident that Dr. Deming, both in private life and in his public interests, was a strong believer in the value of good blood-lines.

John Davidson

The Nomenclature of Nut Varieties

GEORGE H. M. LAWRENCE, *Bailey Hortorium, Ithaca, N. Y.*

This article is intended to introduce to you the International Code of Nomenclature for Cultivated Plants[6] and to point out the ways in which that Code serves the interests and needs of members of the Northern Nut Growers Association.

[6] The International Code of Nomenclature for Cultivated Plants, formulated and adopted by the International Botanical Congress Committee for the Nomenclature of Cultivated Plants and the International Committee on Horticultural Nomenclature and Registration at the Thirteenth International Horticultural Congress, London, September 1952.

Copies of the full text of the Code are available from the Secretary, American Horticultural Council, Inc., Bailey Hortorium, Cornell University, Ithaca, N. Y., (25c postpaid).

The Code as published by the Royal Horticultural Society is a booklet of about 30 pages, containing an excellent historical introduction by W. T. Stearn, a summary or abridged version of the Code, and the full text. It is of necessity somewhat technical in its phraseology, and in places its jargon is overwhelming. Recently, Dr. John S. L. Gilmour, Director of the Cambridge Botanic Gardens, and formerly Director of the R. H. S. Trial Gardens at Wisley, published a very lucid and down-to-earth interpretation of the principle provisions of the Code. It is reproduced with permission at the conclusion of this introduction.

The questions asked about the Code include,

"What's it got that earlier codes did not have?"
"What's new about it?"
"How does it affect me?"

There are several answers to the first query, but the most significant is that here for the first time we have a Code that represents the thinking of horticulturists from all leading horticultural centers of the world. I was a member of a committee of thirteen (representing 6 countries), that met for nine days in Stockholm in 1950 to prepare and edit the first international draft of this Code. Those of each nationality had met in their country previously, with their own leaders, and had come to this round-table session with fixed ideas of what they wanted. By mid-evening of the first session it became apparent that the Swedes, the Dutch, the British, and the Americans had sent some of their most persuasive, vocative, and determined countrymen to represent them. The Swiss representative restrained himself admirably until after the initial lines had been drawn. It looked then as if there might be several codes, but before recessing several hours later some concessions had been made, and discussion on the more volatile points had been deferred. The differences of opinion were well founded and held with good reason. Some reflected an unawareness of situations in an unrelated horticultural field, e.g., a nurseryman did not know the problems encountered by the Danes in developing so-called varieties of vegetables, or by the American in producing hybrid-corn—each calling for different provisions in the Code, nor could the rose specialist be expected to comprehend the genetic situations encountered in many types of hybridity. One botanist in the group had no appreciation of the intricacies of problems and situations found when trying to name some complex groups of cultigens. Add to these reasons the fact that most of these men were representatives or spokesmen for larger groups or national organizations "back home" and were not authorized to act independently from earlier decisions by those groups, and one can only marvel that at the end of the 9-day period we came up with a detailed and workable draft accepted unanimously, and which was modified in no major respect at the more recently International Horticultural Congress in London.

The period between the Stockholm meeting and the London Congress was utilized to distribute mimeographed copies of the Stockholm draft to horticultural leaders in all countries, to provide opportunity for suggesting changes and new provisions for the Code, and to hold one committee meeting of international level, at London in November 1951. As a result, the present Code is the first truly international regulating guide that has been produced as an aid to persons concerned with the nomenclature of cultivated plants. The individual who may object to some provision of the Code must remember that its presence in the Code has had the support of scores of specialists, representing no less than 16 nations and that there is no part of the Code that was not acceptable to a majority of the delegates of each nation and to the groups representing them. The significance of this international character of the Code is not to be discounted.

It is only proper at this point to tell you how the U. S. A. was represented at the International Horticultural Congress in London. Practically every country except the United States has a national horticultural organization, comparable in some respects to the Royal Horticultural Society, with which you are surely familiar. This country had none. When the "call" went out for representatives and delegates to the International Horticultural Congress, that for this country was delivered to the agricultural attache at our Embassy in London. It is reported that he referred it to his home office and attended some preliminary meetings in London. The matter was referred to the U. S. Department of Agriculture in Washington and there sat, apparently, for months. In the interim, private communications were flying across the Atlantic in both directions between interested horticulturists and finally the matter was referred by the Honorable Secretary of the Congress direct to American societies of horticulture to ensure that American views and interests would be adequately represented. It boiled down to the United States being represented by those persons who were going to be at the Congress anyway and by men who were not specialists in nomenclatural matters. Appraised of their lot, these persons made every effort to be briefed and informed on as many aspects of the provisions of the proposed Code as possible. As stated later by Dr. S. L. Emsweller (a member of the committee, representing U. S. D. A. and the American Society for Horticultural Science), this situation brought into sharp focus the need in this country for a single horticultural organization of organizations that could serve as authorized in matters at the international level. The American Horticultural Council, to which the Northern Nut Growers Association belongs may become that organization, but only when authorized by its membership.

Many readers may be familiar with earlier codes, as that adopted by the American Pomological Society (which dates from 1847), that by the American Society of Agronomists (formulated in 1917-18), and with a third code adopted at the sessions of the Botanical Congress meeting in Cambridge, England, in 1935. Knowing of the provisions of these codes, you may ask, "What has the new one got that is different?" There are many new features of which the more significant are given below.

Perhaps foremost is the recognition that, for the most part, the so-called varieties of garden plants are not uniform in their behavior. Some are nothing more than transplants of variations found in the wild, such as the Japanese *Juglans Sieboldiana* var. *cordiformis*, a population having its own geographic range and distinguished from the typical element of the species by several morphological characters. It is a botanical variety that is cultivated. It is not a product of domestication.

The code distinguishes from this botanical variety those so-called varieties that are the result of domestication, variants that have been produced in cultivation but are not known to occur and perpetuate themselves in the wild, such as Schwedler's maple known as *Acer Platanoides* var. *Schwedleri*. Plants of this group, that may be grown from seed and which do show a limited variability, are distinguished from botanical varieties by placing them in a new category called cultivar (a name coined long ago by L. H. Bailey and meaning, a variety from cultivation). The abbreviation for the category is cv. Furthermore, in an effort to differentiate cultivar names from botanical names, it is provided that they be treated as are vernacular or fancy (common) names. That is, that the name be placed in single quotes and not italicized e.g., *Acer Platanoides* cv. 'Schwedleri'.

A third category is that of clone. A clone is an individual propagated not from seeds but by asexual means, as by grafting, budding, cuttings, etc. Most so-called varieties of nuts are clones. A clone may be selected from a species population, from a botanical variety, from a cultivar, or from anyone of several types of hybrid complexes. It may appear as a mutant of another clone. The name for it may, where there is need for precision, be attached to the name of the species (or hybrid) from which it was selected, as *Corylus Avellana* cl. 'DuChilly' or reduced to *Corylus* cl. 'DuChilly.'

For the hybridizer naming and introducing new hybrids to cultivation, the Code is more helpful than any previous set of rules and the needs of hybridizers of various groups have been considered. Many examples illustrating application of each provision are given in the unabridged version.

The person naming plants will find much helpful guidance with regard to the selection of names which should be considered if international usage is to prevail. The Code is just that, a set of dicta provided for guidance by horticulturists throughout the world that there may prevail

a greater uniformity, accuracy, and fixity of names,
a lessening of procedures that would lead to confusion and error if adequately supported or widely adopted, and
a provision for change and revision.

One section of the code ("C") dealing with Registration might well come within the framework of interest and activity of the Northern Nut Growers Association. This section, which suffers materially by condensation in the abbreviated text that follows, occupies nearly a page in the unabridged edition. It envisages the establishment of an international registering body, with headquarters for different groups located in different countries, e.g., that for tulips in the Netherlands, for rhododendrons in Britain, for roses in the United States, etc. The task of compiling, maintaining, and publishing such a registrar (and rejecting names not in conformance with the Code) will fall in many cases on the special plant societies concerned. When societies for a given group of plants exist in 2 or more countries, they will be expected to collaborate. Insofar as I know the Northern Nut Growers Association has not set up any mechanism for the registration of names given to cultivars, hybrids, and clones of nuts. To do so would be to perform a very real service for your membership, for the industry in this country, and would place the Association in a key spot when the proposal for an international registry is activated. The agitation for this phase of international application of the Code is considerable and is more evident in Europe than here. If the Association takes an active stand in the matter and develops a center of registry of nut names for this continent, it may very well display a quality of initiative and service that will make it pre-eminent on the international level and will cause others to look to it for guidance, information, and leadership.

A careful consideration of this Code is commended to all and those interested in the topic are urged to procure a copy of the booklet giving the unabridged edition and Stearn's excellent historical account of the subject.

The New Code For the Naming of Cultivated Plants[7]

[7] Reproduced with permission from Journal of the Royal Horticultural Society, vol. 69, pp. 12-21, 1954.

J. S. L. GILMOUR, *Chairman of the International Committee on Horticultural Nomenclature and Registration*

Anyone who deals in any way with cultivated plants uses plant names. This glimpse of the obvious ought to mean that the appearance of a new Code for the naming of cultivated plants should be a memorable event for all gardeners. I say "ought to mean" advisedly, because there is no doubt that, in the past, Codes of Nomenclature have made little appeal to gardeners, the great majority of whom have been blissfully unaware of their existence. As a consequence, many horticulturists—on hearing of "a new Code"—will, no doubt, raise a respectful (or contemptuous) eyebrow and get on with reading their latest catalogue. The aim of this article is to persuade readers of the Society's JOURNAL, not only that this attitude is against their own interests, but that a good deal of quiet entertainment can be extracted from trying to use plant names correctly—if only the entertainment of putting their neighbour right!

It is true that, hitherto, there has been ample excuse for ignorance. Although Codes for the naming of cultivated plants have existed, in some shape or form, for nearly 100 years, they have been printed in obscure publications, quite inaccessible to ordinary mortals. This excuse no longer holds. The full text, plus a summary, of the new Code can be purchased for one shilling and three pence (post free) from the Society, and, in addition, reference to it has been made in many horticultural and other periodicals. What, then, is the new Code about, how does it affect gardeners, and what should they do about it?

In the first place it is *not* about the Latin names of wild species or varieties—names like *Rhododendron ponticum* and *Aesculus octandra* var. *vestita*. Whether wild plants are grown in gardens or not, their Latin names are governed by quite a separate Code, devised by botanists for the purpose. It is true, of course, that gardeners have to use Latin names for many of the wild plants that they grow, and equally true, unfortunately, that the frequent changes in these names are a source of annoyance to everyone; but that is quite another story, with which we are not concerned here. At the last International Horticultural Congress steps were taken which we all hope will result, with the cooperation of botanists, in the stabilization of widely used Latin names of cultivated plants.

No, the new Code deals, not with wild species, varieties and hybrids, but with what are commonly called "garden varieties"—namely, forms which have been brought into existence by selection, hybridization, or other similar processes devised by man, and are maintained in cultivation as clones or pure lines by man's care and skill—such plants as Rose 'Peace,' Apple 'Beauty of Bath,' and thousands of others. The distinction between "wild" plants and "garden varieties" is not absolutely clear cut, and in the Code a closer definition is attempted—but for our present purpose the difference is obvious enough.

In the Code, it is recommended that "garden varieties" should technically be called "cultivars." This has been their official name for many years and it is clearly desirable, if the two categories of "wild" and "cultivated" varieties are to be recognized, to have a short and internationally current word for each of them. "Variety" and "cultivar" serve this purpose admirably, but it is not to be expected that all gardeners will make the distinction and adopt the word "cultivar" in ordinary parlance, at any rate immediately. Personally, however, I hope and believe that eventually "cultivar" will find favour. It is a clear and easily understood word and will, I think, prove useful to those gardeners who care for accuracy and precision in their craft, and especially to those who have dealings with fellow-gardeners in other countries.

The Code, then, deals with the *names of cultivars*. It may be helpful, I think, to consider its rules and regulations under three headings: firstly, those of interest to all gardeners who have occasion to write the names of cultivars; secondly, those which are concerned with the coining of new names; and thirdly, those more technical provisions which are of interest primarily to horticulturists studying a particular group and trying to establish what are the correct names for its cultivars.

1. The Writing of Cultivar-Names

The most important point, perhaps, concerning the names of cultivars is that they should not be in Latin, but in any modern language using the so-called Roman alphabet (*i.e.* the alphabet in which English, French, German, etc., are written). The reason for this is, of course, to distinguish, at a glance, names of cultivars from names of wild varieties, which are in Latin. In the future, Latin names for cultivars will definitely not be allowed by the Code, but we are faced with the *fait accompli* of hundreds of existing Latin names which have been widely used for many years. For example: the dwarf conifers abound with cultivars called 'nana,' 'prostrata,' 'compacta' and the like, and such names as 'albus,' 'variegatus,' and 'plenus' occur in almost every cultivated genus. It would clearly be foolish to try to alter all these, and the Code accepts such legacies from the past as permanently with us. (As we shall see, however, a distinction is made between the writing of Latin cultivar-names and Latin varietal-names.)

The vast majority, then, of cultivar-names are vernacular or "fancy" names like 'Winston Churchill.' How should the full name of a cultivar be written? The complete and technically correct form, including the Latin name of the species from which the cultivar has been derived, is illustrated in the following example: *Sedum spectabile* Boreau, cv. 'Brilliant,' but I hasten to add that this lengthy designation is for use only on full-dress occasions! In the example, *Sedum* is the name of the genus, *spectabile* the name of the species (technically called the "specific epithet"), Boreau the name of the man who first described the species under that name (technically called "the authority" for the name), cv. the abbreviation of cultivar, and 'Brilliant' the cultivar-name for the particular cultivar concerned. It should be noted (1) that the generic name and specific epithet are printed in italics, (2) that the cultivar-name begins with a capital letter, is printed in ordinary Roman type, and is enclosed in *single* quotation marks, and (3) that there is no "authority" after the cultivar-name. These three points are important, and apply to the writing of all names of cultivars. If we wish to cut down this rather formidable string of words for every-day use, the authority can be omitted in all except very technical writing. Secondly, the abbreviation cv. can normally be left out, as the vernacular form, single quotes, and Roman type of 'Brilliant' indicate quite clearly that it is a cultivar-name, and not the name of a wild variety. In this shortened form, therefore, the name would read simply:

Sedum spectabile 'Brilliant'

and this is the normal method of writing the names of cultivars.

There are, however, two additional refinements that should be mentioned—apart from the special case of cultivars derived from hybrids, which I will deal with later. The first concerns those Latin cultivar-names which are left over from the past. These should be printed in Roman type and enclosed in single quotes to distinguish them from Latin varietal names; thus one would write *Thuja orientalis* 'elegantissima,' where 'elegantissima' is a cultivar-name, but *Aesculus octandra* var. *vestita*, where *vestita* is the name of a wild variety.

The second point refers to the omission of any authority after a cultivar-name. Many cultivars are first described in reports of trials, in catalogues, and other anonymous publications; this makes the quoting of an authority impractical, but there is provision in the Code for writing the raiser's or introducer's name in brackets after the cultivar-name if so desired, thus: *Weigela* 'Avalanche' (Lemoine).

2. Naming New Cultivars

General

Let us suppose that a nurseryman, park superintendent, or amateur gardener has just flowered a batch of seedlings of, say, *Helenium*, and that he spots one as being of a new type and worthy of propagation. In due course he shows the plant at a fortnightly show, under a number, and an Award of Merit is given to it. He must now find a cultivar-name for his new plant. His first problem, of course, is to choose a name that has not been used before in the genus *Helenium*. If he picks on a very unusual personal name he can be fairly certain that he is the first to use it. If, however, he prefers a more general name, like 'Innocence' or 'Venus,' there is a danger that it has been used before. If there existed a registration authority for *Helenium* names, as there does, for example, for names of Daffodils, he could, of course, consult this authority, but in its absence he must do his best to comb the likely literature—for example the Index to this JOURNAL, nurserymen's catalogues, etc.—and to assure himself that his chosen name has not already been used. His next step is to make certain that the name is in accordance with the best practice for coining such names. Here is where the Code will help him. In it he will find (under Section F) a series of rules for his guidance, based on the accumulated experience of horticulturists of many nations. I will not repeat these rules here—they can be read in the Code—but perhaps the most important, apart from the rule already quoted that the name must not be in Latin, is that it should be short (not more than two words), should avoid forms of address liable to be confused (*e.g.* Mr., Mrs. and Miss) and, as far as possible, should be easily pronounceable by all nationalities. As the Code says, 'Centenaire de Rozain-Bourcharlat,' 'Diplomagartenbauinspektor,' and 'Eldwyth Cholmondeley' are not looked upon with favour as cultivar-names! Having chosen a name, it is essential to ensure that it is published, together with an adequate description, since the Code does not recognize names that are not published, or are published with no description. The Code lays down what is meant by publication, and by adequate description. The names of plants which receive an Award at Vincent Square are automatically published, with a description, in this JOURNAL, but for other methods of publication the Code should be consulted.

Hybrids

The naming of cultivars derived from crosses between two or more different species, belonging to the same or different genera, involves rather special problems. By "derived from" I mean not only the *first* generation from a species cross, but all subsequent generations and back-crosses with the original parents or with members of the first or later generations. Any cultivar which is the progeny of a species cross, however remote, comes, for the purpose of the Code, under the heading of a hybrid and its naming is subject to definite rules. The full name of a "hybrid cultivar" must be regarded as consisting of three distinct parts: (1) the name of the genus (or "hybrid genus" if a hybrid between two or more genera is concerned); (2) a "collective" name or phrase covering *all* the progeny resulting from the particular species-cross concerned; and (3) a cultivar-name for the particular form (cultivar) under consideration. In the name *Viburnum* x *bodnantense* 'Dawn,' *Viburnum* is the generic name, x *bodnantense* is the collective name for all progeny of the cross *V. fragans* x *V. grandiflorum*, and 'Dawn' is the cultivar-name for a particular seedling of this parentage. It is essential always to bear in mind these three distinct parts of the name of a garden hybrid, even if, as it often done, one or other of the parts is omitted in actual usage; the three parts broadly correspond, of course, to the generic name, specific epithet, and cultivar-name of a non-hybrid cultivar (see above).

Let us consider these three parts in turn and see what types of name can be used in each part in actual practice—and how this affects the naming of new "hybrid cultivars."

The *first part*, the generic name, presents no difficulties, except in the case of new hybrids between two or more genera. Names of such "hybrid genera" are usually "manufactured" from a combination of the names of the parent genera (*e.g.* x *Heucherella*, from *Heuchera* and *Tiarella*); in the case of hybrids between more than two genera, however, where a "combination" name would be unwieldy, it is permissible to make a new name by adding the termination *ara* to the name of a person connected with the plant concerned (*e.g.* x *Sanderara* for a tri-generic orchid hybrid). Before making a new "hybrid generic" name, a botanist should be consulted, as a Latin description in proper, botanical form must be provided. It will have been noticed that a multiplication sign is placed *in front of* the names of "hybrid genera," but *after* the generic name in the case of hybrids within a single genus; further details on this point are given later.

The *second part* of the full name of a hybrid is a more tricky business. It is, as I have said, a collective designation for all the progeny of the particular cross concerned, and it may take one or all of three possible forms:—

(1) If the parentage is known, a bare *formula* consisting of the names of the parents, in alphabetical order, connected with a multiplication sign, *e.g.* *Lewisia cotyledon* x *rediviva*.

(2) A name in Latin form (corresponding to a non-hybrid specific epithet), preceded by a multiplication sign, *e.g.* *Viburnum* x *bodnantense*.

(3) A vernacular phrase containing the word "hybrid," "cross," "grex" (Latin for flock or group), or other similar word, making evident the

collective nature of the phrase, *e.g. Lilium* 'Bellingham Hybrids.'

Why do we have to have these three different forms for the second part of the full name, and when, to put it briefly, should we use which?

I must admit that practice on this particular point is as yet by no means fixed, and no doubt it will be modified considerably during the next few years, as more horticulturists try to apply the Code to the groups in which they are interested.

In order to understand the alternatives available, it will be helpful, I think, to consider first the *third part* of the full name, as the use of this, and of the second part, are very closely connected. The third part, as I have said, is a cultivar-name of a particular form of the hybrid concerned, and, in the first place, it is extremely important to realize that *every cultivated hybrid that is considered worth naming at all should receive a cultivar-name from the outset of its "career," even if, at the time, only one form is known, or is considered worth naming.* To take an actual example, at the time when the new hybrid *Viburnum* x *bodnantense* was described in the *Botanical Magazine*, only one form, or clone, of it had been "put into circulation," and yet that single clone was given a cultivar-name, 'Dawn.' The object of doing this was to be able, in the future, to refer by name to this particular clone and so avoid confusion with any later, and possibly inferior, forms of the same cross that might be produced. In the absence of the name 'Dawn,' less desirable clones could legitimately be passed off under the collective name *V.* x *bodnantense* without the acquirer realising that he was not getting the original and superior form.

Bearing in mind, then, the principle that *all* cultivated hybrids should have a third part, cultivar-name, we can turn again to the use of the various possible designations for the second part listed above. The choice of designations depends, broadly speaking, on *convenience*, that is to say, on what, in practice, will be found the most useful method of referring to the plants concerned. In principle, of course, every hybrid whose parentage is known has an appropriate formula-designation, and in certain cases such a formula would be sufficient as a second part designation without inventing either a Latin name or a vernacular collective name. For instance, when there are only a very few forms of a particular cross, it might not be considered necessary to be able to refer to the forms collectively, and a second part name would not then be given. For example, *Rubus* 'Merton Thornless,' when raised, was the only form of the hybrid *R. rusticanus inermis* x *thyrsiger* to be put into circulation, and it was not given a second part designation other than the appropriate formula. On the whole, however, it is usually desirable to have a second part, collective designation, rather than a formula only. Whether such designation should be a Latin name or a vernacular phrase, or both, depends on a number of factors which, as I have said, are not yet fully worked out or appreciated. Broadly speaking, if there are a large number of cultivars of a particular cross and these cultivars form a well-marked group, distinct from other hybrid groups in the genus, it is useful to have a vernacular designation for general use, *e.g.* 'Bellingham Hybrids' for all cultivars of the cross *Lilium Humboldtii* x *pardalinum*. This purpose can, however, be equally well served by a Latin name, *e.g.* *Camellia* x *Williamsii* for all cultivars of the cross *C. japonica* x *saluenensis*. Whichever method is chosen, two points are of great importance. Firstly, if a Latin name is given, it *must* be accompanied by a Latin description. Secondly a vernacular designation *must* contain some such word as Hybrids, Crosses, or the like, to distinguish it from a cultivar-name. The practice, in some groups, of giving "straight" cultivar-names as second part collective designations, followed by "var. so-and-so," is condemned by the Code. Existing names of this kind, however, can easily be brought into line by adding a word such as Hybrids or Grex to the old, second part, cultivar-name, and omitting the "var." thus, *Cattleya* 'Fabia' var. 'Prince of Wales,' would become *C.* ('Fabia Grex') 'Prince of Wales' (Grex can he abbreviated to G. if desired). This alteration may seem over-pedantic, but if, in the naming of cultivated hybrids, a clear distinction is not kept between second and third part names, confusion, as I have pointed out, is likely to result.

I will finish this section on the naming of hybrids with one or two additional points on the correct method of writing their names. Here are the full names of three cultivated hybrids:—

Rubus (*rusticanus inermis* X *thyrsiger*) 'Merton Thornless.' *Camellia* X *Williamsii* 'Donation.' Rose (Hybrid Tea) 'Richmond.'

The following points should be noted:—

(*a*) If a formula or a vernacular designation is used as a second part name, it should be placed in brackets between the generic name and the cultivar-name.

(*b*) If a Latin name is used as a second part name, and the hybrid is between plants belonging to the same genus, a multiplication sign should be placed between it and the generic name.

(*c*) The third part cultivar-name follows directly on the second part name and is placed in single quotes, as for cultivar-names of non-hybrids.

In many contexts it would not be necessary to write the second part name at all, but if it is omitted, the cultivar-name, strictly speaking, should be preceded by a multiplication sign, thus: *Camellia* X 'Donation,' but it would be no great crime to omit it, except perhaps in technical publications.

3. What Is the "Correct" Name for a Cultivar?

It is a painfully familiar fact that many cultivars are known by more than one name, and that many cultivar-names have been applied to more than one cultivar—although the position is not so bad as it is in the case of botanical names! This multiplication of names is the inevitable result of many people naming many plants over a period of many years. It is a situation which we must accept and do our best to mitigate. The Code has a number of necessarily rather complicated provisions aiming at selecting the correct name for any cultivar. These provisions are important mainly to the comparatively few horticulturists and botanists who take on the unenviable job of sorting out the nomenclature of cultivated plants, though the results of their labours affect us all. The rules are set out fully in the Code, and here I will attempt only to pick out one or two of the more important.

The Code lays down two basic principles; though, as we shall see, these may be modified if undesirable name-changing can be avoided thereby.

(*a*) No name can be used if it does not conform to the various rules laid down in the Code (including proper publication of the name, with a description).

(*b*) If there are two or more names that conform with the Code, then the *earliest* name is the correct one; this is known technically as the principle of priority.

Now, obviously, as happens with botanical names, the strict application of priority might mean that a universally used name—say, Plum 'Victoria'—would have to be given up and replaced by a completely unknown one. This is unthinkable, and, as in the Botanical Code, there are clauses to prevent it happening. With cultivar-names, however, there is a particular complication, which does not apply to botanical names. The latter are all in Latin, whereas cultivar-names may be in many languages. This means that a cultivar, raised in one country under a name acceptable in that country, may be introduced into another country where the original name is quite unpronounceable or otherwise unsuitable. A new name is, of course, immediately invented by the introducer, and clearly, in many cases, it is useless to try to make the second country adopt the earlier, strictly correct, but unsuitable name. The Code, therefore, allows the retention of the second name as what it calls a "commercial synonym." Thus, Rose 'Permanent Wave' is a commercial synonym in the United States for the Rose raised in Holland as 'Mevrouw van Straaten van Nes.' In any formal list of Roses, both names should be given, together with any other commercial synonyms that may exist. The coining of commercial synonyms is not, of course, to be encouraged, and should only be done if the original name is clearly unsuitable for the new country. Frequently names are translated or transliterated when a cultivar is introduced into another country, and such a translation or transliteration is not regarded in the Code as a *new* name, but as the original name in another form; no difficulty, therefore, arises as to priority in these particular cases.

Perhaps the most important section of the whole Code deals with the *Registration* of cultivar-names. In certain groups (*e.g.* Daffodils) international registration schemes already exist, and it is urged that further schemes, covering all important groups of cultivated plants, should be established as soon as possible. The function of such authorities would consist, primarily, of (1) registering new names and ensuring that they are in accordance with the Code, and (2) preparing, and keeping up to date, lists of cultivars in their groups. In addition, the authorities would choose a particular publication as the "starting-point" of the nomenclature in the group (so as to avoid dipping too deeply into the past in search of ever earlier cultivar-names), and would act as arbiters when a decision has to be made between two or more widely used names for the same cultivar. There is no doubt in the minds of those responsible for the Code that the existence of internationally trusted and respected registration authorities would do more than anything else to stabilize and simplify the naming of cultivated plants. It will obviously take some time before authorities can be set up for all—or even the majority—of important groups, but the International Committee is doing its best to push ahead quickly with this very important side of their work.

At the end of the Code there are sections dealing with certain special categories such as re-selected and improved cultivars, con-varieties, clones, apomicts and line-hybrids, which are of interest mainly to specialists in breeding and taxonomic work on cultivated plants.

I hope I have said enough, in this short article, to convince readers of the JOURNAL that the Code is of some interest to them if they want to use the names of plants so that other gardeners, both here and abroad can understand what they mean. The next step is to read the Code itself—first the Summary and then the full text that follows it. No doubt many who do so will be put off at first by the somewhat legal language used. One can only repeat the time-honoured defence by lawyers when similarly attacked —that to avoid ambiguity experience has shown that a certain amount of jargon is necessary! We have tried hard, in the Code, to keep it to a minimum.

The International Committee is anxious to receive suggestions for improving the Code, so that they can be discussed at the next Horticultural Congress. All such suggestions should be sent to the Secretary of the Committee (DR. H. R. FLETCHER), c/o The Royal Horticultural Society, Vincent Square, London, S.W.1.

In the Historical Introduction to the Code (written by MR. W. T. STEARN, Secretary of the International Committee, during the production of the Code), Fellows of The Royal Horticultural Society will note, I trust with pride, the important part played by their Society, in cooperation with many other bodies and individuals in many parts of the world, in the preparation, drafting and publication of the Code. I hope that they will feel it their not unpleasant duty to make themselves and others familiar with the provisions of the Code, to follow its rules and recommendations when they use the names of cultivated plants, and to let the Secretary of the Committee know how they think it might be improved. Article I of the Code states that its aim is "to promote uniformity, accuracy and fixity … with the minimum disturbance of existing nomenclature"—an aim surely close to the heart of every Fellow of The Royal Horticultural Society.

I am grateful to MR. A. SIMMONDS, MR. W. T. STEARN AND MR. P. M. SYNGE for help in the preparation of this article.

[Illustration: Exhibit at the Harvest Show of the Massachusetts Horticultural Society—Boston, Mass. October, 1953.]

Attendance Register, Rochester, N. Y., 1953

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